

Heads in the Sand: Theory and Experiment on the Interdependence of Information Avoidance

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Abstract

People sometimes prefer not to learn the truth, even when information is readily available, simple to process, and allows better decision-making. While information avoidance has been studied in individual settings, its interaction within groups remains poorly understood. We study whether information avoidance is interdependent in settings where one person's ignorance imposes costs on others. We develop a model of anticipatory utility induced by information, predicting heterogeneous strategic responses to others' ignorance. We test this prediction in a novel experiment where group members decide whether to avoid information about a potentially unpleasant event, at the cost of increasing the event's severity for all group members. We find that most individuals fall into four strategic types: 42% unconditionally acquire information, 21% exhibit strategic complementarity, 19% strategic substitutability, and 9% unconditionally avoid. Exogenously increasing the expected prevalence of avoidance among others increases own avoidance on average, driven by Strategic Complements. Mediation analysis shows that higher expectations about others' avoidance worsen anticipatory utility upon learning bad news, but not good news. Simulations varying the mix of types reveal sharp threshold effects: small changes in group composition can induce large equilibrium responses towards collective ignorance. Our findings highlight that information avoidance can exhibit contagion in groups, with implications for designing information acquisition incentives that account for social interdependence and tipping points.

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1 Introduction

Information avoidance—the deliberate choice to avoid readily available and instrumentally valuable information—is well documented in individual decision-making (Thornton, 2008; Ho et al., 2021).¹ For instance, people often choose not to learn whether they carry a potentially fatal disease, despite the availability of accurate tests (Oster et al., 2013) and investors have been found to behave like “ostriches”, avoiding financial news (Galai & Sade, 2006; Karlsson et al., 2009; Fong & Hunter, 2022; Olafsson & Pagel, 2025), even though learning this information would enable them to make better decisions and achieve better outcomes. In many situations, however, individual payoffs are interlinked, such that avoidance of decision-relevant information by one agent imposes negative externalities on other agents’ material payoffs. Examples include global warming, infectious diseases, or a company’s success (or failure). Consider, for instance, a bank investing in assets that may be either of high quality or junk. Each employee understands that the bank’s future depends not only on their own decisions but also on the choices of others—whether his peers stick to the current strategy (optimal if the assets are of high quality) or liquidate the assets (optimal if they are junk). Making the right decision, however, requires assessing the quality of the assets and potentially acknowledging that the bank may be on the brink of bankruptcy (and that all employees might be facing imminent unemployment). Importantly, if the assets turn out to be junk, the severity of the situation depends on the choices of others. If others continue as usual rather than taking action, the outcome will be much worse. Therefore, each employee’s decision to bury his head in the sand or not impacts not only his own payoff but also everyone else’s.

This paper studies information avoidance in a group setting where remaining ignorant imposes a negative payoff externality on other members. People choose to remain ignorant or acquire information (and act on it) in an environment where, as in the example above, their decision impacts both their own payoff and the payoffs of others. When more individuals remain ignorant, the outcome in the bad state of the world becomes worse for everyone. In this context, a key question arises that set it apart from information acquisition in individual decision-making. Are there interdependencies in information avoidance? In other words, do individuals base their decisions to acquire or avoid information on the choices of others in their group, and if so, how?

These questions are crucial to understand whether and how information avoidance self-sustains within organizations and groups. In a seminal theoretical paper, Bénabou (2013) describes what he refers to as contagious ignorance—a situation in which individuals become more likely to avoid information if others are doing the same. Intuitively, if everyone else chooses to bury their head in

¹For a comprehensive review on information avoidance, see Golman et al. (2017). Hertwig & Engel (2016) define the related concept of *deliberate ignorance* as “the conscious individual or collective choice not to seek or use information”, specifically “in situations where the marginal acquisition costs are negligible and the potential benefits potentially large, such that—from the perspective of the economics of information (Stigler, 1961)—acquiring information would seem to be rational (Martinelli, 2006)”.

the sand, the severity of the bad state will be so overwhelming that individuals may prefer to stay unaware rather than confront such a potentially disastrous reality. History is filled with examples of collective information avoidance that appear consistent with the notion of contagious ignorance, from Easter Island to Lehman Brothers. However, the existing empirical literature lacks a systematic study of whether and how information avoidance self-reinforces in groups. This work aims to fill this gap.

This paper answers these questions by combining theory, experiment, and simulations. Our investigation begins with a theoretical model inspired by Bénabou (2013), which studies the strategic interdependencies that may arise in group settings. As well as contagious ignorance, we show that in our setup there is also an alternative possibility. Depending on the nature of their utility function, individuals may exhibit strategic substitutability in information avoidance: when others bury their heads in the sand, these individuals become less inclined to shun information. These findings indicate that the dynamics of information avoidance in groups may vary considerably. Furthermore, the theory also shows that some individuals may not base their choices on the actions of others at all, and may instead always avoid or always acquire information, regardless of what others do.

Informed by this theory, we design a novel lab experiment to assess the interdependence of information avoidance empirically.² Participants are matched in groups and informed that there are two possible states of the world: *Good* and *Bad*. In the *Bad* state, everyone in the group will be exposed to an aversive future event: at the end of the experiment, they will hear a series of distressing and unpredictable screams, a consumption event known to produce anticipatory anxiety.³ In the *Good* state, no noises will be heard. At the beginning of the experiment, each participant chooses whether to discover the state immediately (“Now” option) or to wait until the future event unfolds (“Later” option). Importantly, the volume of the screams in the *Bad* state decreases in the number of members who acquire information about the state (that is, who choose Now), reflecting the decision value of information. In other words, each participant understands that by selecting Now, they can mitigate the intensity of the noises not only for themselves but also for everyone else in their group, should the state turn out to be *Bad*.

To assess the interdependence, we aim to characterize how one’s demand for information depends on others’ information decisions. For this, we employ two empirical strategies. First, using an information provision experiment, we exogenously vary subjects’ expectations about the prevalence of information avoiders (people who choose Later) in their group by informing them about an analogous share observed in a pilot study. We randomize two treatments at the level of experimental groups: in the *Many Avoided* condition, subjects are told that 80% of subjects in a prior

²The experiment was preregistered, AsPredicted #153871 and #175588.

³This is shown in [Beaurenaut et al. \(2020\)](#). We further discuss the rationales for the use of screams as a consumption event in Section 4.2.

group avoided information, whereas in the *Few Avoided* condition, subjects are told only 20% did. The resulting variation in posterior beliefs allows us to test whether, on average, a higher expected prevalence of avoidance among peers increases the likelihood of own's avoidance. Second, using the strategy method (Selten, 1967; Fischbacher et al., 2001), we elicit each subject's best-response *schedule*—his conditional response to a range of shares of avoiders within the group. This allows us to identify individual heterogeneity in interdependence beyond the average effect.

We find evidence that, on average, information avoidance is interdependent, and in particular, contagious. Our treatment intervention strongly shifts participants' expectations about their peers' information decisions: beliefs about the share of avoiders increase by 22 percentage points on average, from 37% in *Few Avoided* to 59% in *Many Avoided*. The treatment also affects behavior: consistent with contagious ignorance, the share of subjects avoiding information increases by one-third, from 24% to 33%. 2SLS regressions further confirm that the average relationship between beliefs and behavior (among subjects who respond to the treatment) is positive: for every 10 percentage points of increase in beliefs about the prevalence of avoidance among others, the likelihood of avoidance increases by about 5 percentage points. Thus, on average, information avoidance exhibits strategic complementarity, in line with the prediction of Bénabou (2013).

We provide supportive evidence that a key mechanism driving the interdependence of information choices is a deterioration of anticipatory utility upon finding bad news in response to exogenous changes in expectations about the prevalence of avoidance. To show this, we develop and validate a novel and portable elicitation of anticipatory utilities, as expected by the subjects at the moment of the choice between acquiring or avoiding information.⁴ Its key characteristic is its granularity: instead of simply eliciting the anticipatory utility upon acquiring information, we further condition on the events of finding out good news and bad news, separately. In line with the theoretical channel, subjects in the *Many Avoided* condition (relative to those in *Few Avoided*) report worse anticipatory utility when imagining the event of acquiring information and discovering *bad* news about the state, but we find no differences across treatments in the anticipatory utility upon discovering *good* news. This pattern lends support for the focal mechanism of this paper and, in contrast, is hard to square with more traditional channels such as conformity, herding, and social norms, which predict that belief about others' behavior—shifted by the treatment—affects the utility associated to acquiring information, without distinguishing between the events of discovering bad versus good news.

Importantly, beyond the average complementarity, individuals differ substantially in the type of interdependence they exhibit. Consistent with theoretical predictions, we find substantial heterogeneity in how subjects best-respond to varying shares of information avoiders in their group. The key finding is that we document strategic interdependencies in information decisions at the

⁴Note that it is the expectation at that time, and not the actual anticipatory utilities, what matter for the information decisions.

individual level. Among those influenced by others' choices (about 40% of the sample), half exhibit a pattern consistent with contagious ignorance (strategic complementarity), as in [Bénabou \(2013\)](#), while an equally sizeable share display the opposite tendency (strategic substitutability), imposing limits to the spread of ignorance. There is also a substantial share of participants whose information choice is independent of their teammates' decisions. The larger portion consists of those who choose to unconditionally acquire information (approximately 42%), while just under 9% of participants consistently choose to avoid information. Therefore, we document that, empirically, individuals exhibit substantial variation in their best-response types.

The diversity of such types of best-response schedules suggests that their composition within a group matters for the equilibrium prevalence of avoidance. Motivated by this intuition, we employ simulations to show how the interaction of strategy types determines the equilibrium prevalence of information avoidance in a group. Groups primarily composed of individuals who exhibit strategic complementarity in information decisions generate multiple equilibria, and risk getting trapped in a so-called *Mutually Assured Ignorance* (MAI) equilibrium (see Section 3), where everybody prefers to remain ignorant, at the cost of worsening the future. The inclusion of unconditional avoiders precipitates this outcome: with 30% unconditional avoiders, the MAI equilibrium arises with 75% probability, and with 50% it emerges with certainty. Likewise, unconditional information getters can drive the group toward a virtuous equilibrium where everybody acquires information (*Mutually Assured Awareness* equilibrium). In contrast, groups composed of individuals who exhibit strategic substitutability never reach a MAI equilibrium but are also unable to sustain an equilibrium characterized by universal information acquisition.

Furthermore, the relationship between group composition and the equilibrium prevalence of information avoidance is monotonic and nonlinear. For example, when the participation of Always Avoiders increases from 0% to 10% in a group conformed otherwise by Strategic Complements, the median equilibrium share of information avoiders increases from 0% to 22%. However, when the participation increases by a similar amount from 10% to 20%, the median jumps sharply from 22% to 100%. Similar nonlinear dynamics are observed for other quantiles and for the average of the distribution of equilibria.

The remainder of the paper is organized as follows. In Section 2, we discuss related literature. In Section 3, we lay out our theoretical framework. In Section 4, we present the experimental design. In Section 5, we report the empirical findings. In Section 6, we use our data to inform the relationship between group compositions and equilibrium levels of avoidance. Finally, in Section 7, we conclude.

2 Related Literature

Our study relates mainly to two strands of literature: attitudes towards information and suboptimal decision-making in groups.

First, we contribute to the literature on attitudes towards information from both theoretical and empirical perspectives. On the theoretical side, standard economic theory recognizes that information is a valuable resource for decision-making and accordingly predicts that, if available at no cost, agents will typically acquire information about payoff-relevant states of the world to improve their subsequent decisions. However, a growing literature documents and acknowledges the possibility that economic agents may have intrinsic preferences for information beyond and above the instrumental, decision-value of information. Starting from [Kreps & Porteus \(1978\)](#), the theoretical literature has generated a variety of models in which agents may prefer to remain uninformed, or delay the acquisition of information, despite information being available at no or low cost. [Golman et al. \(2017\)](#) provides a review of models of information avoidance, defined as the deliberate choice to avoid freely available information when the agent is aware of its existence. [Bénabou \(2013\)](#), the model closest to our work, extends the analysis of information avoidance decisions to group settings. It shows theoretically that when willful blindness to information makes other agents worse off, it can become contagious. This rises the possibility of multiple equilibria, including a pervasive one in which all agents remain ignorant. We contribute to this literature by extending [Bénabou \(2013\)](#) to show that, under a broader class of preferences, strategic substitutability can also arise in that setting. In addition, we provide an empirical test of the prediction of interdependence.

On the empirical side, several studies both in the lab and in the field documented that individuals often prefer to remain ignorant despite information being available at no or low cost and being useful for subsequent decision-making (e.g., [Dana et al., 2007](#); [Karlsson et al., 2009](#); [Oster et al., 2013](#); [Ganguly & Tasoff, 2017](#); [Pagel, 2018](#); [Ho et al., 2021](#); [Meissner & Pfeiffer, 2022](#); [Masatlioglu et al., 2023](#); [Momsen & Ohndorf, 2023](#); [Engelmann et al., 2024](#); [Falk & Zimmermann, 2024](#)). Across studies and contexts, rates of avoidance range from 5% ([Ganguly & Tasoff, 2017](#)) to more than 90% ([Oster et al., 2013](#)). The vast majority of empirical studies have focused on information avoidance in individual decision-making settings. Most closely related to our paper, [Falk & Zimmermann \(2024\)](#) studies intrinsic preferences for information about a potentially adverse future event in an individual setting. In their experiment, information confers no instrumental value: the decision-maker's future payoff is independent of whether information is acquired. We build on their experimental design to study information avoidance decisions in groups. Departing from them, we introduce a positive private return and externalities to information acquisition: becoming informed benefits the decision-maker and other members in her group. Therefore, we contribute to this literature in several ways. First, we document that, even in cases where information avoidance makes others worse off (negative payoff externalities), a substantial proportion of subjects still prefers to remain

ignorant. Second, we provide, to the best of our knowledge, the first test of the interdependence of information decisions. Third, we document substantial heterogeneity of strategy types, and study its implications for the equilibrium levels of avoidance in groups. Fourth, we document how social preferences correlate with information decisions, which affect others' wellbeing.

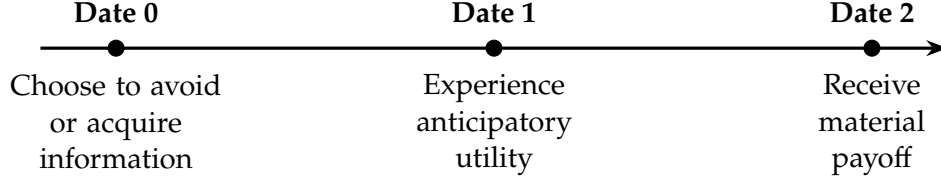
Second, our study also relates to the literature on non-Bayesian information processing, overconfidence, and suboptimal decision-making in groups. A recent literature suggests that decisions in groups can become correlated through the social transmission of biases. [Suvorov et al. \(2024\)](#) shows that members of a group share noisy feedback about their own ability to other group members in a selective and asymmetric way: group members are more likely to share positive feedback about their ability to perform in an investment task than negative feedback. This selective information sharing leads to groups becoming overconfident about the group's ability to perform in investment decisions, and this overconfidence leads to suboptimal investment decisions by low ability groups. A closely related study by [Oprea & Yuksel \(2022\)](#) shows that when individuals can socially exchange ego-relevant beliefs, their initial biases amplify because subjects respond to their counterparts' beliefs in an asymmetric way. Such studies have focused on ex-post processing of information, after its content has been revealed to the decision-makers. Complementing these studies, we focus on ex-ante avoidance of information, and show that groups can also attain suboptimal (material) outcomes by deliberately avoiding learning from freely available signals, when signals reveal the state perfectly, and without any communication among group members.

3 Theoretical Framework

We now describe our theoretical framework and key results. All proofs are in the Appendix. Consider the following setup, inspired by [\(Bénabou, 2013\)](#).⁵ There are two states of the world: G (good), realized with probability $q \in (0, 1)$, and B (bad), realized with probability $1 - q$. Material payoff is received at date 2 and generates utility u_i . Letting $U_i := \mathbb{E}[u_i]$, at date 1 agents experience utility $v_i(U_i)$, where $v_i(\cdot)$ is strictly increasing. We can think of v_i as capturing *anticipatory feelings* about future (date 2) utility, or more generally as reflecting preferences over the timing of uncertainty resolution, in the spirit of [Kreps & Porteus \(1978\)](#). At date 0, agents evaluate lotteries over date-1 anticipatory utility according to $\mathbb{E}[v_i(U_i)]$ and choose whether to avoid ($y_i = 1$) or acquire ($y_i = 0$) information that perfectly reveals the state of the world. If they avoid, at date 1 they experience anticipatory feelings over expected date-2 utility, since they are uncertain about which state will realise. If they acquire, the uncertainty is resolved and they experience anticipatory feelings over the good or bad state accordingly. The timeline is summarized below.

Denote by g the utility obtained at date 2 when the state of the world is good, and by $\beta < g$ the utility obtained when the state is bad. A key feature of the model is that individual payoffs

⁵The correspondence between the notation used in our model and [Bénabou \(2013\)](#) is provided in Appendix A.1.



depend not only on one's own action but also on the actions of others, giving rise to externalities among group members. Accordingly, utility in the bad state worsens as more group members avoid information. This reduced-form approach mirrors a natural real-world feature: one cannot respond appropriately to a problem without being aware of its existence.⁶ Suppose the number of agents in a group is N , and let $\lambda_{-i} \equiv \sum_{j \neq i} y_j / (N - 1)$ denote the share of group members other than i who avoid information. Agent i 's utility in the bad state is a function of this share, as follows.

$$\beta = \begin{cases} b(\lambda_{-i}) & \text{if } i \text{ avoids information} \\ b(\lambda_{-i}) + \Delta & \text{if } i \text{ acquires information} \end{cases} \quad (1)$$

with

$$b(\lambda_{-i}) \equiv \bar{b} - \Delta[1 + (N - 1)\lambda_{-i}] \quad (2)$$

where $\bar{b} < g$ is the utility obtained when the state is bad and all group members acquire information (and undertake corrective actions), and Δ captures the *instrumental value* of information: it reflects how much each agent can mitigate the adverse effects of the bad state, which we set to $\Delta = \frac{1}{N}$.⁷ Throughout the analysis, we assume that the expected value of a lottery that delivers $\bar{b}$ with probability $1 - q$ and g with probability q exceeds the value of obtaining $\bar{b} + \Delta$ for sure.⁸ That is,

$$q(g - \bar{b}) > \Delta. \quad (3)$$

This assumption simplifies the exposition but is not necessary for our results. Our equilibrium concept is Nash equilibrium. An individual's *net* expected utility from avoiding information taking

⁶The assumption that informed agents take corrective action while uninformed ones do not is imposed exogenously for simplicity, but would arise endogenously if corrective action is cheap enough for informed agents to act yet costly enough that uninformed agents optimally abstain. In our experiment, removing this choice allows participants to focus on the core decision of whether to acquire or avoid information based on their expectations of how many teammates do so, avoiding potential confounds from adding an extra layer of uncertainty about how many agents act on their information.

⁷Generally, the private return and the externality from acquiring information may differ: $\beta = \bar{b} - \Delta_{\text{private}}y_i - \Delta_{\text{spillover}}(N - 1)\lambda_{-i}$. For simplicity, here and in our experiment we focus on the symmetric case $\Delta_{\text{private}} = \Delta_{\text{spillover}} = 1/N$.

⁸Formally, $(1 - q)\bar{b} + qg > \bar{b} + \Delta$.

everybody else's decisions as given is equal to

$$\begin{aligned} \varphi_i(\lambda_{-i}) &\equiv \mathbb{E}[v_i(U_i)|y_i = 1] - \mathbb{E}[v_i(U_i)|y_i = 0] \\ &= \underbrace{v_i((1-q)b(\lambda_{-i}) + qg)}_{\text{Anticipatory utility when avoiding information}} - \underbrace{[(1-q) \cdot v_i(b(\lambda_{-i}) + \Delta) + q \cdot v_i(g)]}_{\text{Expected anticipatory utility when acquiring information}} \end{aligned} \quad (4)$$

We can now describe i 's *best response function*. This is

$$y_i^*(\lambda_{-i}) = \begin{cases} 1 & \text{if } \varphi_i(\lambda_{-i}) > 0 \\ 0 & \text{if } \varphi_i(\lambda_{-i}) < 0 \end{cases} \quad (5)$$

3.1 Best Response and Strategy Types

We now explore how the nature of v_i affects an individual's best response function. The first two results consider v_i that is everywhere convex or everywhere concave.

Lemma 1 *If v_i is everywhere convex, individual i is an “Always Getter”. At date 0, she acquires information independently of the information acquisition choices of other agents in her group.*

If v_i is convex, the utility boost obtained from discovering that the state is good outweighs the utility drop when the state is bad. Consequently, the agent always prefers to acquire information. Consider now the case where v_i is concave, as in Bénabou (2013).

Lemma 2 (Bénabou, 2013) *If v_i is everywhere concave, individual i (i) is an Always Getter (ii) is an “Always Avoider” or (iii) is affected by the choices of others in her group in the following way: there exists an interior threshold value λ_i^* such that i strictly prefers to avoid information if $\lambda_{-i} > \lambda_i^*$ and strictly prefers to acquire information if $\lambda_{-i} < \lambda_i^*$ (Strategic Complementarity).*

In the Appendix, we characterize the necessary and sufficient conditions for each of (i), (ii) or (iii) to apply. Always Getters and Always Avoiders make the same information acquisition choice independently of the choices of others in their group. Alternatively, the agent may condition her decision on the choices of others in her group, exhibiting strategic complementarity. The intuition for this is as follows. If acquiring information had no instrumental value, the agent would always prefer to avoid it, since the loss in anticipatory utility (distress) from discovering that the state is bad outweighs the gain from discovering the state is good — a direct consequence of the concavity of anticipatory utility. Once instrumental value is introduced, as in our setup, the agent may opt to acquire information. However, as more agents choose to avoid information, the bad-state outcome deteriorates, amplifying the distress associated with learning that the state is bad. Meeting the

conditions for information acquisition therefore becomes increasingly demanding. This may give rise to strategic complementarity: the agent acquires information only if a sufficient share of her peers does so; otherwise, she opts for ignorance.

We now show that strategic substitutability may also arise in our setting. This is a novel result relative to [Bénabou \(2013\)](#): in that paper, when remaining ignorant imposes a cost on others — as in our setup — interdependencies necessarily take the form of strategic complementarity. To see how strategic substitutability may arise, consider a reference-dependent v_i that is largely insensitive to expected date-2 utility except around some reference utility level. Denoting i 's reference expected utility as $\hat{U}_i$, suppose that (i) $v_i''(U) < 0$ for $U > \hat{U}_i$ and (ii) $v_i''(U) > 0$ for $U < \hat{U}_i$.

Lemma 3 (Strategic substitutability) *If v_i is reference dependent, under some conditions the following applies: there exists an interior threshold value λ_i^{**} such that i strictly prefers to avoid information if $\lambda_{-i} < \lambda_i^{**}$ and strictly prefers to acquire information if $\lambda_{-i} > \lambda_i^{**}$ (Strategic Substitutability). The value of λ_i^{**} satisfies $(1 - q)b(\lambda_i^{**}) + qg = \hat{U}_i$, decreasing in $\hat{U}_i$.*

In the Appendix, we provide sufficient conditions for strategic substitutability. To illustrate, suppose that in the bad state utility at date 2 always falls below the reference level, while in the good state it always exceeds it. Consider first a situation in which many agents in i 's group avoid information. Date 2 utility in the bad state is then so low that the expected utility under ignorance also falls below the reference level. In this case, relative to remaining ignorant, learning that the state is bad entails only a modest loss in anticipatory utility at date 1, whereas learning that the state is good yields a substantial gain. Agent i therefore prefers to acquire information. Now consider a situation in which many agents in i 's group acquire information. Date 2 utility in the bad state is no longer so low, and as a result expected utility under ignorance rises above the reference level. Under mild conditions, agent i now prefers to remain ignorant: the gain in date-1 anticipatory utility from learning that the state is good is small relative to the loss from learning that it is bad.⁹

Note that here the locus of reference dependence is *anticipatory* utility rather than *consumption* utility. This represents a meaningful departure from the standard approach. Conventional reference-dependent models — following Kahneman and Tversky (1979) and Kőszegi and Rabin (2006) — apply gain-loss thinking to realized consumption: utility depends on the gap between what an agent consumes and their reference consumption level. Here, by contrast, reference dependence operates one stage earlier. When contemplating what will occur at date 2, people experience anticipatory feelings that depend on how expected date 2 utility fares relative to a reference utility level.

A further relevant observation is that anticipatory utility may be shaped at least partially by social preferences. To fix ideas, at the cost of some slight abuse of notation, we may think of v_i

⁹For example, suppose $b(\lambda_{-i} = 0) = -10$, $b(1) = -90$, $g = 50$, $\Delta = 5$, $p = 0.5$, $\hat{U}_i = 0$, and $v_i(U) = U^{1/3}$ if $U > 0$ and $= -(-U)^{1/3}$ if $U < 0$. Then $\varphi^i(0) \approx 2.58$ and $\varphi^i(1) \approx -2.87$, so the agent exhibits strategic substitutability.

as comprising two components: one reflecting anticipatory feelings over own date-2 consumption utility, and one reflecting altruistic anticipatory feelings over the date-2 consumption utility of others. We explore this extension fully in Appendix A.3, where we show that the effects of altruism depend on the curvature of the altruistic anticipatory utility function. When this function is concave, altruism amplifies strategic complementarity; when linear, it has no effect on interdependence; when convex, it dampens complementarity. We also study the role of reciprocity concerns, showing that agents with stronger reciprocity concerns are more prone to strategic complementarity in their information choices.

The following proposition summarizes our key results in this section.

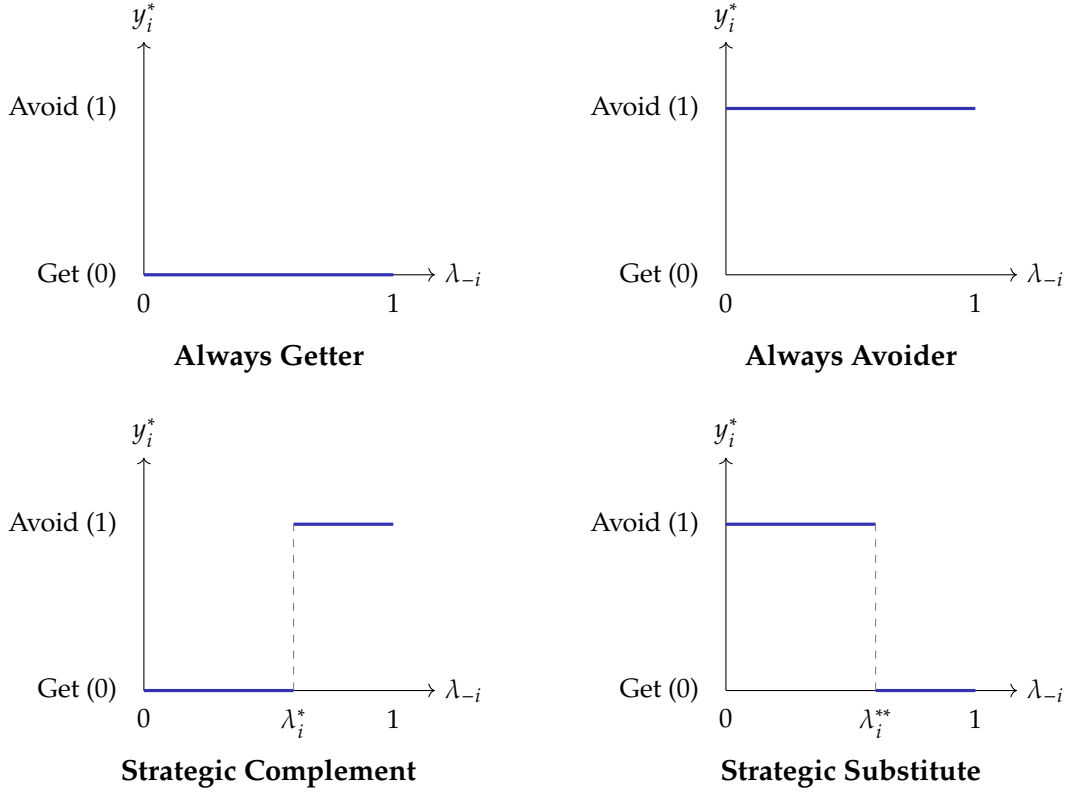
Proposition 1 (Heterogeneity of interdependence) *Depending on the nature of v_i , i 's propensity to avoid information may be (i) independent of, (ii) increasing (strategic complementarity) or (iii) decreasing (strategic substitutability) in the share of group members who avoid information.*

In what follows, we will use the term “strategy type” to indicate whether an individual is an Always Getter, an Always Avoider, or alternatively exhibits preferences featuring strategic complementarity (in short, is a “Complement”) or substitutability (is a “Substitute”). Figure 1 illustrates examples of each strategy type.

Naturally, the specific information avoidance equilibria that may be reached vary depending on the agents' strategy types. Bénabou (2013), for example, examines the risks that emerge when agents are Complements. In that case, groups may become trapped in a “Mutually Assured Ignorance” (MAI) equilibrium where no one seeks out information, and as a result, no corrective actions are taken to shield the group from adverse outcomes in the face of a bad state of the world—potentially with disastrous consequences.¹⁰ By the same token, a “Mutually Assured Awareness” (MAA) equilibrium, where everyone acquires information, is also possible. The path taken depends critically on the agents' expectations about the actions of others in their group. Conversely, a group of Substitutes cannot fall into either a MAI or MAA equilibrium, as its agents are inherently inclined to resist such dynamics. More generally, let $\pi_k \in [0, 1]$ denote the share of type $k \in \{C, S, AA, AG\}$ in the group, where $\pi_C + \pi_S + \pi_{AA} + \pi_{AG} = 1$. Among Complements (resp. Substitutes), suppose that λ_i^* (resp. λ_i^{**}) is distributed according to a distribution function F_C (resp. F_S). Let r_C (resp. r_S) denote the share of Complements (resp. Substitutes) who avoid information in equilibrium.

¹⁰Bénabou (2013) uses the acronym MAD (Mutually Assured Delusion) equilibrium. However, in the context of information avoidance MAI provides a more accurate description of the equilibrium. The same holds for the acronym MAA (Mutually Assured Awareness).

Figure 1: Strategy types



Notes: The figure shows examples of each strategy type. Each panel shows the best response function $y_i^*(\lambda_{-i})$ where λ_{-i} is the share of other group members who avoid information.

Proposition 2 *In equilibrium, the shares r_C and r_S of information-avoiders among Complements and Substitutes satisfy the following conditions*

- (i) $F_C(r_C\pi_C + r_S\pi_S + \pi_{AA}) = r_C$ and
- (ii) $1 - F_S(r_C\pi_C + r_S\pi_S + \pi_{AA}) = r_S$.

Corollary 1

- (i) *In a group composed entirely of Complements, both a Mutually Assured Ignorance (MAI) equilibrium and a Mutually Assumed Awareness (MAA) equilibrium can emerge. When Always Avoiders (resp., Always Getters) are introduced into such a group, there exists a threshold π_{AA}^* (resp., π_{AG}^*) such that if the share of unconditional agents exceeds that threshold, the unique equilibrium is MAI (resp., MAA).*
- (ii) *In a group that contains a positive share of Substitutes, neither a MAI nor a MAA equilibrium are possible.*

In our experiment we use the strategy method to elicit full best response schedules; results are discussed in Section 5. We also assess how the mixture of types impacts aggregate equilibrium avoidance by employing simulations in Section 6.

4 Experimental Design and Procedures

4.1 Experimental Design

Informed by the model in Section 3, we design an experiment to investigate the strategic interdependence of information avoidance in groups in a context where ignorance is socially costly. Specifically, we ask: do individuals condition their decisions to avoid information on whether others avoid it? How heterogeneous are the best responses $y_i^*(\lambda_{-i})$?

4.1.1 Decision Environment

Our experimental design features the key theoretical ingredients of the model in Section 3: a future consumption event that depends on a state of the world and induces anticipatory utility, a prior decision to acquire or avoid information about the state, and payoff externalities induced by the information decision. We describe each element below. Our design builds on [Falk & Zimmermann \(2024\)](#), who study information avoidance in an individual setting. We extend their setting to groups with payoff externalities and employ two methodological innovations: the strategy method to elicit complete best-response schedules, and a novel measure of anticipatory utility.

Future consumption event. After being assigned to groups, participants are told that a state of the world (Good or Bad) will be randomly drawn. The state determines whether the group will experience a negative consumption event during a 4-minute period at the end of the experimental session (labeled to participants as the “Risk Period”). If the state is Bad (labeled as the “Screams” state), the group will hear a series of aversive screams at random times during the Risk Period. If the state is Good (“Quiet”), the group will not hear any scream. The prospect of hearing such screams have been shown to sustain anticipatory anxiety over long periods ([Beaurenaut et al., 2020](#)), are scalable to large groups, and allow fine-grained variation in severity through volume levels.¹¹

Information decision. Before the consumption event occurs, every group member has free access to information that perfectly reveals the state of the world (Screams or Quiet). Each member chooses between two options: acquiring information (labeled as option “Now”) or avoiding it (“Later”). If she chooses “Now”, she learns the state immediately. If she chooses “Later”, she defers learning until the consumption event occurs.¹² This choice —denoted $y_i = 0$ for “Now” (acquire) and $y_i = 1$ for “Later” (avoid)— is referred to as the “information decision”.¹³

From a standard economic perspective, in our setting the acquisition of information is costless. First, choosing Now or Later involves no monetary cost. Second, cognitive costs are negligible, as

¹¹Section 4.2 provides more details on the rationale for using screams.

¹²Participants are aware that if they choose Later, they will be informed about the state just before the consumption event occurs.

¹³In each group, information decisions are simultaneous and without communication, precluding social learning.

participants learn simple information revealing a binary state (Good or Bad). Third, information decisions are simultaneous. Thus, a standard rational agent is predicted to choose Now (acquire information).

Externality. Unlike studies of single-agent information avoidance, our design features a payoff externality from information decisions. Acquiring information generates a positive return for both oneself (standard instrumental value) and other group members (externality). Concretely, the *volume* of the screams in the Screams state decreases with the number of members who choose Now: each member who acquires information lowers the volume of screams by 2 points; avoiding information is socially costly.¹⁴

4.1.2 Conditional Choices: Elicitation of Best-Response Schedules

Proposition 1 predicts heterogeneity in how individuals respond to others’ information decisions. To empirically identify this heterogeneity, we use the strategy method (Selten, 1967) to elicit each participant’s complete best-response schedule $y_i^*(\lambda_{-i})$ across the full range of shares of avoiders among other members.

Following the theoretical classification of strategy types in Section 3.1, we classify participants as Strategic Complements if they choose Later only when the share of avoiders is sufficiently large, and Strategic Substitutes if they choose Later only when the share of avoiders is sufficiently small. Participants who always choose the same option regardless of others’ choices are classified as Always Getters (always acquire) or Always Avoiders (always avoid). The remaining cases (participants switching multiple times) are classified as “Others”. The elicitation of individual best-response schedules allows us to empirically test the existence and estimate the distribution of the theoretical types.

4.1.3 Treatments

The strategy method identifies individual-level heterogeneity in best responses. Complementing this, we implement a between-subjects treatment that exogenously shifts participants’ expectations about the prevalence of avoidance among other members in their group (λ_{-i}). The resulting variation in beliefs allows us to identify the average causal effect of beliefs on own information decisions, examine whether effects vary across strategy types, and test whether the mechanism operates through anticipatory utility as theory predicts.

Specifically, we implement an information provision treatment (Haaland et al., 2023), randomizing participants at the group level into two conditions: *Many Avoided* and *Few Avoided*. In *Many*

¹⁴Each member’s contribution to volume reduction is independent of others’ choices (additive separability). Therefore, we examine whether strategic interdependence in information decisions can arise even in the absence of any in-built interdependencies in the technology. See Section 4.2 for details.

Avoided (Few Avoided), participants are informed that 80% (20%) of participants within a particular group of a similar, past pilot study chose Later. To help participants understand the implications of others' choices, participants are also shown the scream volume that would result if their own group members behave like the pilot participants.¹⁵ This treatment is designed to generate exogenous variation in beliefs about the share of avoiders among other members, λ_{-i} . By comparing information decisions and anticipatory utilities across conditions, we can identify both the causal effect of beliefs and test the focal mechanism through which beliefs affect decisions, as explained in Section 4.1.4.

To check whether the treatment manipulation works as intended, we elicit the participants' posterior beliefs about the fraction of avoiders among their group mates after the information provision treatment.

4.1.4 Additional Measurements

Anticipatory utility. To shed light on the mechanism, we introduce a novel elicitation designed to measure anticipatory utility over the future consumption event. A crucial feature of our elicitation is that we measure anticipatory utility *conditional on each information set* —namely, upon acquiring information and discovering a positive signal (“good news”), upon acquiring information and discovering a negative signal (“bad news”), and upon avoiding information and discovering no signal (“no news”). Specifically, we ask participants:

- a) “How happy would you be *thinking about the outcome ahead* if you chose Now and then discovered that the lottery outcome is Screams?”;
- b) “How happy would you be *thinking about the outcome ahead* if you chose Now and then discovered that the lottery outcome is Quiet?”; and
- c) “How happy would you be *thinking about the outcome ahead* if you chose Later?”

These elicitations are designed to measure, respectively, the anticipatory utilities $v_i(b(\lambda_{-i} + \Delta))$, $v_i(g)$, and $v_i((1 - q)b(\lambda_{-i}) + qg)$. To infer the net utility of avoiding information φ_i , we also elicit $(1 - q_i)$, the participant's subjective belief that the state is Screams (i.e., the prior belief about finding out bad news upon choosing Now). Together, these four questions measure the components of expression (4), which defines the net utility of avoiding versus acquiring information and theoretically determines the information decision. We call this set of questions the “Kreps-Porteus questions” (or “KP questions”).

¹⁵Specifically, we inform each participant that, if their group mates behave like pilot participants, the volume in the Screams state would be 89 or 91 in *Many Avoided* (60 or 62 in *Few Avoided*), depending on whether she herself avoids or acquires information. The 2-point difference within each condition reflects the focal participant's choice; the difference between conditions reflects others' choices.

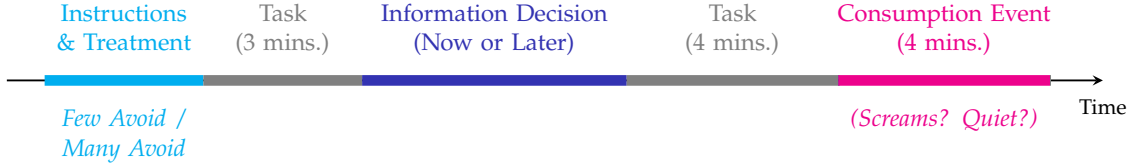


Figure 2: Timeline of experiment

The KP questions allow us to test mechanisms. Theory predicts an asymmetric effect: when the share of avoiders λ_{-i} increases, the payoff in the bad state worsens, decreasing the anticipatory utility upon learning bad news ($v_i(b(\lambda_{-i} + \Delta))$) but leaving anticipatory utility upon learning good news ($v_i(g)$) unchanged (Section 3). Since the KP questions elicit anticipatory utility separately for good and bad news, we evaluate the prediction by testing whether the treatment lowers only $v_i(b(\lambda_{-i} + \Delta))$.

Baseline attitudes towards information. Individuals may vary in their overall inclination to learn potentially undesirable information. To measure such baseline preferences, we administer the “Information Preferences Scale” (“IPS”; [Ho et al., 2021](#)). The IPS captures an individual’s willingness to obtain information that may be unpleasant but is instrumentally valuable, across multiple domains. Higher IPS scores indicate greater general tendency to seek information. Measuring these general preferences serves three purposes: (i) to verify balance between treatment groups on a trait potentially relevant for our outcome variable (choosing Later); (ii) to control for it to improve the precision of our estimates; and (iii) to use as a benchmark in the validation of our elicitation of anticipatory utility.

Social preferences. Since information decisions generate payoff externalities on other members, social preferences potentially predict information avoidance and interact with its interdependence (shown theoretically in Appendix A.3). To empirically assess the relationship between social preferences and information decisions, we elicit individual measures by administering the Global Preferences Survey (GPS) ([Falk et al., 2018](#)), which captures altruism and reciprocity, as well as trust, risk attitudes and time preferences.

4.1.5 Task and Earnings

Following [Falk & Zimmermann \(2024\)](#)’s design, we include an incentivized task for participants to complete during two intervals of time: one before the information decision stage and another between the information decision stage and the consumption event stage. The experimental timeline is summarized in Figure 2.

4.2 Experimental Procedures

4.2.1 Recruitment and Sample

The experiment was conducted at the Centre for Decision Research and Experimental Economics (CeDEx) Lab, University of Nottingham.¹⁶ We ran 17 experimental sessions (each constituting an independent group) across two waves of data collection (December 2023 and May–June 2024), recruiting participants from the CeDEx subject pool. In total, the sample for analysis comprised 465 participants, with an average of 27.35 (± 3.66) subjects per session.¹⁷ Each session lasted about one hour, and participants earned on average £12.52 (± 2.00), including a £5 show-up fee. Full descriptive statistics of baseline characteristics are reported in Table B.1 in the Appendix.

4.2.2 Session Structure and Timeline

At the start of each experimental session, participants were matched into groups, and each group was randomly assigned to a treatment condition.¹⁸ Participants were then provided with detailed instructions on computer screens.¹⁹ The instructions included a description of the state of the world lottery, the negative consumption event (screams), the timeline of the experiment, the information decisions with its implications, and the task. In order to ensure understanding of the decision environment, we included control questions about the timing of learning the outcome and about the nature of returns and externalities from information acquisition/avoidance. During the instructions phase, we also implemented our information provision treatment.

A computerized lottery determined the outcome (either Screams or Quiet) for each group. If the outcome was Screams, during the Risk Period all members of the group listened through headphones (worn throughout the session) to a series of unpleasant screams, drawn from the validated stimulus set in [Beaurenaut et al. \(2020\)](#).²⁰ To ensure hearing safety, the maximum possible volume level (100), was calibrated at 80dB in accordance with UK hearing safety regulations. The calibration was carried out separately for each headphone-computer pair, ensuring the hearing safety of every subject. To ensure exposure to the potential screams, participants were informed at the outset that headphone use was mandatory throughout the session and that attempts to remove them

¹⁶The experiment was preregistered, AsPredicted #153871 and #175588. The experiment was approved by the University of Nottingham’s School of Economics Research Ethics Committee. At the start of the session, participants were informed about the nature of the experiment and provided written consent to participate.

¹⁷Originally, we recruited 493 participants in 18 sessions. However, due to technical issues, data (including the main outcomes of interest) was not recorded for 22 subjects within a single session of 28 participants and is excluded from the analysis; our results are robust to the inclusion of the 6 individuals from this session for which data is available. In a separate incident, one participant left early (about an hour after the session started) reducing observations for final-stage variables by one.

¹⁸The random assignment of the treatment was blocked by week of year and time of session (morning and afternoon).

¹⁹Full instructions are available in Appendix D. The experiment was programmed in Lioness ([Giamattei et al., 2020](#)).

²⁰Participants did not hear any sample screams before making their information decisions, but were informed that “[p]revious research has shown that people consider similar screams as *more aversive (disturbing)* than the sound of *nails sliding on a chalkboard*”. Evidence of this statement can be found in [Kumar et al. \(2009\)](#).

would be a basis for expulsion. Compliance and correct headphone use were monitored by two invigilators per session.²¹ Participants were told they could leave the session at any time.

The use of screams instead of electric shocks (employed in experimental studies on information avoidance in individual decision making) is based on four rationales. First, one of the motives that may lead to information avoidance are anticipatory emotions, such as anxiety about a future outcome, and studies show that the screams are able to sustain anxiety for longer periods than electric shocks (Beaurenaut et al. (2020)). This feature is especially desirable for experiments with group decision-making, which typically take longer compared to individual decision-making. Second, the provision of human scream sounds is more scalable than electric shocks, since the only required equipment are computers and headphones, as opposed to devices specialized for the provision of shocks.²² Non-scalability imposes a natural limit to group sizes, which increase the likelihood of the emergence of complementarities in information decisions (Bénabou (2013)). Third, screams allow a more granular manipulation and more intuitive understanding of variations in severity (i.e., volume levels as opposed to electric shock intensity); this granularity in severity is important as group sizes become larger and the marginal return of acquisition of information (Δ) by any individual member becomes smaller. Fourth, the use of human screams alleviates ethical concerns compared to electric shocks.²³

The experimental and model assumption is that screams are a bad. To assess whether this is the case, we elicit the participants' utility over two different levels of volume of screams. We ask: "Imagine that the Outcome is Screams, and that the volume of the screams will be level 100. How happy would you be thinking about the Outcome ahead?"; we ask a similar question for volume level 50. This allows us to identify subjects who, contrary to our assumption, derive higher utility from higher levels of volume of scream sounds ("volume loving", as opposed to "volume averse" subjects).

For groups whose lottery outcome was Screams, the volume of the screams was determined by the information decisions of the members of the group. Specifically, the volume of the screams in group g was determined by the following function:

²¹We are grateful to the people who generously helped as invigilators: Pierce Gately, Kieran Stockley, Tong Fang, Xue Wang, Matías Golman, Jesús Rodríguez, Yifan Li, Adrian Brown, Thomas Barnes, Lara Suraci, and Niramay Chugh.

²²The lower implementation requirement also facilitates replicability.

²³Recently, experimental literature in Economics has started using the prospect of money losses as a negative future event (Pagel, 2018; Engelmann et al., 2024). In particular, Engelmann et al. (2024) finds that money losses induce anticipatory anxiety and wishful thinking about the future event. The literature on intrinsic preferences for information has focused attention on future *consumption* events as distinct from *income* events, because knowing the latter introduces a planning advantage that motivates the acquisition of information about the future income event. To avoid introducing this extra planning advantage, in our experiment, we follow the latter tradition and focus on a consumption event. We argue that, theoretically, the relevant feature of the event is that the prospect of hearing human screams, much like the prospect of money losses, is perceived as a bad event by participants. In Section 5.1.1, we show evidence that, in our experimental sample, this is the case.

$$vol_g = 50 + (100 - 50) \left(\frac{1}{N_g} \sum_{i=1}^{N_g} y_{ig} \right), \quad (6)$$

where $vol_g \in [50, 100]$ is the volume of screams for group g conditional on the lottery outcome being Screams, N_g is the number of members in group g , and $y_{ig} = 1$ if subject i in group g chooses Later and 0 otherwise. The acquisition of information about the state, $y_{ig} = 0$, thus partially mitigates the volume of the screams.

Equation (6) operationalizes the payoff in the bad state from Equation (2): here, $\bar{b} = 50$, $\Delta = (100 - 50)/N_g$. Each group was composed of an entire experimental session, with group sizes of approximately $N_g = 30$. This implies that the return to each participant's information acquisition (in terms of volume reduction) was $\Delta = 50/N_g \approx 1.67$ volume points.²⁴

While the volume production function in Equation (6) appears related to a public good game (PGG), our decision environment differs in several ways that reverse the standard theoretical predictions. First, the nature of the cost is distinct: in a PGG, the cost of contributing is a known monetary amount, identical for all players and independent of others' actions. Here, there is no monetary cost of contributing: the cost is instead psychological (from potentially learning bad news about the future) and is heterogeneous across individuals. Moreover, the psychological cost plausibly varies with the choices of others (which influence how severe the bad state is). These differences lead to a divergence in the prediction for a purely material-payoff maximizing agent: such an agent would optimally free-ride in a standard PGG but would always prefer to "contribute" (i.e., acquire information) in our setting. Consequently, the central puzzle in each setting is inverted: whereas the literature on PGGs seeks to explain why people *do* contribute, in our setting the question is why people *do not* (i.e., why they avoid information) and how this reluctance to contribute is conditional on others' decisions.

The information decision (the binary decision between options Now and Later) is our main outcome of interest. To obtain an alternative, continuous measure of preferences, we also ask participants, after they chose either Now or Later, to rate the strength of preference for their selected option on a Likert scale ranging from 0 ("I am indifferent") to 10 ("I have a very strong preference for the selected option"). From this measure, we construct the "Information Avoidance Scale" (IA scale) as follows. For those choosing Later, the IA scale equals the reported strength; for those choosing Now, it equals the negative of that strength. Thus, the IA scale ranges from -10 (very strong preference for Now) to +10 (very strong preference for Later), with 0 indicating indifference. The IA scale thus provides more variation than the binary outcome. We also ask participants to write down the reason for the selected option in an open text box.

To explore individual heterogeneity in best responses, we use the strategy method (Selten, 1967)

²⁴We described to participants the return to information acquisition as being equal to "around 2 out of 100 volume level points".

to elicit each participant’s complete schedule of choices between options Now and Later as a function of others’ choices. This schedule captures their choice conditional on every possible share of information avoiders among others in the group, specifically for the shares: 0%, 20%, 40%, 60%, 80%, and 100%.

We make the strategy method elicitation incentive-compatible by assigning a strictly positive probability that these choices are implemented. Specifically, before the elicitation, we inform participants about the following procedure. After the group members make their (unconditional) informational decisions (between Now and Later), we randomly select $N_g - 1$ subjects and calculate the proportion of information avoiders *among these $N_g - 1$ subjects*, based on their unconditional decisions. For the remaining participant, we implement her selected best response to this share.

To measure general preferences for information, we administer the “Information Preferences Scale”, or “IPS” (Ho et al., 2021). This scale measures an individual’s desire to obtain or avoid information that may be unpleasant but is instrumentally valuable, across several domains.²⁵ The scale ranges from 1 = “Definitely not want to know” to 4 = “Definitely want to know”, so that higher IPS scores indicate greater general willingness to obtain information.

To measure social and economic preferences, we administer the Global Preferences Survey (GPS) (Falk et al., 2018). The GPS measures risk attitudes, time preferences, altruism, reciprocity, and trust. Each dimension of preferences is elicited through the responses to two questions.²⁶ Following Falk et al. (2018), we standardize the answers, transforming them into z-scores. Higher z-scores indicate, respectively, higher risk seeking, patience, altruism, reciprocity, and trust. We also elicited demographics (e.g., age, gender), prior work experience, ratings about the perceived difficulty of instructions and quiz questions. Participants were given the opportunity to feedback on their experience in the experiment in open text boxes.

Following Falk & Zimmermann (2024), we include an incentivized task before the Now or Later decision and between such decision and the Risk Period. Participants are asked 90 general knowledge quiz questions, with earnings increasing in the number of correct answers. Each participant is paid according to her own performance, which is the only source of monetary earnings in our experiment, apart from the show-up fee.

²⁵The IPS includes 18 questions that ask the subject’s willingness to obtain information in hypothetical scenarios across five domains: health, finance, social relations, ego-relevant characteristics, and occupation. For each question, the possible answers are “Definitely don’t want to know” (encoded as 1), “Probably don’t want to know” (2), “Probably want to know” (3), and “Definitely want to know” (4). Following Ho et al. (2021), we construct the “overall” scale by averaging the answers to all the 18 questions, and “domain-specific” subscales by averaging all questions in a given domain.

²⁶For risk attitudes and time preference, one of the questions used by Falk et al. (2018) involve the staircase method. Since administering this could be time-consuming and cognitively burdening in the middle of our experiment, we did not include such questions.

5 Results

This section presents the analysis of our experimental data. We begin by establishing two necessary preconditions for our study. In Section 5.1.1, we verify that, consistent with our theoretical and experimental assumption, screams are perceived as a bad by most participants. In Section 5.1.2, we document the existence of information avoidance in our decision environment (a context where the positive instrumental value and externalities from acquisition of information introduces an additional incentive to *acquire* it). Further, we establish that the decision to avoid information in our environment is likely not random nor driven by indifference, as information avoiders report strong preferences for their choice.

Following these prerequisite validations, we turn to our primary results. In Section 5.2, we leverage experimental variation in beliefs about others' information decisions to test for strategic interdependence, assessing whether the likelihood of information avoidance increases (strategic complementarity) or decreases (strategic substitutability) on average. In Section 5.3, we move beyond the average to document and characterize the individual heterogeneity of the interdependence. Using the strategy method, we document the empirical distribution of strategy types and report heterogeneous treatment effects by type. In Section 5.4, we explore the mechanisms driving the average treatment effect. Specifically, we use the KP questions to decompose the effect by identifying which components of the expression for the net incentive to avoid information (4) are most impacted by the treatment. We also discuss the plausibility of alternative mechanisms. Finally, the next section is devoted to analyzing how the group composition, in terms of strategy types, shapes the distribution of equilibrium share of information avoiders.

While one of our empirical exercises focuses on estimating the average effect of exogenously shifted beliefs on information avoidance, the main objective of this section is not to document a large aggregate treatment effect. Rather, our goal is to identify the structure of strategic interdependence in information avoidance decisions. As we show below, individuals differ systematically in how they respond to others' information choices, exhibiting both strategic complementarity and strategic substitutability. These heterogeneous best-response types generate offsetting effects at the aggregate level, naturally limiting the size of the average treatment effect, but imply large and highly non-linear differences in equilibrium information avoidance depending on the composition of the group. We therefore interpret the average treatment effect as a gateway result, establishing the existence of causal interdependence, and place primary emphasis on individual heterogeneity and its implications for group-level outcomes.

5.1 Preliminary Checks

This section reports a set of preliminary checks that validate key assumptions underlying our experimental design and delineate the scope of our subsequent analysis. Specifically, we verify that

the negative future event used in the experiment is indeed perceived as undesirable by most participants, and we document that deliberate information avoidance arises in our decision environment despite the presence of positive private returns and externalities from information acquisition. These checks are not intended to establish our main substantive findings, but rather to ensure that the environment operates as theorized and that observed information avoidance reflects deliberate choice rather than indifference or confusion.

5.1.1 Are Screams a Bad?

We first conduct a validation check of a foundational assumption underlying our study: that participants actually perceive the prospect of listening to screams as a bad. We provide evidence confirming that the assumption holds for our sample. First, we asked subjects how “(un)happy” they would be if screams were played at volume 50 and, separately, at volume 100. The difference in these ratings provides a proxy of each participant’s disutility of listening to screams at higher volume. Comparing these ratings reveals that a large majority of subjects report lower utility levels for screams at higher volumes: specifically, 82% of subjects report that screams at volume 100 would be strictly worse than at volume 50.²⁷ Second, using the KP questions, we asked subjects to rate their “(un)happiness” upon discovering that the state was Screams after choosing Now, and separately, if it was Quiet. The difference in these ratings provides a measure of each participant’s utility for listening to screams relative to silence. If the screams are perceived as a bad, we would expect lower ratings for the Screams state than for the Quiet state. As expected, this pattern holds for the majority of subjects: 81% report that discovering that the Screams state is strictly worse than discovering that the Quiet state.²⁸ The distribution of types of preference screams and volume is shown in Figure B.2 and Figure B.1 in the Appendix. The evidence hence supports the view that the vast majority of subjects perceive screams as a bad.

Furthermore, the fact that higher-volume screams are perceived as worse than lower-volume ones confirms that reductions in volume are meaningful along the relevant utility margin.

For the main analysis, we retain the full sample, including participants with weak preferences over screams and volume. We refer to such subjects respectively as “scream-lovers” and “volume-lovers”.²⁹ Analyses that exclude these individuals are presented in the Appendix Section B.6. They confirm that our findings are robust to the exclusion of volume- and scream-lovers, and, in some cases, become even stronger.

²⁷Among the remaining subjects, 3% are indifferent between volume levels, and 9% would prefer volume 100. We call the later type of subjects “volume-lovers”.

²⁸Among the remaining subjects, 6% are indifferent between the two states and 14% would prefer discovering the Screams state. We call the later type of subjects “scream-lovers”.

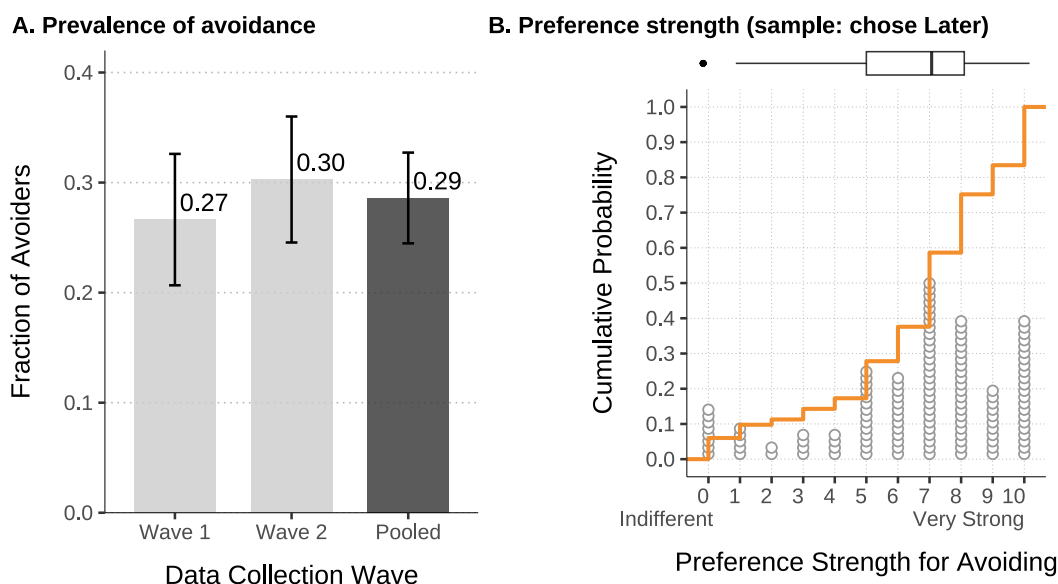
²⁹There are 111 subjects who are either scream lovers or volume lovers or both. The distribution of preferences is reported in Appendix Section B.2.

5.1.2 The Prevalence of Information Avoidance

In order to study the interdependence of information avoidance, it must first be established that avoidance occurs.

We find a positive and statistically significant share of information avoiders. Panel A of Figure 3 shows the prevalence of avoidance by data collection wave and for the pooled sample. In the pooled data, the prevalence is 0.286, significantly greater than zero at 1% level. The proportion is stable across data collection waves.

Figure 3: Information avoidance: prevalence and preference strength



Notes: The figure displays the prevalence of and preference for information avoidance. Panel A shows the fraction of subjects who chose Later (avoided information), by data collection wave (brown bars) and in the full sample (blue bar). Whiskers indicate 95% confidence intervals. Pairwise comparisons using two-sample proportion tests yield p-values greater than 0.4. Panel B restricts attention to the subset of subjects who chose Later (avoided information) and shows the distribution of the reported strength of preference for this choice (0 = “I am indifferent” between Now and Later; 10 = “I have a very strong preference” for the chosen option). Gray dots represent the absolute frequency of each preference strength (one dot per subject). The orange line shows the empirical cumulative distribution function (CDF). The boxplot shows the median and the interquartile range, with whiskers extending to the most distant point within 1.5 times the interquartile range.

This result relates to prior findings in the literature. In an individual decision making context, [Falk & Zimmermann \(2024\)](#) find that 52% of subjects avoid information. The higher proportion in their study could be attributed to a series of factors. In addition to some differences in implementation, in their setup information provides no material benefit (i.e., it does not affect the severity of the future negative consumption event), eliminating instrumental and prosocial motives for information acquisition.

The choice to avoid information does not appear to be driven by indifference nor trembling. Panel B of Figure 3 plots the distribution of preference strength among avoiders (people who chose

Later), elicited on a 0-10 scale (0=“indifferent”; 10=“very strong preference”). The average strength of preference is 6.6, well above indifference. Moreover, the cumulative distribution (orange line) shows that only 6% of avoiders report indifference, while more than 80% report a strength of at least 5. Thus, evidence suggests that most information avoiders strongly prefer to remain uninformed, rather than choosing randomly or out of indifference.

5.2 The Interdependence of Information Avoidance

We now turn to testing whether information avoidance decisions are causally interdependent across individuals. The purpose of this section is to establish the existence and sign of interdependence by leveraging exogenous variation in beliefs about others’ information choices induced by the information provision across conditions. Importantly, our aim is not to document a large aggregate response, but to verify that individual information decisions react systematically to expectations about peers’ behavior. As we show below, while the resulting average effect is positive and statistically significant, its magnitude reflects the aggregation of heterogeneous and offsetting individual responses, which we characterize in detail in the subsequent section.

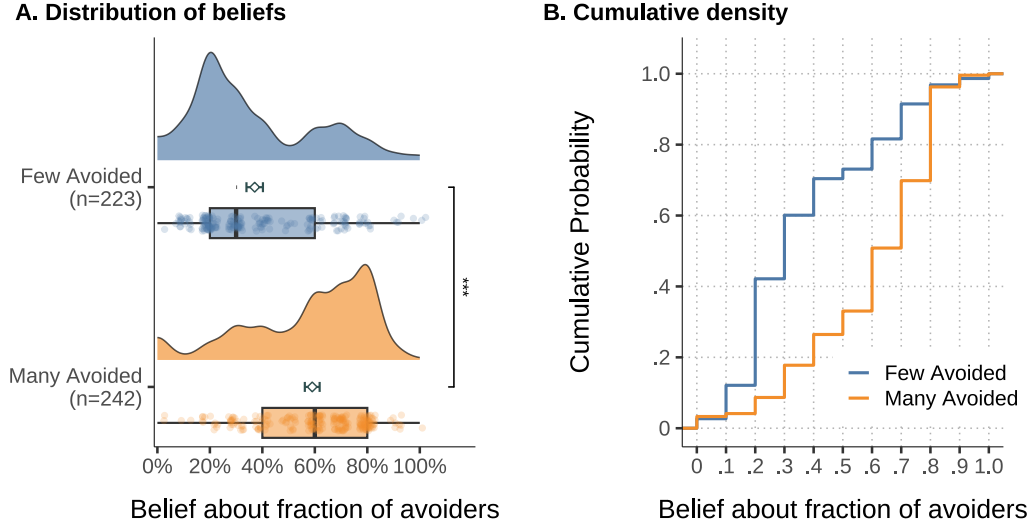
As a preliminary step, we verify that treatment groups are balanced in all key covariates. Balance checks are reported in the Appendix Table B.3. Importantly, IPS scores are balanced between treatment groups, both for the overall scale and for each of the domain-specific subscale. Hence, differences in information avoidance between treatments cannot be attributed to baseline differences in general or domain-specific attitudes toward information. The representation of some fields of study is somewhat imbalanced; to account for this, in subsequent analyses we report results both with and without field-of-study fixed effects in the regressions.³⁰

Our treatment manipulation successfully shifts expectations about the share of avoiders among other members in the group. Figure 4 reports the effect of the information-provision treatment on beliefs about others’ information decisions. Panel A displays the distribution and summary statistics of beliefs by treatment condition, while Panel B shows the corresponding empirical CDFs. Mean beliefs about the share of avoiders (indicated by hollow diamonds) were 37% in the *Few Avoided* condition versus 59% in the *Many Avoided* condition, a highly significant difference of 22 percentage points ($p < 0.001$). Appendix Table B.4 reports OLS estimates from a linear regression of beliefs on the treatment dummy and shows that the estimate of the average treatment effect is robust—both in magnitude and statistical significance—to the inclusion of different sets of controls. Moreover, the distributional evidence in Panel B indicates a first-order stochastic dominance of beliefs in the *Many Avoided* condition relative to *Few Avoided*. In sum, the information-provision treatment performs as intended in shifting beliefs about others’ behavior.

We now examine whether the treatment-induced shift in beliefs translates into differences

³⁰*Many Avoided* has fewer Business and of Economics students and more Humanities students than *Few Avoided*.

Figure 4: Manipulation check



Notes: The figure shows the treatment effect on beliefs about the share of avoiders among other group members. Panel A shows the distribution of beliefs, by treatment condition. Dots (jittered) represent individual observations; violin plots show kernel density estimates; box plots show median (center line) and interquartile range (box); diamonds show mean beliefs with error bars representing 95% confidence intervals; p-value from a two-sample t-test. Panel B shows the empirical CDF of beliefs.

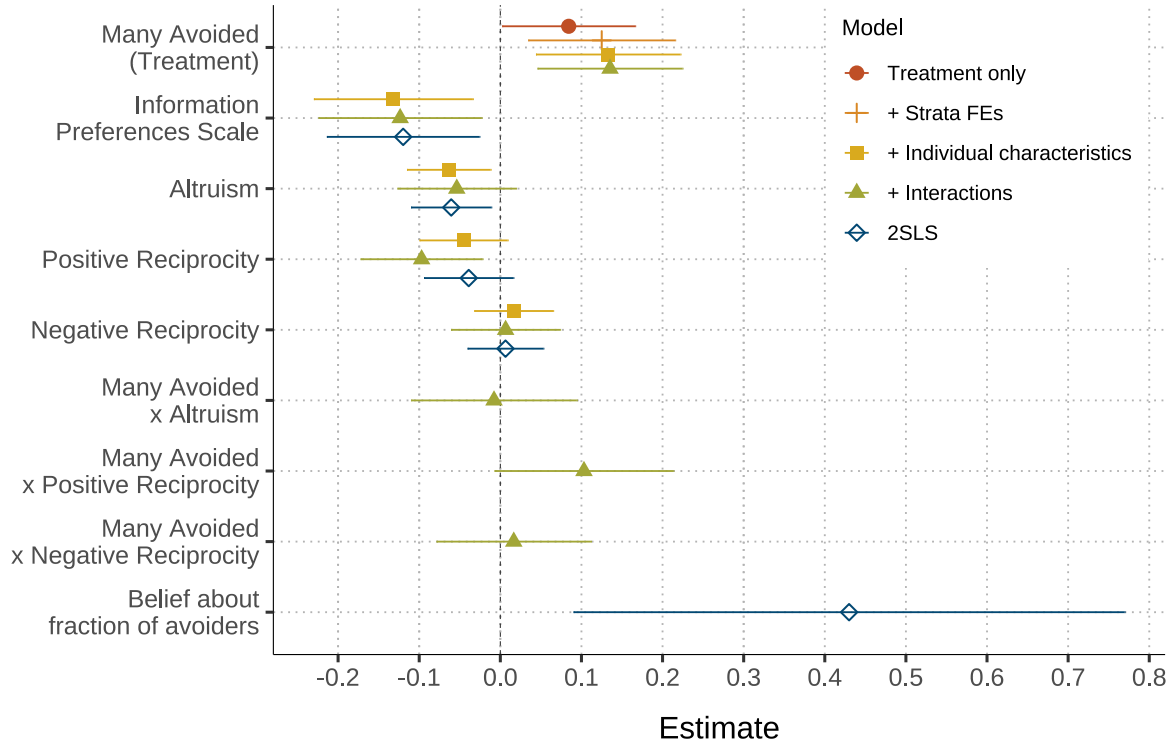
in actual information decisions. Figure 5 reports regression-based estimates of the effect of the information-provision treatment on information avoidance across alternative specifications. In the baseline specification, exposure to the *Many Avoided* condition significantly increases the likelihood of information avoidance relative to the *Few Avoided* condition. While the estimated average effect is economically modest, it is precisely estimated and robust across specifications, indicating that information avoidance decisions respond systematically to expectations about others' behavior. Raw treatment-control differences in avoidance rates are reported in Appendix B.5. As discussed above, we examine this heterogeneity directly using the strategy method in the next section.³¹

To quantify the relationship between beliefs and behavior (the interdependence), we move beyond the reduced-form results and estimate a 2SLS model instrumenting beliefs with the treatment assignment (Appendix Table B.10).³² The estimated coefficient ranges from $\hat{\beta}^{2SLS} \approx 0.4$ to 0.6, implying that a 10 percentage point increase in expected avoidance by others raises an individual's probability of avoiding information by about 5 percentage points. This supports the conclusion

³¹The positive average effect is robust across a wide range of alternative specifications, sample restrictions, and outcome measures. In particular, the effect remains statistically significant when excluding participants with non-standard preferences (Scream- and Volume-Lovers), under alternative sets of controls, and when using a continuous measure of information avoidance. Detailed robustness analyses are reported in Appendices B.6 and B.7.

³²The exogeneity assumption is satisfied by design. The monotonicity assumption—requiring that for every subject potential beliefs are weakly higher in *Many Avoided* relative to *Few Avoided*—is likely satisfied, as suggested by the first-order stochastic dominance of the belief distribution in *Many Avoided* relative to *Few Avoided* shown in Figure 4. The exclusion restriction requires that the effect of the information provision treatment operates only through posterior beliefs about others' information decisions. While this assumption is not directly testable, we discuss mechanisms in Section 5.4.

Figure 5: Information avoidance: treatment effect and predictors



Notes: The figure reports regression-based estimates of the effect of the information-provision treatment on information avoidance. Coefficients are shown for alternative specifications of a linear probability model and for a 2SLS specification instrumenting beliefs with treatment assignment, where the dependent variable is the “Avoided information” dummy (chose Later = 1, Now = 0). The first model (red dots) includes only the treatment dummy. The second model (orange plus signs) adds strata and field of study fixed effects. The third model (yellow squares) further controls for individual characteristics: the Information Preferences Scale (Ho et al., 2021; with higher values indicating a stronger overall preference for *knowing* potentially undesirable information), gender, age, social preferences (altruism, positive and negative reciprocity), and economic preferences (risk aversion, patience, and trust; omitted from the figure for conciseness). The fourth model (green triangles) adds interactions between treatment and social preferences. The last model (blue hollow diamonds) instruments beliefs with the treatment assignment and shows 2SLS estimates. Error bars represent 95% confidence intervals computed based on robust standard errors.

that, on average, information avoidance is a strategic complement in our sample. As with the reduced-form estimates, this average causal response should be interpreted as establishing the presence of interdependence rather than summarizing its underlying structure.

We also examine how the average treatment effect correlates with individual characteristics, including social preferences. These patterns should be interpreted descriptively rather than causally, and serve primarily to motivate the richer characterization of individual best-response behavior developed in the next section. In particular, general preferences for information (as captured by the IPS), reciprocity, and altruism are each associated with systematic differences in information decisions across treatment conditions.³³ Taken together, these correlations suggest that individual characteristics help organize variation in information decisions, but do not by themselves capture the strategic structure of interdependence that we analyze below.

Taken together, the results in this section establish the existence of causal interdependence in information avoidance decisions, but leave open how this interdependence operates at the individual level—a question we address by directly characterizing best-response behavior in the next section.

5.3 The Heterogeneity of Interdependence

Section 5.2 established that information avoidance decisions are causally interdependent on average: increasing expectations about others' avoidance raises the likelihood of avoiding information. However, the average effect alone does not reveal how individuals respond to others' information choices, nor whether these responses are uniform across people. As Proposition 1 shows, theory allows substantial heterogeneity in best-response behavior: some individuals may be more likely to avoid information when others do so (strategic complementarity), while others may react in the opposite direction (strategic substitutability), or not condition their decisions on others at all.

In this section, we move beyond average effects to characterize the structure of interdependence at the individual level. Using the strategy method, we elicit each participant's full best-response *schedule*—namely, how their information decision varies with the expected shares of avoiders in their group. This allows us to classify individuals into distinct strategy types and to quantify the prevalence of strategic complements, strategic substitutes, and unconditional types.³⁴ We then show how this heterogeneity underlies the average effect documented above and motivates the analysis of group composition and equilibrium information avoidance that follows.

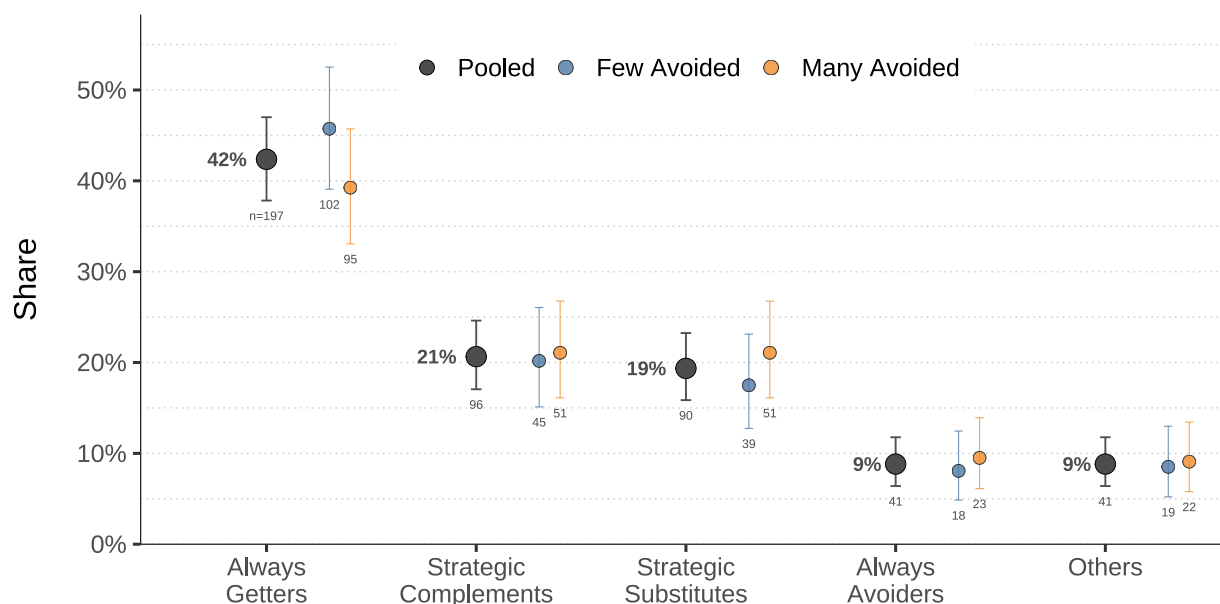
The data reveal substantial heterogeneity in how individuals respond to the information avoid-

³³Specifically, individuals with a stronger tendency to seek potentially undesirable information (higher IPS scores) are less likely to avoid information; more reciprocal individuals are on average less (more) likely to avoid in the *Few Avoided* (*Many Avoided*) condition; more altruistic individuals are less likely to avoid information ($\beta = -0.068$).

³⁴As detailed in Section 4.1, we categorize these empirical schedules based on whether they are increasing in the share of avoiders (Strategic Complements), decreasing (Strategic Substitutes), or constant (either Always Getters or Always Avoiders).

ance decisions of others. Figure 6 presents the empirical distribution of individual strategy types in our sample. Black dots indicate the observed shares of each type in the full sample. The most prevalent type is the Always Getter (42.4%), who seeks information regardless of the share of avoiding peers. At the opposite extreme, the least common type is the Always Avoider (8.8%), who avoids learning the state irrespective of the share of avoiders.

Figure 6: Distribution of strategy types



Notes: Proportion of each strategy type in the full sample (Pooled) and by treatment condition. Points represent the observed proportion; error bars show 95% confidence intervals. Percentage labels indicate shares in the the pooled sample. Sample sizes displayed below each estimate.

Beyond these unconditional types, we document the existence of conditional—or interdependent—types, whose behavior depends on that of others. Approximately 40% of participants exhibit interdependent strategies. Among them, 21% are Strategic Complements: they avoid information only if a sufficiently high share of their group members also avoids it. The proportion of Strategic Complements is statistically different from zero, with a 99% confidence interval of (0.171, 0.246). A similar proportion (19.4%) are Strategic Substitutes, who avoid information only if a sufficiently *low* share of their group members avoids it. The remaining minority (8.8%) are subjects who switch their information decisions multiple times and, hence, whose best responses oscillate. Therefore, we find substantial heterogeneity in strategy types, with a significant share of individuals exhibiting dependence on the choices of others.

We now assess the treatment effect heterogeneity by strategy type, by examining both the extensive margin (whether the treatment affects the distribution of types) and the intensive margin (how the treatment changes the probability of avoidance conditional on the type).

On the extensive margin, we find, reassuringly, that the treatment has no statistically signifi-

cant effect on the distribution of types. Figure 6 shows the distribution of types split by treatment condition (blue and orange dots). The proportions of each type appear remarkably similar across treatments, with overlapping confidence intervals. A regression analysis confirms this graphical intuition. Appendix Table B.11 reports estimates from regressions where a dummy for each strategy type is regressed on the treatment indicator. In these regressions, the treatment coefficient captures the change in the proportion of each type between *Many Avoided* and *Few Avoided*. No statistically significant differences emerge for any type.

The distributional invariance across treatments is consistent with theory. Best-response *schedules* themselves should not change unless some (perceived) parameter of the environment—beyond the expected behavior of others—changes. Since the treatment was designed to change only the expectation about others’ behavior, but no other feature of the environment, there is no theoretical reason to expect the distribution of types to shift across treatments. Accordingly, we find that, empirically, the distribution of strategy types remains similar between treatments.

We now turn to the intensive margin. In Figure 7, Panel A compares the share of avoiders (who chose Later) between treatment conditions conditional on the strategy type. Table 1 reports a linear regression of the avoidance dummy on the treatment for each type. In these regressions, the treatment coefficient captures a Conditional Average Treatment Effect (CATE): the difference in the likelihood of avoidance between the *Many Avoided* and *Few Avoided* conditional on the strategy type. We find that the *Many Avoided* treatment significantly increases the likelihood of information avoidance for Strategic Complements. This is in line with theoretical intuitions: since the treatment successfully raised beliefs about others’ avoidance—the argument of best-response schedules—and Complements’ schedules are increasing, they should respond by avoiding more often. By contrast, however, we found no significant response among Strategic Substitutes.

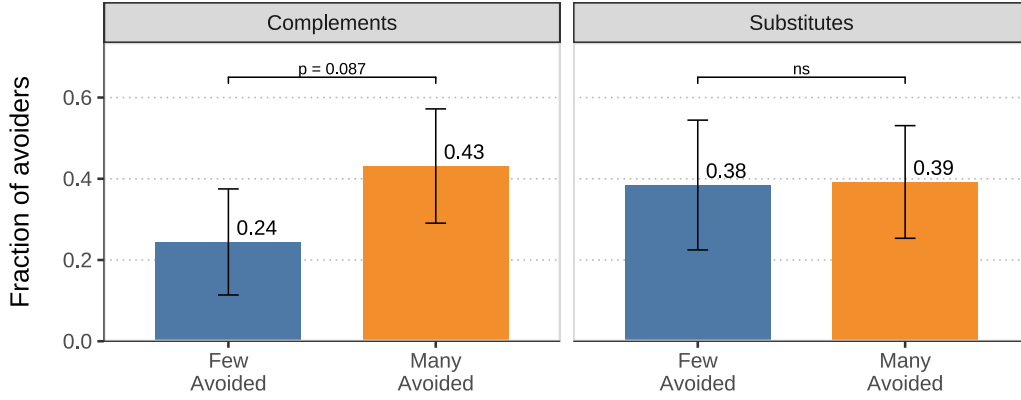
The difference in responsiveness between Strategic Complements and Substitutes can be explained by the interaction between switching threshold distributions and the treatment-induced shift in beliefs, shown in Panel B of Figure 7. Average beliefs are shown in pink lines and average switching threshold in black lines. While the treatment leaves the average switching thresholds unchanged for both types, it substantially increases average beliefs.³⁵ For Strategic Complements, the belief shift crosses the threshold, triggering increased avoidance. For Strategic Substitutes, however, the new beliefs remain on the same side of the threshold, resulting in no significant change in average avoidance rates.

Finally, columns (1) and (2) of Table 1 show that, as expected, the treatment does not affect the avoidance rate for Always Getters and Always Avoiders, who, by definition, follow strategies that are irresponsive to the behavior of others. Taken together, the pattern of intensive margins of treat-

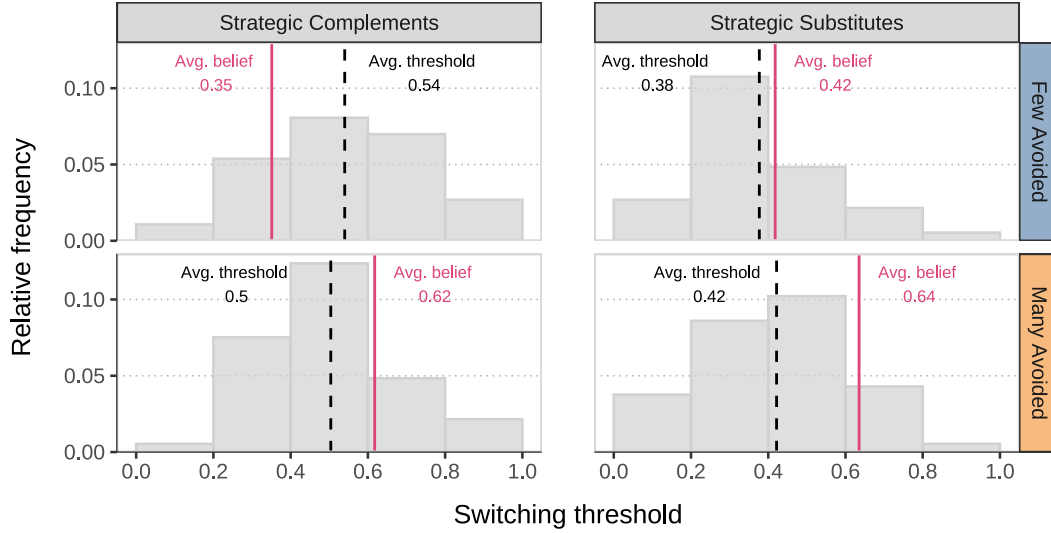
³⁵As shown in Figure 7, average switching thresholds are approximately 0.5 for Strategic Complements and 0.4 for Strategic Substitutes, while average beliefs increase from 0.35 to 0.62 for Complements and from 0.42 to 0.64 for Substitutes.

Figure 7: Treatment effect heterogeneity by strategy type

A. Treatment effect on avoidance



B. Switching thresholds and beliefs



Notes: The figure shows the heterogeneity of treatment effects on information avoidance across interdependent types. Panel A shows the fraction of avoiders by treatment condition, separately for Strategic Complements and Strategic Substitutes. Error bars represent 95% confidence intervals. Panel B shows the switching thresholds—the fraction of others avoiding information at which a subject switches her own information decision—and beliefs about that fraction by strategy type (columns) and treatment condition (rows). The distribution of switching thresholds is shown by the histogram (gray bars); the average switching threshold by black, dashed lines; and the average belief about the fraction of others who avoid information by pink, solid lines.

ment effects by types suggest that the positive average treatment effect documented in Section 5.2 is primarily driven by the responsive behavior of Strategic Complements.

Our analysis reveals substantial heterogeneity underlying aggregate patterns of information avoidance. The average increase in information avoidance is not uniform across individuals, but is driven by Strategic Complements, while Always Getters, Always Avoiders, and Strategic Substitutes are largely unresponsive to the belief shift induced by the treatment. This pattern follows

	Dependent variable: Avoided information			
	(1) Always Getters	(2) Always Avoiders	(3) Complements	(4) Substitutes
Many Avoided	0.014 (0.033)	0.056 (0.057)	0.187* (0.096)	0.008 (0.106)
Control group mean	0.049** (0.022)	0.944*** (0.057)	0.244*** (0.066)	0.385*** (0.080)
Observations	197	41	96	90

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Notes: The table presents estimates from linear probability models where the dependent variable is an indicator equal to 1 if the subject avoided information (chose Later). Each column reports a separate regression estimated on the subsample of a distinct strategy type. The independent variable is the treatment dummy (equal to 1 for the *Many Avoided* condition and 0 for the *Few Avoided* condition).

Table 1: Treatment effect on information avoidance, by strategy type (intensive margin)

directly from the interaction between switching thresholds and beliefs about others' behavior.³⁶ These results highlight that aggregate information avoidance depends critically on the composition of strategy types within the group. Before studying how this heterogeneity aggregates into equilibrium outcomes, we first examine the mechanisms underlying strategic complementarity in information avoidance.

5.4 Mechanisms Underlying Strategic Complementarity

Section 5.3 documented substantial heterogeneity in best-response behavior and showed that the positive average effect of higher expected avoidance by others is driven entirely by Strategic Complements. In this section, we investigate the mechanisms underlying such strategic complementarity in information avoidance. While the strategy method directly identifies how individuals respond to others' behavior, it is silent on *why* some individuals exhibit increasing best responses. We therefore examine a set of psychological and preference-based channels that may give rise to strategic complementarity, and assess whether these mechanisms help distinguish Strategic Complements from other types.

The strategic complementarity in information avoidance could be driven by a variety of mechanisms. First, we focus on the "Mutually Assured Ignorance" (MAI) channel, formalized by [Bénabou \(2013\)](#) and central in our model: the idea that when a subject expects others to remain ignorant about the state, her own payoffs in the bad state become worse, making the event of discovering bad news about the state more aversive. In Section 5.4.1, we provide evidence supporting this mechanism by using our KP measures of expected anticipatory utilities conditional on finding out good,

³⁶Although Strategic Substitutes are unresponsive to the belief shift induced by this intervention, the strategy method allows predicting their behavior under wider counterfactual changes in beliefs.

bad, and no news. We then discuss the plausibility of alternative mechanisms in Section 5.4.2.

5.4.1 The Mutually Assured Ignorance Mechanism

Anticipatory utility and information avoidance. To investigate the mechanisms underlying strategic complementarity in information avoidance, we use the KP questions, which elicit participants' anticipatory utility from acquiring versus avoiding information. First, we assess whether anticipatory utility predicts information avoidance beyond established measures of information preferences. Second, we examine whether changes in anticipatory utility (as captured by the KP measures) mediate the treatment effect. Third, and crucially, we show that the treatment increases the incentives of avoiding information by lowering the anticipatory utility of discovering *bad* news, while leaving the utility of discovering *good* or *no* news unaffected, in line with the MAI mechanism in Bénabou (2013) and with our model.

The KP questions elicit participants' expected anticipatory utility *conditional on each possible information set*: discovering good news ($\hat{v}_i(g)$), bad news ($\hat{v}_i(b + \Delta)$), and no news ($\hat{v}_i((1 - \hat{q}_i)b + \hat{q}_i g)$), as well as their subjective belief that the state is bad ($1 - \hat{q}_i$). Because these questions are elicited prior the information decision, we interpret them as *expected* anticipatory utilities, evaluated before any information is acquired.

Based on the KP responses, we compute for each participant the expected anticipatory utility of acquiring information as $U_{0,i}^{Get} \equiv (1 - \hat{q}_i) \cdot \hat{v}_i(b + \Delta) + \hat{q}_i \cdot \hat{v}_i(g)$ and the expected anticipatory utility of avoiding information as $U_{0,i}^{Avoid} \equiv \hat{v}_i((1 - \hat{q}_i)b + \hat{q}_i g)$. We then define the net expected anticipatory utility of avoiding information as

$$\begin{aligned}\hat{\varphi}_i &\equiv U_{0,i}^{Avoid} - U_{0,i}^{Get} \\ &= \hat{v}_i((1 - \hat{q}_i)b + \hat{q}_i g) - [(1 - \hat{q}_i) \cdot \hat{v}_i(b + \Delta) + \hat{q}_i \cdot \hat{v}_i(g)]\end{aligned}$$

which serves as our empirical estimate of the theoretical construct φ_i .

The model predicts that an individual i should avoid information whenever the net anticipatory utility is positive ($\varphi_i > 0$). We test this prediction by regressing the avoidance dummy y_{ig} on the empirical counterpart $\hat{\varphi}_i$, either as a continuous measure or as a dummy variable for $\hat{\varphi}_i > 0$, and benchmark these specifications against the IPS.³⁷ All regressions control for field of study, strata fixed effects, demographic characteristics, and economic preferences.

Consistent with the theoretical prediction, anticipatory utility is a strong predictor of information avoidance. A unit increase in the continuous measure $\hat{\varphi}_i$ is associated with a sizable increase in the probability of avoiding information ($\hat{\beta} = 0.023$, std.err. = 0.003, $p < 0.01$). Similarly, when focusing on the specification with the dummy variable for individuals with $\hat{\varphi}_i > 0$, individuals with

³⁷The dummy is equal to 0 if $\hat{\varphi}_i < 0$, and encoded as a missing value for boundary cases where $\hat{\varphi}_i = 0$ (12 individuals in our sample). Therefore, the number of observations drops by 12 in regressions that include the dummy.

a positive net anticipatory utility are substantially more likely to avoid information than those with $\hat{\phi}_i < 0$ ($\hat{\beta} = 0.372$, std.err. = 0.046, $p < 0.01$). By comparison, the IPS is a statistically significant but much weaker predictor of avoidance ($\hat{\beta} = -0.129$, std.err. = 0.052, $p < 0.05$). Including the net utility measures increases the R^2 from 0.059 to 0.185 (continuous measure) and 0.211 (binary measure), compared to only 0.071 when including the IPS. Importantly, when both the IPS and KP-based measures are included in the same specification, the coefficients on the KP measures remain stable in magnitude and significance, indicating that anticipatory utility captures decision-relevant, environment-specific variation beyond general preferences for information.

Consistent with this interpretation, when the KP measures are included alongside the treatment indicator, the estimated treatment effect attenuates markedly. In particular, the baseline treatment effect of approximately $\hat{\beta} = -0.122$ (std.err. = 0.049, $p < 0.05$) shrinks toward zero and loses statistical significance ($\hat{\beta} = -0.081$, std.err. = 0.045, $p < 0.10$) once $\hat{\phi}_i$ is controlled for, whereas inclusion of the IPS alone leaves the treatment coefficient largely unchanged ($\hat{\beta} = -0.123$, std.err. = 0.049, $p < 0.05$). While standard causal mediation assumptions may not hold, this pattern suggests that the treatment operates primarily through changes in anticipatory utility. Complete regression results are reported in Appendix B.11.

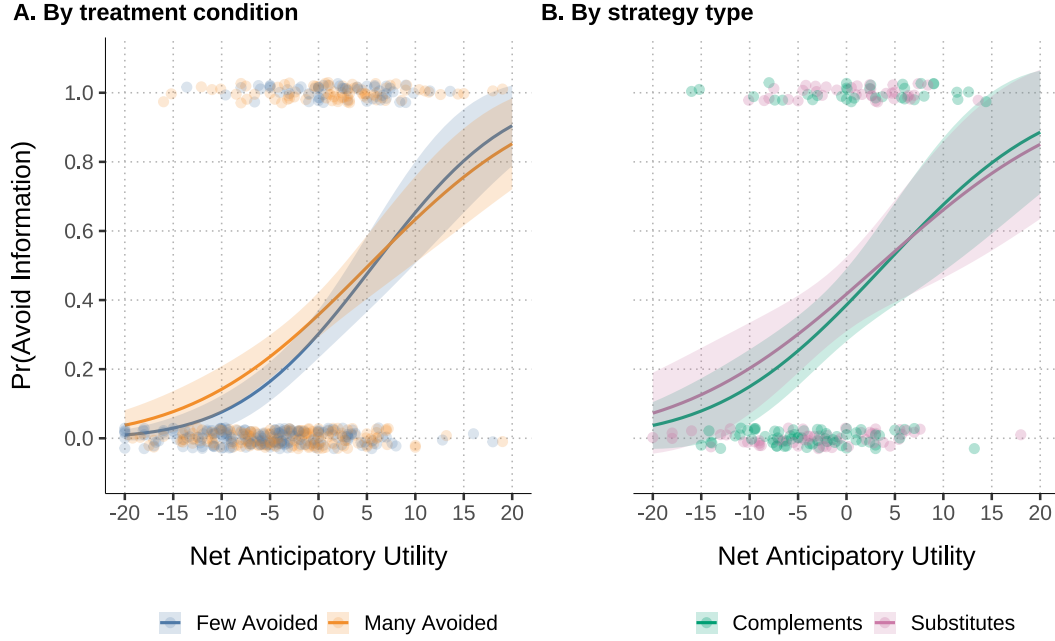
To further illustrate this point, Figure 8 plots the relationship between the probability of avoiding information and the empirical net anticipatory utility $\hat{\phi}_i$, using a probit specification. The probability of avoidance increases with $\hat{\phi}_i$. Crucially, this relationship is statistically indistinguishable across treatment conditions (*Many Avoided* and *Few Avoided*). We find the same pattern when estimating the model separately for Strategic Complements and Substitutes, where there is variation in behavior. This pattern of invariance is consistent with the theoretical prediction that, conditional on anticipatory utility, neither treatment status nor strategy type should have additional explanatory power for information avoidance.

Overall, these results indicate that the KP measures proxy relevant determinants of information avoidance and provide support for their role as a key channel through which the treatment affects behavior.

Prevalence of avoidance among others, anticipatory utility, and contagion. We now decompose the treatment effect on anticipatory utility to assess the plausibility of the MAI mechanism.

Figure 9 reports the effects of the treatment on the net anticipatory utility of avoiding information and on of its components, based on separate regressions with and without controls. The (*Many Avoided*) treatment significantly increases the net utility of avoiding information. This effect is driven by a deterioration in the anticipatory utility of discovering *bad* news: in the *Many Avoided* condition, learning bad news becomes significantly more unpleasant relative to the *Few Avoided* condition. By contrast, the anticipatory utility of discovering *good* news is unaffected, consistent with the fact that the payoff in the good state (silence, g) is invariant with respect to the number

Figure 8: Anticipatory utility and information avoidance (Probit)



Notes: The figure illustrates the relationship between the measured net expected anticipatory utility of avoiding information and the propensity to avoid information. The net utility is calculated as the difference between the utility of avoiding information and the expected utility of acquiring information, elicited from the KP questions. Each point represents an individual participant's data, jittered vertically. The solid line shows the predicted probability of information avoidance from a probit regression model, estimated separately for each treatment condition (panel A) and for each strategy type (panel B). The shaded bands indicate the 95% confidence intervals.

of avoiders in the group. This pattern of asymmetric effects provides evidence for the theoretical MAI mechanism: when others avoid, the bad state deteriorates, making learning bad news more distressing and driving information avoidance.

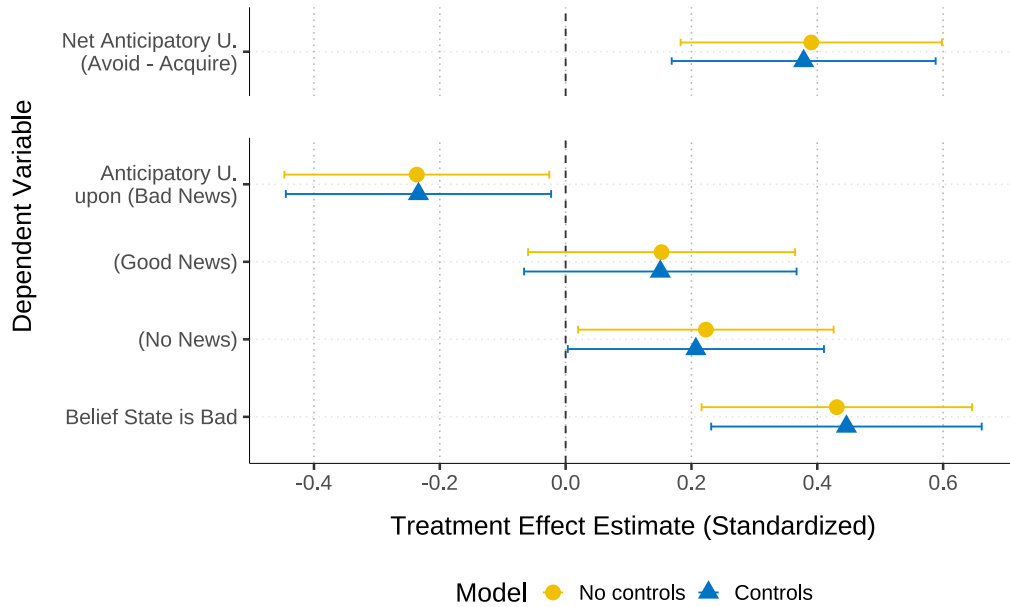
The treatment also increases subjects' belief that the state is bad. In principle, this raises the possibility that belief updating may interact with and amplify the MAI mechanism. However, existing experimental evidence suggests that exogenous shifts in prior beliefs about the bad state do not, by themselves, generate substantial changes in information avoidance (Falk & Zimmermann, 2024). Accordingly, the dominant channel operated by the treatment in our setting appears to be the heightened disutility of discovering bad news when others avoid information.

Taken together, the decomposition exercise in Figure 9 shows stronger incentives to avoid information in groups arise primarily from the expected deterioration of anticipatory utility in the event bad news, consistent with the MAI mechanism of Bénabou (2013) and with our model.

5.4.2 Alternative Mechanisms

While the evidence is consistent with the MAI channel, the interdependence of information avoidance could, in principle, arise from other behavioral mechanisms. We discuss some of the main

Figure 9: Mechanisms: treatment effects on anticipatory utility



Notes: The figure reports the treatment effects on net anticipatory utility and each of its components from separate regressions. Yellow circles show estimates from models without further controls; blue triangles correspond to models that include controls for demographics, preferences, and fixed effects (see notes in Figure 5). The top row reports effects on the net anticipatory utility (of avoiding information relative to acquiring it); lower rows report effects on its components: the anticipatory utilities upon learning bad, good, or no news, and the subjective belief that the state is bad. Coefficients are standardized; whiskers indicate 95% confidence intervals (robust standard errors).

possibilities below.

Conformity and herding. A natural alternative explanation is conformity or information herding, whereby individuals derive direct utility from matching others' behavior or infer the optimal action from others' past choices. While these mechanisms are grounded in distinct psychological motives, they share the same empirical prediction: an individual's likelihood of information avoidance increases with the perceived share of avoiders in the group. We therefore assess whether the observed treatment effects are consistent with *conformity*-driven behavior (for brevity we use "conformity" as a handle for both mechanisms). We implement four tests that yield little support for conformity or herding. First, under conformity, the utility of acquiring information —particularly when discovering *good* news, whose material payoff is invariant to others' choices— should differ across conditions. Consistent with Figure 9, we find no such effect. Second, conformity predicts that beliefs about others' behavior should affect information avoidance *independently* of beliefs about the state. In contrast, we find that beliefs about others' avoidance predict own avoidance only when interacted with beliefs that the state is bad (Appendix Table B.13); once this interaction is included, beliefs about others' information decisions alone lose explanatory power. This pattern aligns with the groupthink mechanism rather than to conformity. Third, if conformity were op-

erative, the treatment should shift the switching thresholds of Strategic Complement. As shown in Figure 7, we observe no such threshold shifts across treatment conditions. Fourth, a pure conformity channel would not predict interaction between treatment and reciprocity; however, we find mild evidence of such interaction (Table B.5). Taken together, these tests provide little support for conformity or herding, and in several cases reinforce the interpretation based on strategic complementarity and groupthink.

Social learning. Another alternative is social learning, whereby individuals update their beliefs about their own preferences based on others' past behavior. As a result, this may generate differences in information avoidance across treatment conditions. While we do not directly test for this channel, it is unlikely to play a major role in our settings, as subjects have prior experience with the noise and individual payoffs are transparent.

Non-linear preferences over volume. A final alternative is that subjects may perceive a two-point reduction in volume (out of 100 points) as less valuable in the *Many Avoided* condition relative to the *Few Avoided* condition. While possible in principle, casual experience suggests that such small changes are only weakly perceptible, making it unlikely that this channel generates the observed treatment effects unless subjects systematically exaggerate the magnitude and asymmetry of volume changes across conditions.

In summary, the evidence in this sub-section indicates that strategic complementarity in information avoidance is primarily driven by changes in anticipatory utility, consistent with the MAI mechanism. Having clarified why some individuals amplify others' information avoidance, we now turn to the aggregate implications of this heterogeneity. In the next section, we study how the composition of strategy types within a group shapes equilibrium levels of information avoidance.

6 Equilibrium Implications of Interdependent Avoidance

The analysis in Section 5.3 shows that individuals differ substantially in how they respond to others' information decisions. While some amplify others' avoidance, others either counteract it or do not respond at all. The presence of strategic complements, strategic substitutes, and unconditional types suggests that the equilibrium level of avoidance in a group depends on its composition of types, and that small compositional changes may generate large differences in aggregate information avoidance. In this section, we study how this heterogeneity maps into group-level outcomes. Using simulations based on the empirically estimated distribution of strategy types, we examine how group composition determines *equilibrium* levels of information avoidance.

We refer to the equilibrium prevalence of information avoidance as an “information equilibrium”. Formally, an information equilibrium is a share λ^* of information avoiders such that a proportion λ^* of group members best-responds to it by avoiding information, while the remaining proportion best-responds by acquiring it.

6.1 The Heterogeneity of Equilibrium Avoidance across Strategy Types

6.1.1 Method

To derive the set of information equilibria (“equilibrium set”) of a given group, we proceed in two steps: 1) from the individual best response schedules $y_i^*(\lambda_{-i})$, derive the group-level (“aggregate”) best response schedule $Y^*(\lambda)$, and 2) find the equilibrium set as the set of fixed points of the schedule Y^* .

We derive the aggregate best response schedule Y^* as follows. Each individual best response schedule is a function $y_i^* : [0, 1] \rightarrow \{0, 1\}$ mapping each share of avoiders to a decision to avoid (1) or acquire (0) information. For each λ , we compute and add up each individual best response, $\sum_i y_i^*(\lambda)$, giving the total number of best responding avoiders at λ . We divide by the number of group members to compute the proportion of best responding avoiders $Y^*(\lambda)$. For example, consider a simple group composed of one Always Getter (with $y_1^*(\lambda_{-1}) = 0$ for all λ_{-1}) and one Always Avoider (with $y_2^*(\lambda_{-2}) = 1$). Then the aggregate best response function is $Y^*(\lambda) = 0.5$ for all λ : for any proportion of avoiders λ , half of the group members best-respond by avoiding information.

Since, in practice, the measurement of y_i^* is coarse (possible only at a finite set of values of λ), the derived aggregate schedule Y^* is also coarse, with the risk of missing equilibrium points. Estimating its full shape requires an assumption about the missing intervals of λ . Our approach is to impute missing values by linear interpolation. This approach has a number of desirable properties: it is simple and intuitive, provides good linear approximations around observed points under the assumption that Y^* is continuous, and ensures that the estimated schedule varies smoothly with λ . While we acknowledge the possibility of discontinuous jumps in Y^* , we are uncertain about the location of such jumps, and the linear interpolation does not presuppose specific locations.

Once the shape of Y^* is estimated, we identify the equilibrium points as the fixed points of Y^* , namely, the set of values λ^* such that $Y^*(\lambda^*) = \lambda^*$; graphically, it is the set of points where the schedule Y^* crosses the identity line.

6.1.2 Result

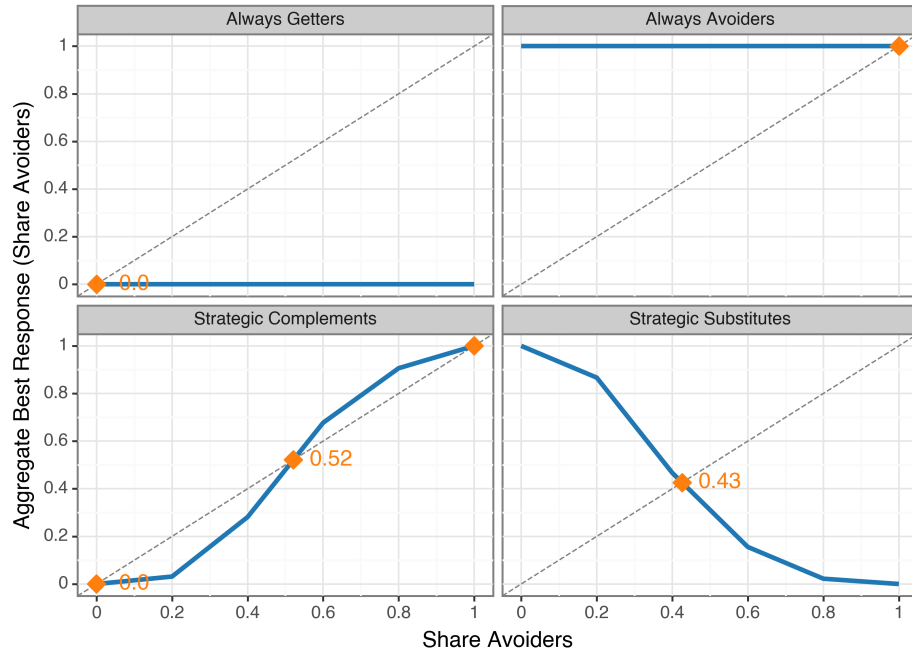
To build intuition, we start our analysis by considering homogeneous groups, namely, groups composed entirely of a single strategy type. We construct these groups using all participants of each type. Figure 10 plots the best-response function $Y^*(\lambda)$ for each type. Blue lines show the estimated

schedule $Y^*(\lambda)$. The equilibrium set is shown as orange diamonds at the intersection(s) of $Y^*(\lambda)$ and the identity line (dotted).

Groups of unconditional types (Always Getters or Always Avoiders) display a flat aggregate schedule, reflecting the fact that the best-responses of all its members are independent of others' decisions. Those groups have a unique information equilibrium, where everybody is either informed (Always Getters) or uninformed (Always Avoiders).

In contrast, homogeneous groups of conditional types generate monotonic schedules: negatively sloped for Strategic Substitutes, and positively sloped for Strategic Complements.³⁸ Substitutes feature a unique interior equilibrium. Complements, however, exhibit multiple equilibria—including fully informed (MAA), fully uninformed (MAI), and an interior equilibrium—providing empirical evidence for the theoretical prediction of [Bénabou \(2013\)](#). The multiplicity of equilibria captures the risk of collective ignorance, even when full awareness is also sustainable. The plot therefore illustrates Corollary 1: Complements can sustain both the MAA and the MAI equilibrium, while Substitutes cannot reach either.

Figure 10: Aggregate best-response functions and information equilibria (homogeneous groups)



Notes: Aggregate (group-level) best response schedules $Y^*(\lambda)$ for homogeneous groups. Each panel shows the aggregate schedule constructed using all experimental participants of the indicated strategy type (not bootstrap samples). Blue lines show $Y^*(\lambda)$, the share of group members who best-respond by avoiding information at each share of avoiders λ . The dotted diagonal line is the identity line. Orange diamonds mark the information equilibria. Always Getters and Always Avoiders have flat schedules with corner equilibria. Strategic Complements have an upward-sloping schedule with three equilibria ($\lambda^* = 0, 0.52, 1$). Strategic Substitutes have a downward-sloping schedule with one interior equilibrium ($\lambda^* = 0.43$).

³⁸The precise piece-wise slopes will depend on the distribution of individual switching thresholds.

6.2 Group Composition and Threshold Effects in Equilibrium Avoidance

The diversity of shapes in Y^* schedules suggests that equilibrium sets may vary greatly with group composition. To study this relationship systematically, we simulate groups with different compositions of strategy types based on empirical individual schedules and characterize the resulting equilibrium outcomes.

6.2.1 Method

The simulation algorithm is summarized in Table 2. In short, define a group size n and a sequence of compositions $\Pi \equiv \{\pi_t\}_{t=1,\dots,T}$ where π_t is the vector $\pi_t \equiv (\pi_{AG}, \pi_{AA}, \pi_{SC}, \pi_{SS})_t$ whose elements are probabilities of Always Getters, Always Avoiders, Complements, and Substitutes, respectively, summing up to 1. For each step t , bootstrap B groups of size n and composition π_t . For each bootstrap group $b = 1, \dots, B$, construct the aggregate best response Y_b^* (including the linear interpolation), and find the set of fixed points of Y_b^*, λ_b^* , classifying their stability. At the composition level (step t), compute summary statistics of the union (across the B bootstrap groups) of the equilibrium sets.

Algorithm 1: Simulation of Equilibrium Sets Across Group Compositions

Input: Group size n , sequence of group compositions $\Pi = \{\pi_t\}_{t=1}^T$, bootstrap iterations B , individual schedules $\{y_i^*\}_{i=1}^N$

Output: Distribution of equilibrium sets and statistics for each composition

foreach composition $\pi_t \in \Pi$ **do**

- Initialize $\pi_t = (\pi_{AG}, \pi_{AA}, \pi_{SC}, \pi_{SS})_t$
- for** $b \leftarrow 1$ **to** B **do**
 - // Bootstrap sampling
 - Draw n individuals with replacement from $\{y_i^*\}_{i=1}^N$, probabilities π_t
 - // Construct aggregate best response
 - foreach** $\lambda \in [0, 1]$ **do**
 - $Y_b^*(\lambda) \leftarrow \frac{1}{n} \sum_{i=1}^n y_i^*(\lambda)$
 - Apply linear interpolation for unmeasured λ
 - // Find equilibria
 - $\Lambda_b \leftarrow \{\lambda^* : Y_b^*(\lambda^*) = \lambda^*\}$
 - Classify stability of each $\lambda^* \in \Lambda_b$
- // Aggregate across bootstrap samples
- Compute mean and 95% CI of $Y_b^*(\lambda)$ for each λ
- Compute quantiles of $\{\Lambda_b\}_{b=1}^B$

return $\{Y_t^*(\lambda), \Lambda_t\}_{t=1}^T$

Table 2: Simulation algorithm

We simulate over a grid of compositions where each type's proportion varies from 0 to 1 in

increments of 0.1. Given the constraint that the probabilities sum to 1, the grid includes 286 distinct compositions. For each composition, we bootstrap $B = 100$ groups of $n = 13$ members.³⁹ We focus on small groups because composition manipulation remains policy-relevant at this scale. The simulation thus involves 367,600 sampled individuals across 28,600 groups, across 286 group compositions.

6.2.2 Results

Among the large number of sequences based on the 286 compositions, we restrict attention to simple sequences that vary only two types: one conditional ($C \in \{SC, SS\}$) and one unconditional ($U \in \{AG, AA\}$), with $\pi_U + \pi_C = 1$ and π_U varying from 0 to 1.⁴⁰ These restricted sequences reveal the effect of adding unconditional types to groups initially composed entirely of conditional types. This design is motivated by the observation that groups of Strategic Complements can support multiple equilibria; introducing unconditional types (whose behavior is independent of others' choices) allows us to ask questions such as "How many Always Avoiders does it take to tip the equilibrium toward full avoidance?"

We show how the schedule Y^* responds to changes of group composition in Appendix Figure C.1. Starting with homogeneous groups of conditional types (whose schedules slope upward or downward), increasing the participation of unconditional types progressively flattens the schedule and shifts it towards the extremes: $Y^* = 0$ for Always Getters and $Y^* = 1$ for Always Avoiders.

The aggregate best-response schedules (which vary across bootstrap groups) generate each a set of equilibrium points. The distribution of information equilibria by group composition is shown in Figure 11. Each panel starts with groups composed entirely of one conditional type (Strategic Complements in Panels A and C; Strategic Substitutes in Panels B and D). Moving rightward, each facet gradually increases the participation of one unconditional type (Always Avoiders in Panels A and B; Always Getters in Panels C and D).

Homogeneous groups of Strategic Complements (facets "AAs: 0%" and "AGs: 0%") exhibit a wide range of equilibria. All such groups feature a full-avoidance and a full-acquisition equilibrium (both typically stable), plus an interior equilibrium (typically unstable). As the participation of Always Avoiders increases (Panel A), the distribution shifts rightward: full avoidance remains as an equilibrium, while full acquisition gradually disappears.

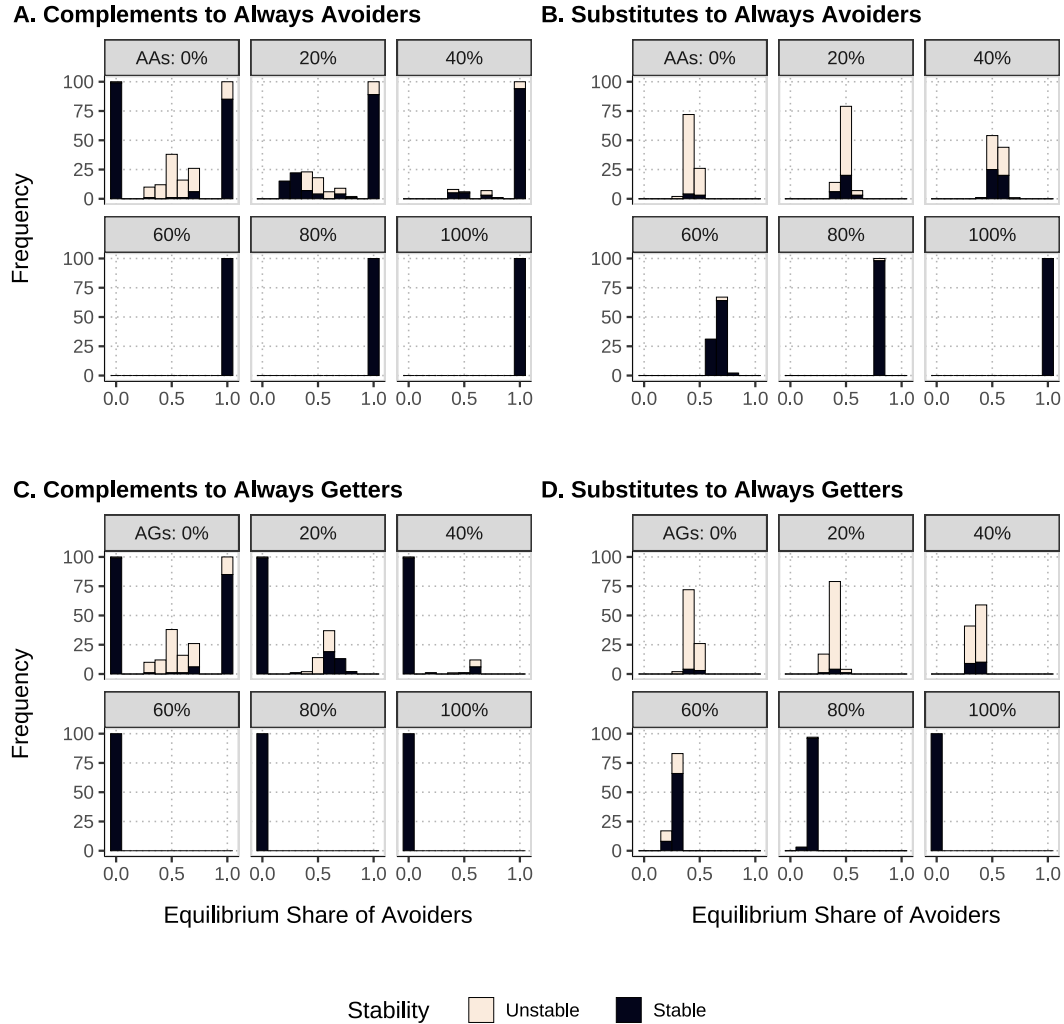
Strategic Substitutes (Panels B and D) show a very different picture. Homogeneous groups of Substitutes ($\pi_{SS} = 1$) feature a single interior equilibrium, typically unstable. As expected, in-

³⁹When n divides evenly into round numbers (e.g., $n = 10$), continuous portions of Y_b^* can lie exactly on the identity line, creating infinitely many equilibrium points and apparent non-monotonicities in the distribution of equilibria as composition varies. This introduces technical details about how to count and adjust for these when calculating the relative frequency of equilibria. We use a non-round number (e.g., $n = 13$) to ensure that discrete steps in Y_b^* do not align with the identity line.

⁴⁰Higher-dimensional compositions are analyzed in Appendix C.

Introducing Always Avoiders shifts this equilibrium rightward (toward higher avoidance), while introducing Always Getters shifts it leftward (toward lower avoidance). Eventually, as the participation of unconditional types becomes sufficiently high, only corner equilibria remain: either full-avoidance or full-acquisition equilibria.

Figure 11: Distribution of information equilibria across group compositions



Notes: The figure shows how the distribution of information equilibria (i.e., the equilibrium prevalence of avoidance within each bootstrapped group) varies with group composition. Within each panel, each facet corresponds to a given share of unconditional types —Always Avoiders (“AA”, panels A and B) or Always Getters (“AG”, panels C and D)—and a remainder share of a conditional type (Strategic Complements or Substitutes). Bars report the absolute frequency (across bootstrapped groups) of the equilibrium share of avoiders indicated on the x -axis. Stable equilibria are shown in dark bars, and unstable equilibria in light bars.

Since unstable equilibria are unlikely to persist in practice, in what follows we restrict attention to stable equilibria.

Policy-makers may be interested in specific features of the equilibrium distribution. For example, they may want to know the expected avoidance rate, or the maximum possible avoidance

(a worst-case scenario). While the histograms in Figure 11 show the full distribution of equilibria, they obscure how summary features (such as the mean, median, or other quantiles) vary with group composition. To illustrate these relationships more clearly, Figure 12 plots how such features change as the proportion of unconditional types increases.

Figure 12 shows how the distribution of equilibrium outcomes varies with group composition. Each panel plots quantiles of the equilibrium share of avoiders (colored lines) across 100 simulated groups at each composition level. Gray vertical bars indicate the range (minimum to maximum) at each composition.

Across all panels, the relationship is approximately monotonic—adding Always Avoiders increases equilibrium avoidance, while adding Always Getters decreases it—but the nature of this monotonicity contrasts markedly between types. For Strategic Complements (Panels A and C), the relationship exhibits sharp threshold effects: small increases in the participation of unconditional types trigger discontinuous jumps in equilibrium outcomes. For example, in Panel A, when Always Avoiders’ participation increases from 10% to 20%, the median equilibrium share of avoiders jumps from approximately 0.2 to 1 (full avoidance). Similarly, in Panel C, a threshold around 40-50% participation of Always Getters causes the maximum avoidance in the equilibrium distribution to collapse from 0.6 to 0 (full acquisition). These thresholds reveal critical tipping points where small compositional changes produce large shifts in equilibrium behavior.

In contrast, Strategic Substitutes (Panels B and D) exhibit smooth relationships. As the participation of unconditional types increases, equilibrium outcomes adjust gradually without discontinuous jumps. The spread between quantiles remains relatively stable, indicating that all quantiles respond similarly to compositional changes.

To quantify the threshold effects more precisely, Figure 13 plots the slope of each curve in Figure 12—the rate of change in equilibrium outcomes per unit increase in unconditional type participation. For Strategic Complements (Panels A and C), slopes exhibit dramatic spikes at specific compositions, reaching values of 5-8. These spikes identify the thresholds where equilibria shift abruptly: in Panel A, slopes peak at 10-20% Always Avoider participation (median and Q1) and at 50% (minimum); in Panel C, they peak at 30-50% Always Getter participation (maximum and Q3). At these critical compositions, a 10-percentage-point increase in unconditional type participation can shift quantiles of equilibrium avoidance by 50-80 percentage points. Outside these threshold regions, slopes approach zero, indicating stable quantiles that are insensitive to small compositional changes.

In stark contrast, Strategic Substitutes (Panels B and D) exhibit no threshold effects. Slopes remain modest and stable across all compositions, fluctuating between -2 and +2, with no sharp spikes. Equilibrium outcomes respond gradually and uniformly to compositional changes, with no critical tipping points.

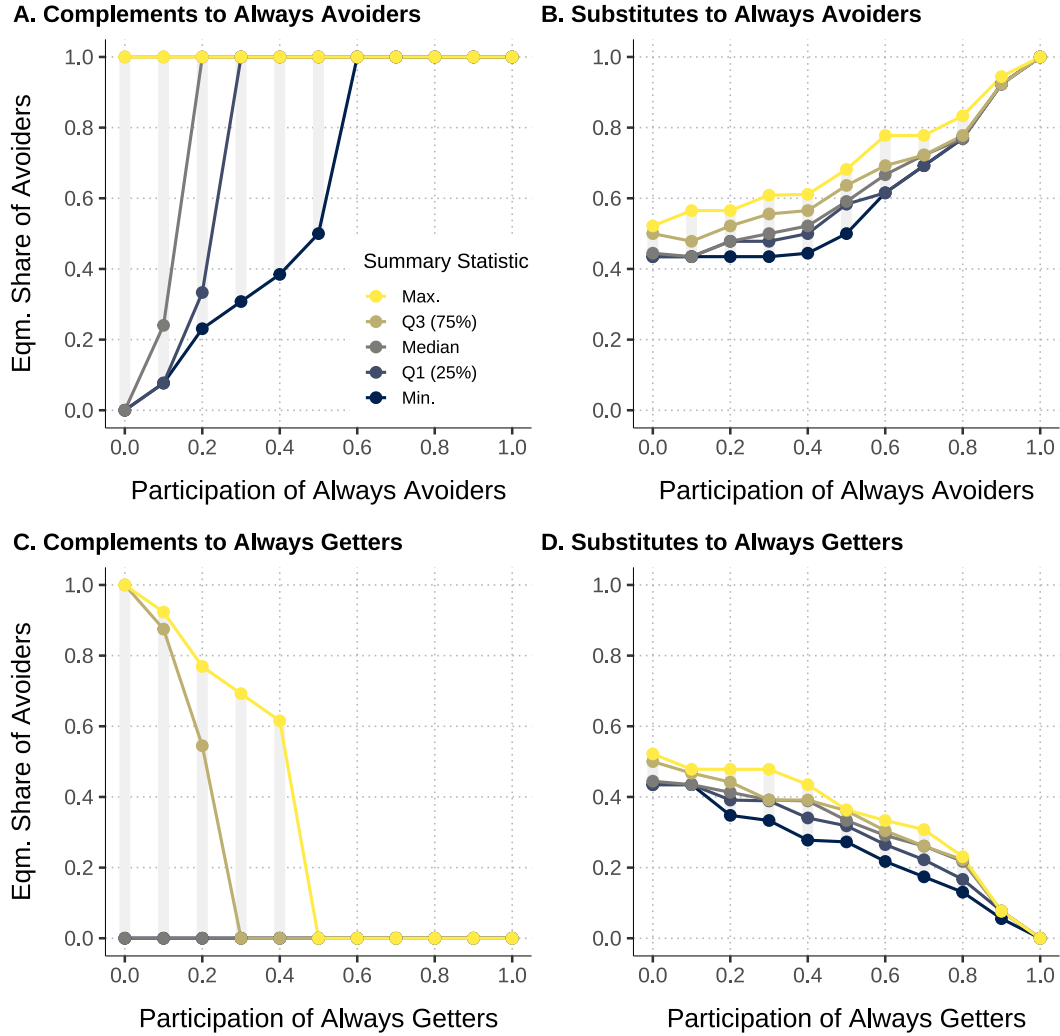
In sum, aggregate responses to the composition of heterogeneous types are approximately

monotonic but can be highly non-linear.

Our findings have policy implications for organizational design. In groups of Strategic Complements, small interventions —seeding a few information seekers, or conversely, tolerating a few more avoiders— can tip entire groups between widespread information acquisition and collective ignorance. In groups of Strategic Substitutes, the same interventions produce only incremental effects.

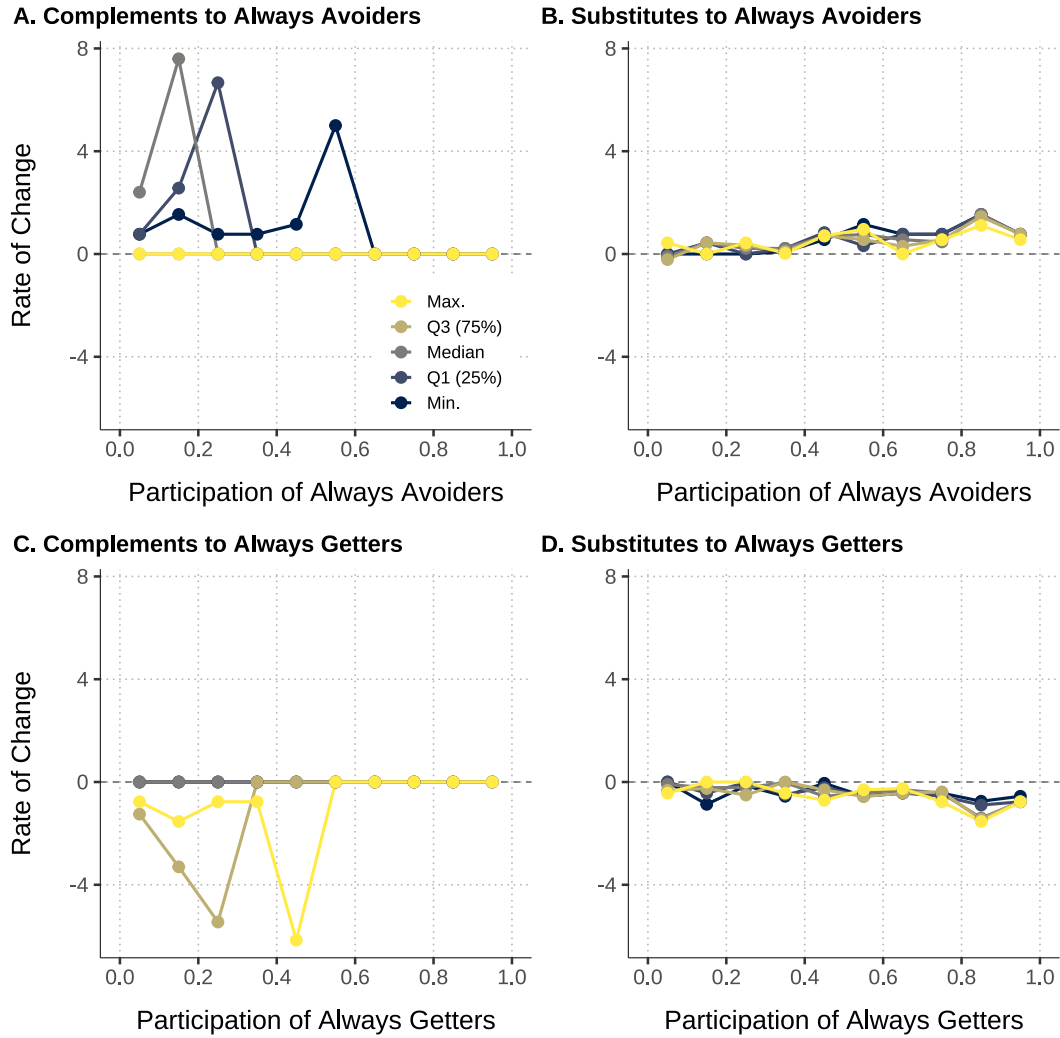
Overall, our findings show that the composition of strategy types strongly determines equilibrium outcomes—different mixtures produce widely different equilibrium sets—and that critical thresholds exist where small compositional changes trigger large behavioral responses.

Figure 12: Effect of group composition on equilibrium distribution



Notes: The figure shows the effect of group composition of strategy types on the equilibrium prevalence of information avoidance. Each panel focuses on mixtures of two strategy types: Panels A and B start with groups of Strategic Complements and gradually increase the participation of Always Avoiders (Panel A) and of Always Getters (Panel B); Panels C and D start with groups of Strategic Substitutes and gradually increase the participation of Always Avoiders (Panel C), and of Always Getters (Panel D). The x -axis represents the participation of the unconditional type; y -axis measures the share of information avoiders. Gray vertical bars indicate the support of the distribution of information equilibria (over bootstrapped groups) of a given composition; dots indicate summary statistics of the distribution of equilibria (maximum in yellow, median in dark gray, and minimum in dark blue).

Figure 13: Equilibrium sensitivity to group composition: threshold effects



Notes: Slope of curves in Figure 12 (i.e., rate of change in quantiles of the equilibrium set as unconditional type participation increases). Each colored line represents the slope of a specific quantile. For Strategic Complements (Panels A, C), sharp spikes identify threshold locations where equilibria shift abruptly. Threshold locations differ by quantile. For Strategic Substitutes (Panels B, D), slopes remain stable (-2 to +2), indicating gradual responses without thresholds. Dashed line marks zero slope.

7 Conclusion

We live in an era of abundance of information in a variety of domains —politics, health, financial, environmental. Despite the wide availability of virtually free information, sometimes individuals prefer to avoid it, choosing, in effect, to remain in a state of willful ignorance. In this paper, we study whether this behavior is interdependent among individuals in a setting where remaining ignorant imposes negative payoff externalities on other members. Our experimental evidence shows that, on average, willful ignorance can be contagious: individuals are more likely to avoid payoff-improving and freely available information when they expect others to do so, and hence, when they expect the outcome in the bad state to be more severe (and news revealing a bad state to be more aversive). Our findings suggest that the interdependence is mediated by an increase in expected anticipatory disutility of discovering bad news as a result of others' information avoidance, and it is moderated by social preferences, particularly, by reciprocity. Beyond average effects, however, there is substantial heterogeneity in individual strategy types: while we empirically find individuals who exhibit strategic complementarity and who amplify willful blindness as in [Bénabou \(2013\)](#), we also document the existence of an equally sizeable share of strategic substitutes, who counter the spread of ignorance. This heterogeneity of reaction functions suggests that the degree of aggregate contagiousness of willful ignorance may depend crucially on both the composition of the group in terms of reaction functions and the distribution of subjective expectations about others' behavior. Simulations show that, in effect, changes in group composition generate a wide variety of information equilibria, ranging from full information acquisition (and maximum mitigation of the bad state outcome) to full willful ignorance (and maximum severity), and that the effects can be strong and non-linear.

Organization designers and policy-makers are often interested in promoting the take-up of information and learning within their organizations and populations. Our study adds nuance to the recommendations from the literature on information preferences, which typically focus on individual decision settings. Our findings suggest that incentivizing a person to acquire information (which may be beneficial if the individual is in isolation) may generate positive or negative externalities in the informational responses other members. Therefore, policy-makers might benefit from taking into account these “cognitive externalities” and general equilibrium effects when designing policies that aim at increasing the aggregate take-up of information in groups when individuals' outcomes are interlinked.

Our results suggest some implications for organization design. Taking the group as given, a potential way to encourage information take-up is by correcting potential misperceptions about how others deal with information. Another way is to design the group composition in terms of strategy types when assembling teams. Our simulations show that the group composition has a strong impact in the distribution of equilibrium shares of informed members, and suggest the

equilibrium responses can be non-linear in the mixture of types, in a way that there are regions in which the same amount of change in composition delivers the strongest responses.

Our investigation has some limitations and motivates avenues for future research. First, because our experimental setup is based on the setup of [Bénabou \(2013\)](#) where there is a theoretical link between agents' decisions and the severity of the outcome, our design cannot disentangle whether (and how much of) the contagiousness arises from variations in the severity of the outcome ("severity effect") or from variations in the proportion of others who choose to remain ignorant ("herding effect"). This limitation applies both to the average treatment effect and the individual best-response functions. We view our study as a first step in establishing the empirical existence of interdependence in information decisions. Further research is needed to disentangle the severity effect from the herding effect. Second, we consider a static decision-making setting, where all individuals choose their information decisions simultaneously. In many applications, such decisions occur sequentially, motivating future studies on willful ignorance in groups in dynamic settings.

Our study highlights that understanding the diverse ways in which individuals react to others' information decisions is crucial for designing environments that promote the acquisition of information, which, usually, improves subsequent decision-making in organizations.

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A Theoretical Appendix

A.1 Mapping of Notation

Table A.1 shows the correspondence between the notation used in Section 3 and Bénabou (2013).

Concept	Bénabou (2013)	This Paper
Good state	$\sigma = H$	$\sigma = G$
Bad state	$\sigma = L$	$\sigma = B$
Date-2 payoff in Good state	$U_2^i \equiv \sum_{j=1}^N a_H^{ji} e^j + b_H^{ji} (1 - e^j)$	$u_i = g$ (independent of $e^j \forall j$)
Date-2 payoff in Bad state	$U_2^i \equiv \sum_{j=1}^N a_L^{ji} e^j + b_L^{ji} (1 - e^j)$	$b \equiv \bar{b} - \Delta \cdot (y_i + \lambda_{-i}(N - 1))$, where $y_i = 1$ if i avoids information (0 otherwise) and $\lambda_{-i} \equiv \frac{\sum_{j \neq i} y_j}{N-1}$
Effort (Bad-state mitigating action)	$e^i \in \{0, 1\}$	(Implicit)
Return to i from j with $e^j = 1$ in state H	a_H^{ij}	0 for all i, j
Return to i from j with $e^j = 0$ in state H	b_H^{ij}	0 for all i, j
Return to i from j with $e^j = 1$ in state L	a_L^{ij}	$-\frac{1}{N}$ for all i, j
Return to i from j with $e^j = 0$ in state L	b_L^{ij}	0 for all i, j
Prior belief that state is Good	q	(Idem)
Damage from others' choices	$d_L^i \equiv \sum_{j \neq i} (b_L^{ji} - a_L^{ji}) \geq 0$	$\frac{N-1}{N} > 0$
Optimal payoff (full info, Good state)	$A_H^i \equiv \sum_{j=1}^N a_H^{ij}$	0
Optimal payoff (full info, Bad state)	$B_L^i \equiv \sum_{j=1}^N b_L^{ij}$	$\bar{b}$

Table A.1: Mapping of notation: Bénabou (2013) and this paper

A.2 Proofs

Proof of Lemma 1 Consider v_i everywhere convex. Agent i prefers to avoid information if $\varphi_i > 0$, i.e.

$$v_i(pb + (1 - p)g) > pv_i(b + \Delta) + (1 - p)v_i(g)$$

Since $v_i(\cdot)$ is strictly increasing, we have $v_i(b + \Delta) > v_i(b)$, hence we must have

$$v_i(pb + (1 - p)g) > pv_i(b) + (1 - p)v_i(g)$$

but this contradicts convexity. Therefore, the agent never avoids information, i.e., the agent is an Always Getter. ■

Proof of Lemma 2 Suppose v_i everywhere concave. First note that the net incentive to avoid

information increases in the share of other group members who avoid information:

$$\begin{aligned}\frac{\partial \varphi_i}{\partial \lambda_{-i}} &= p v'_i (p b + (1-p) g) \frac{\partial b}{\partial \lambda_{-i}} - p v'_i (b + \Delta) \frac{\partial b}{\partial \lambda_{-i}} \\ &= [v'_i (p b + (1-p) g) - v'_i (b + \Delta)] p \frac{\partial b}{\partial \lambda_{-i}} \\ &> 0\end{aligned}$$

by concavity of v_i and assumption 3. It follows that:

- If $\varphi_i (\lambda_{-i} = 0) > 0$ then by monotonicity $\varphi_i (\lambda_{-i}) > 0$ for any λ_{-i} , that is, the agent is an Always Avoider.
- If $\varphi_i (1) < 0$ then by monotonicity $\varphi_i (\lambda_{-i}) < 0$ for any λ_{-i} , that is, the agent is an Always Getter.
- If $\varphi_i (0) < 0$ and $\varphi_i (1) > 0$, then by continuity there exists an interior λ_{-i}^* such that the agent acquires information whenever $\lambda_{-i} < \lambda_{-i}^*$ and avoids it when $\lambda_{-i} > \lambda_{-i}^*$ (strategic complementarity). ■

Proof of Lemma 3 We assume throughout that $b(0) + \Delta < \widehat{U}_i$. Suppose that if everyone else avoids information, then at date 1 the expected utility under ignorance is so low that it falls below the reference level: $p b (1) + (1-p) g < \widehat{U}_i$. Then v_i is convex at $p b (1) + (1-p) g$. By convexity,

$$v_i (p b (1) + (1-p) g) < p v_i (b (1)) + (1-p) v_i (g)$$

so even without instrumental value of information Δ , avoiding information is dominated by acquiring it. Since v_i increasing, adding the instrumental value makes avoiding information even less desirable:

$$v_i (p b (1) + (1-p) g) < p v_i (b (1)) + (1-p) v_i (g) < p v_i (b (1) + \Delta) + (1-p) v_i (g)$$

Therefore, $\varphi^i (1) < 0$. When all other agents acquire information, agent i prefers to avoid. Suppose that if everyone else acquires information, then at date 1 the expected utility under ignorance goes above the reference level: $p b (0) + (1-p) g > \widehat{U}_i$. Then v_i is concave at $p b (0) + (1-p) g$. By concavity,

$$v_i (p b (0) + (1-p) g) > p v_i (b (0)) + (1-p) v_i (g)$$

For avoidance to be optimal, we need

$$v_i (p b (0) + (1-p) g) > p v_i (b (0) + \Delta) + (1-p) v_i (g)$$

Define the difference in expected utilities $d \equiv v_i(g) - v_i(pb(0) + (1-p)g)$. Then, the avoidance condition can be re-expressed as:

$$v_i(g) - d > pv_i(b(0) + \Delta) + (1-p)v_i(g)$$

$$\text{i.e. } p[v_i(g) - v_i(b(0) + \Delta)] > d$$

When $d \rightarrow 0$, the condition above always holds. Therefore, by continuity, there exists a value $d^* > 0$ such that for $d < d^*$ the agent avoids information. If $d < d^*$, the agent avoids information when all other agents in the group acquire it ($\varphi_i(\lambda_{-i} = 0) > 0$) and acquires information when all others avoid it ($\varphi_i(\lambda_{-i} = 1) < 0$). By continuity of $\varphi_i(\lambda_{-i})$, there exists an interior threshold λ^{**} such that agent i strictly prefers to avoid information if $\lambda_{-i} < \lambda_i^{**}$ and strictly prefers to acquire information if $\lambda_{-i} > \lambda_i^{**}$ (strategic substitutability). ■

A.3 Social Preferences and Information Avoidance

In our decision environment, acquiring information generates a positive externality on other agents. This creates a natural role for social preferences. An agent who internalizes others' payoffs faces stronger incentives to acquire information, but also experiences anticipatory utility from others' outcomes, which may introduce countervailing forces. We extend the baseline model to incorporate two forms of social preferences: altruism and reciprocity.

A.3.1 Altruism

First, we consider altruistic preferences. Let $a(x)$ denote the **altruistic anticipatory utility (AU) function**: the direct utility an agent derives from anticipating the outcomes of other agents. The agent's total anticipatory utility is $v(x) + \alpha \cdot a(x)$, where $\alpha \geq 0$ is the weight placed on the anticipatory utility from others' outcomes, nesting the baseline model when $\alpha = 0$. The function $a(x)$ encodes how the agent's anticipatory utility responds to changes in other agents' outcomes. We assume a increasing and continuously differentiable. $a(x)$ may differ from $v(x)$ in its curvature and other properties.

The net incentive to avoid information of an altruistic agent is:

$$\begin{aligned} \varphi^{Altr} &= \underbrace{[v(\mu) + \alpha a(\mu)]}_{\text{AU under ignorance}} - \underbrace{\{(1-q)[v(b+\Delta) + \alpha a(b+\Delta)] + q[v(g) + \alpha a(g)]\}}_{\text{Expected AU with information}} \\ &= \underbrace{v(\mu) - [(1-q)v(b+\Delta) + qv(g)]}_{\equiv \varphi^v} + \alpha \cdot \underbrace{\{a(\mu) - [(1-q) \cdot a(b+\Delta) + q \cdot a(g)]\}}_{\equiv \varphi^a} \end{aligned}$$

where we remove the dependence on λ_{-i} from b for notational simplicity. The second equality

implies that the net incentive to avoid information can be decomposed into the standard incentive φ^v and the altruistic incentive $\alpha\varphi^a$.

Altruism and incentive to avoid information Differentiating the net incentive to avoid with respect to α :

$$\frac{\partial \varphi^{Altr}}{\partial \alpha} = \varphi^a = \underbrace{a(\mu) - a(\mu + (1-q)\Delta)}_{(1)} + \underbrace{a((1-q)(b+\Delta) + qg) - [(1-q) \cdot a(b+\Delta) + q \cdot a(g)]}_{(2)}$$

Altruism increases incentives to avoid to the extent that $\varphi^a > 0$. Then, similarly to the selfish case, there are two forces determining the sign of φ^a . Given a increasing, effect (1) is always negative: information acquisition improves expected payoff from μ to $\mu + p\Delta$, and an altruistic agent values this (feels better) for others; this increases incentives to acquire information. Effect (2) depends on the curvature of a : the effect is positive if a is concave, negative if a convex, zero if a linear. Hence, for linear and convex a , the total effect is strictly positive; for concave a , the two effects oppose each other, so the total effect is ambiguous.

Altruism and strategic interdependence The effect of altruism on the interdependence is captured by the cross-partial:

$$\begin{aligned} \frac{\partial^2 \varphi^{Altr}}{\partial \lambda \partial \alpha} &= \frac{\partial \varphi^a}{\partial \lambda} = (1-q) \{a'(b+\Delta) - a'(\mu)\} \\ &= (1-q) \cdot a''(z^*) \cdot \underbrace{[(b+\Delta) - \mu]}_{<0} \end{aligned}$$

where the third equality follows from the Mean Value Theorem, which guarantees the existence of $z^* \in (b+\Delta, \mu)$ since a' is continuous and differentiable in $(b+\Delta, \mu)$.

Lemma 4 *The sign of the effect of altruism on interdependence (cross-partial) is fully determined by the curvature of the altruistic anticipatory utility function a . Specifically, altruism:*

- (i) *amplifies the interdependence if $a''(z^*) > 0$*
- (ii) *has no effect if $a''(z^*) = 0$*
- (iii) *dampens the interdependence if $a''(z^*) < 0$*

where z^* is implicitly defined by $a''(z^*) = [a'(\mu) - a'(b+\Delta)] / [\mu - (b+\Delta)]$. Moreover, the magnitude of the interdependence is strictly decreasing in $a''(z^*)$.

Proposition 3 (Effect of altruism) *Depending on the curvature of a , the effects of altruism on the net incentive to avoid information and on the interdependence are:*

Shape of a	Net incentive $\partial\varphi^{Altr}/\partial\alpha$	Interdependence $\partial^2\varphi^{Altr}/\partial\lambda_{-i}\partial\alpha$
Concave	≥ 0	> 0
Linear	< 0	$= 0$
Convex	< 0	< 0

Table A.2: Effects of altruism

A.3.2 Reciprocity

We also extend the model to incorporate reciprocity concerns. An agent may derive utility from harming others if they avoid information and worsen her outcome, or from benefiting them if they become informed and help her. Formally, her reciprocity utility is given by $-(1 - q) \cdot \rho_i \cdot (y_i - \lambda_{-i})^2$, where ρ_i measures her sensitivity to reciprocity, $y_i = 1$ if she avoids information (and 0 if she acquires it), and q is the subjective belief that the state is Good. Note that, given that the payoff in state G is independent of avoidance y_j for all j , reciprocity concerns matter only for state B payoffs. Intuitively, a higher belief $(1 - q)$ that the state is Bad implies the agent perceives others' avoidance as more unkind.⁴¹ The net incentive to avoid information, φ_i , now includes an additional term reflecting reciprocity motives equal to $-(1 - q) \cdot \rho_i \cdot (1 - 2\lambda_{-i})$.

Lemma 5 (Effect of reciprocity) *For any $q \in (0, 1)$, reciprocity strengthens the strategic complementarity: $\partial^2\varphi_i/\partial\lambda_{-i}\partial\rho_i > 0$.*

Intuitively, on top of the baseline incentives to avoid information, reciprocity concerns introduce a motive to match the actions of others, creating a tendency towards strategic complementarity.

⁴¹Excluding $(1 - q)$ from the reciprocity term does not qualitatively affect our results.

B Additional analyses

All subsequent analyses are referenced in the document. Brief descriptions are provided for each analysis.

B.1 Descriptive statistics

Descriptive statistics of the experimental sample are reported below.

	N	Mean	SD	Median	Min	Max
<i>Demographic characteristics</i>						
Age	460	22.509	3.713	22.000	18.000	47.000
Gender (Female=1)	465	0.583	0.494	1.000	0.000	1.000
Has work experience	460	0.154	0.362	0.000	0.000	1.000
<i>General information preferences</i>						
IPS (overall scale)	464	3.116	0.414	3.111	1.000	4.000
<i>Social and economic preferences</i>						
Patience	464	-0.000	1.000	0.201	-3.493	1.124
Risk seeking	464	-0.000	1.000	0.296	-2.678	1.570
Positive reciprocity	464	-0.000	0.803	-0.017	-3.911	1.244
Negative reciprocity	464	0.000	0.836	0.117	-1.833	1.946
Altruism	464	0.000	0.836	0.096	-2.163	1.947
Trust	464	-0.000	1.000	-0.178	-1.662	2.048
<i>Field of study</i>						
Medicine	465	0.213	0.410	0.000	0.000	1.000
STEM	465	0.200	0.400	0.000	0.000	1.000
Humanities	465	0.228	0.420	0.000	0.000	1.000
Economics	465	0.108	0.310	0.000	0.000	1.000
Business	465	0.211	0.408	0.000	0.000	1.000
Other fields	465	0.056	0.230	0.000	0.000	1.000

Notes: The table shows descriptive statistics. The number of observations is lower for some variables elicited at the end of the session (e.g., age) because a few participants did not respond. For other variables (e.g., the IPS score and social preferences), one observation is missing due to a participant leaving the session early to attend a lecture.

Table B.1: Sample summary statistics

B.2 Are screams a bad? Preferences over screams and volume

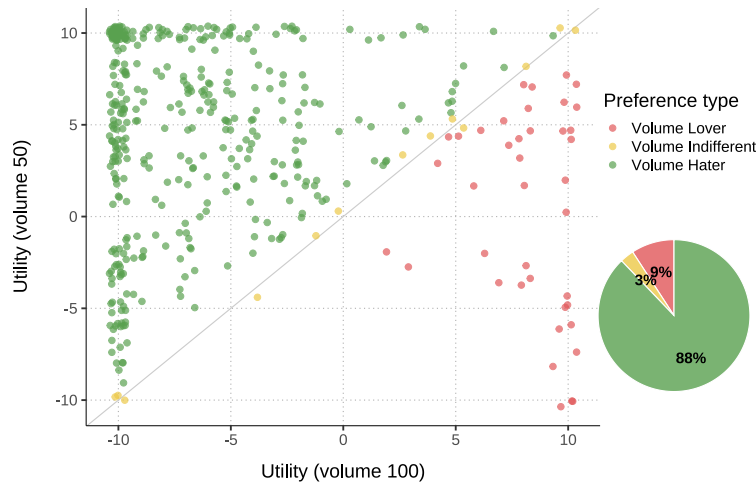
We classify “Volume Haters” (resp., “Lovers”) as subjects who report higher (lower) utility from hearing screams at volume 50 than at volume 100. Similarly, we classify “Scream Haters” (“Lovers”) as subjects who report a higher (lower) utility from learning that the state is Quiet than that it is

Screams. To capture the intensity of these preferences, we compute for each individual the difference between the utility of volume 50 and volume 100, and the difference between the utility of learning that the state is Quiet state relative to Screams.

Figure B.1 shows the joint distribution of subjects' reported utilities of hearing screams at different volumes, distinguishing each type of volume preference. In our sample, 88% of subjects are Volume Haters, 3% are Volume Indifferent, and 9% are Volume Lovers.

Figure B.2 presents the joint distribution of subjects' reported utilities of learning each state, distinguishing each type of scream preference. In our sample, 81% of subjects are Scream Haters, 6% are Scream Indifferent, and 14% are Scream Lovers.

Figure B.1: Preferences over volume of screams: reported utilities and preference types



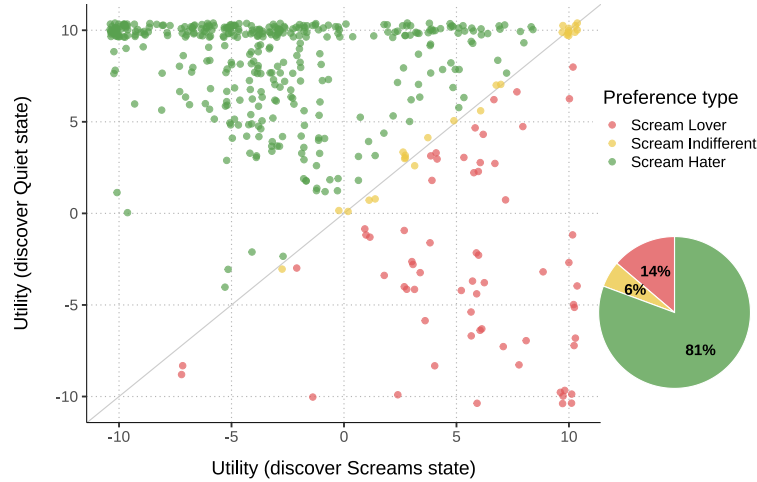
Notes: The figure displays reported utilities when screams were played at different volumes. Each point (jittered) corresponds to a participant's reported utilities for volume 50 (y -axis) and volume 100 (x -axis). The gray 45° line indicates indifference between volume levels. Observations are classified into three preference types: Volume Haters (green), who prefer quieter screams; Volume Indifferents (yellow), who report no difference; and Volume Lovers (red), who prefer louder screams. The pie chart summarizes the prevalence of each preference type in the sample.

The joint distribution of preference types over screams and volume is shown in Table B.2.

B.3 Balance checks

Table B.3 reports the treatment balance checks.

Figure B.2: Preferences over screams: reported utilities upon learning each state



Notes: The figure displays reported expected utilities upon learning the state of the world. Each point (jittered) corresponds to a participant's reported utilities for learning the Quiet state (y -axis) and the Screams state (x -axis). The gray 45° line indicates indifference across states. Observations are classified into three preference types: Scream Haters (green), who prefer to discover the Quiet state; Scream Indifferents (yellow), who report no difference; and Scream Lovers (red), who prefer to discover the Scream state. The pie chart summarizes the prevalence of each preference type in the sample.

Scream Preference Type	Volume Preference Type				Total
	Volume Lover	Volume Indifferent	Volume Hater		
Scream Lover	23	3	34	4	64
Scream Indifferent	1	5	19	1	26
Scream Hater	16	5	330	24	375
Total	40	13	383	29	465
Scream Lover	4.9%	0.6%	7.3%	0.9%	13.8%
Scream Indifferent	0.2%	1.1%	4.1%	0.2%	5.6%
Scream Hater	3.4%	1.1%	71.0%	5.2%	80.6%
Total	8.6%	2.8%	82.4%	6.2%	100.0%

Table B.2: Distribution of Scream- and Volume-preference types

	(1) Few Avoided	(2) Many Avoided	(3) Difference (2)-(1)
<i>Demographic characteristics</i>			
Age	22.53 (3.98)	22.49 (3.46)	-0.04 (0.35)
Gender (Female=1)	0.57 (0.50)	0.60 (0.49)	0.03 (0.05)
<i>Information preferences</i>			
IPS, overall scale	3.10 (0.42)	3.13 (0.41)	0.02 (0.04)
Health subscale	3.04 (0.72)	3.08 (0.68)	0.04 (0.06)
Finance	2.88 (0.67)	2.90 (0.63)	0.02 (0.06)
Personal	3.11 (0.55)	3.14 (0.55)	0.03 (0.05)
General	3.04 (0.70)	3.08 (0.70)	0.03 (0.06)
Occupational	3.29 (0.48)	3.30 (0.50)	0.01 (0.05)
<i>Field of study</i>			
Medicine	0.19 (0.40)	0.23 (0.42)	0.04 (0.04)
STEM	0.18 (0.39)	0.21 (0.41)	0.03 (0.04)
Humanities	0.19 (0.40)	0.26 (0.44)	0.07* (0.04)
Economics	0.14 (0.35)	0.08 (0.27)	-0.06** (0.03)
Business	0.25 (0.43)	0.18 (0.38)	-0.07* (0.04)
Other fields	0.06 (0.24)	0.05 (0.22)	-0.01 (0.02)
<i>Work experience</i>			
Has work experience	0.16 (0.37)	0.15 (0.36)	-0.01 (0.03)
Observations	223	242	465

* p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses.

Table B.3: Balance checks

B.4 Treatment effect on belief about prevalence of avoidance

Table B.4 reports estimates from linear regressions of beliefs on the treatment dummy and different sets of controls. Column (1) reproduces the results in Figure 4: on average, the expected share of avoiders among other group members is 21.9 percentage points higher in *Many Avoided* than in *Few Avoided*. This difference is highly significant and robust to the inclusion of the following controls: strata fixed effects (column 2), field of study fixed effects and individual characteristics (age and gender; column 3), and economic and social preferences (altruism, reciprocity, risk-seeking, patience, trust), as well as the IPS (column 4). Overall, the evidence confirms that the treatment successfully shifted beliefs about the prevalence of avoidance among other members.

	Dep. var: Belief about share of avoiders			
	(1)	(2)	(3)	(4)
Many Avoided	0.219*** (0.021)	0.215*** (0.025)	0.218*** (0.025)	0.221*** (0.025)
Control group mean	0.371*** (0.016)	0.471*** (0.072)	0.394*** (0.137)	0.494*** (0.163)
Strata FEs		✓	✓	✓
Field of study FEs			✓	✓
Individual characteristics			✓	✓
Preferences (economic, social, IPS)				✓
Observations	465	465	460	460

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Notes: The table shows the estimated coefficients of a regression of beliefs about the share of information avoiders among other members on the treatment dummy. The dependent variable is a continuous measure of beliefs, ranging from 0 (0% of the subject's group will avoid information") to 1 (100%). The treatment dummy is equal to one if the subject was in the *Many Avoided* condition and zero if in the *Few Avoided* condition. Individual characteristics include age and gender. "Preferences" include economic and social preferences (risk-seeking, time preferences, altruism, reciprocity, and trust) and the Information Preferences Scale (Ho et al., 2021), measuring an individual's general preference for obtaining information. Field of study fixed effects include dummies for fields of study. Strata-fixed effects include dummies for week of year and time of day. The number of observations drops in columns (3) and (4) because five participants did not provide information about their age.

Table B.4: Manipulation check (regressions)

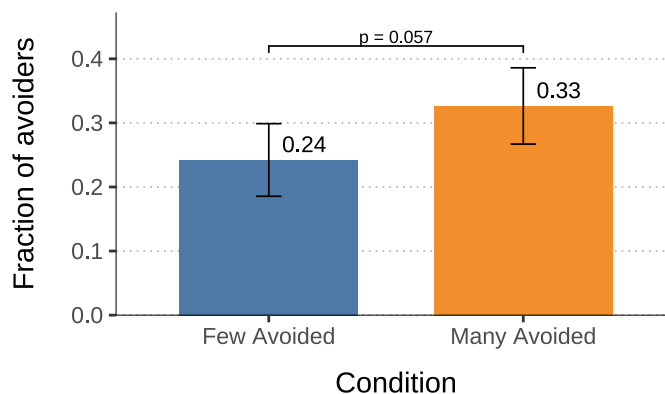
B.5 Treatment effect on information avoidance

This section presents further details on the treatment effect on information avoidance. Figure B.3 reports the raw treatment-control differences in information avoidance rates, showing a that the probability of avoiding information is higher in *Many Avoided* than in *Few Avoided*. This difference is statistically significant at 5%.

Table B.5 reports regression-based estimates of the treatment effect. Column 1 reports the unconditional treatment effect. As columns 2 to 5 show, the treatment effect is robust to the inclusion of individual characteristics, as well as fixed effects by strata and field of study (column 2; $\hat{\beta} = 0.109$, significant at 5%), and increases to 0.123 when all controls are included, significant at the 1% level (column 4). In this specification, the treatment effect corresponds to roughly half of the baseline prevalence of avoiders.

The estimate of the treatment effect is robust to a range of alternative specifications. Excluding Scream- and Volume-Lovers from the sample makes the effect even stronger (Appendix Tables B.6 and B.7). In addition, the estimate remains robust and significant across alternative specifications, regardless of whether the set of controls is selected by the researcher (Simonsohn et al., 2020) or via the data-driven post-double selection procedure (Belloni et al., 2014), as shown in Appendix Section B.7. Finally, we obtain qualitatively similar results when using as the outcome variable the IA scale (the continuous preference strength for information avoidance) (Appendix Section B.8).

Figure B.3: Treatment effect on information avoidance



Notes: The figure shows raw treatment-control differences on information avoidance (chose Later). Bars represent the fraction of avoiders under the *Few Avoided* (blue) and *Many Avoided* (orange) treatments. Error bars represent 95% confidence intervals.

The regressions in Table B.5 reveal that individual characteristics predict information avoidance. First, the Information Preferences Scale significantly predicts choices in our experiment: individuals with a stronger tendency to seek potentially undesirable information (higher IPS scores) are less likely to avoid information in our experiment.

Second, social preferences are also associated with information avoidance. More altruistic in-

	Dependent variable: Avoided information				
	(1)	(2)	(3)	(4)	(5)
Many Avoided	0.084** (0.042)	0.109** (0.049)	0.114** (0.049)	0.123** (0.049)	0.124** (0.049)
Information Preferences Scale			-0.125** (0.051)	-0.130** (0.052)	-0.118** (0.054)
<i>Social preferences</i>					
Altruism				-0.065** (0.028)	-0.052 (0.041)
× Many Avoided					-0.006 (0.057)
Positive reciprocity				-0.046 (0.029)	-0.101** (0.041)
× Many Avoided					0.109* (0.060)
Negative reciprocity				0.013 (0.026)	-0.005 (0.036)
× Many Avoided					0.030 (0.051)
Constant	0.242*** (0.029)	0.126 (0.225)	0.326 (0.301)	0.280 (0.311)	0.273 (0.321)
Strata FEs		✓	✓	✓	✓
Field of study FEs		✓	✓	✓	✓
Individual characteristics			✓	✓	✓
Observations	465	465	460	460	460

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Notes: The table reports estimates from a linear probability model regressing information avoidance on the treatment dummy. The dependent variable is a binary indicator equal to one if the subject avoid information (chose Later) and zero otherwise. The treatment dummy equals one if the subject was in the *Many Avoided* condition and zero in the *Few Avoided* condition. The regressions include strata fixed effects, demographic controls, field of study, and measures of economic and information preferences (see notes in Table B.4). The number of observations drops in column (3) to (5) since five participants did not provide information about their age.

Table B.5: Treatment effect on information avoidance: regressions

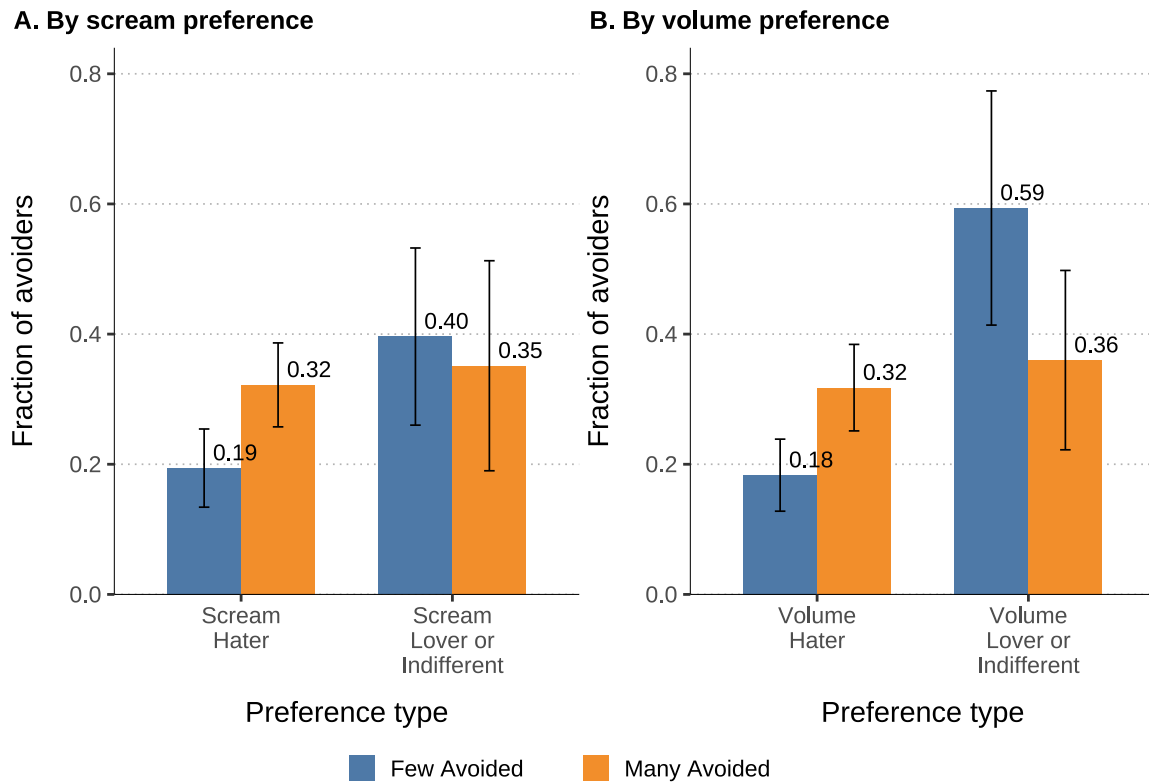
dividuals are less likely to avoid information ($\hat{\beta} = -0.068$), consistent with the idea that prosocial motives encourage information acquisition to benefit others. While the estimated coefficients cannot be interpreted causally, the observed patterns align with the theoretical predictions (Proposition 3 in Appendix A.3). For reciprocity, we observe a significant and positive interaction with the treatment: more reciprocal individuals are on average less likely to avoid in the *Few Avoided*, and more likely in the *Many Avoided* condition, as predicted by Lemma 5 in Appendix A.3. However, as Appendix Table B.6 and Table B.7 show, the estimated coefficients shrink when Scream- and

Volume-Lovers are excluded, becoming statistically insignificant, although the signs of the point estimates remain unchanged.

B.6 Robustness to exclusion of Scream- and Volume-Lovers

Here, we show that the main regression results are robust to the exclusion of Scream- and Volume-Lovers. Figure B.4 shows the treatment effect on information avoidance by type of preference for scream (panel A) and for volume (panel B). The treatment effect is positive for scream- and volume-haters, but the direction of the effect is is, if anything, the opposite for scream- and volume-lovers. This suggests that, by not excluding scream- and volume-lovers from our main regressions, the treatment effect is rather attenuated.

Figure B.4: Treatment effect on information avoidance by preference over screams and volume



Notes: The figure shows treatment effects on information avoidance (chose Later) conditional on individual preference types. Panel A splits participants by scream preference: Scream-Haters versus Scream-Lovers or -Indifferents. Panel B splits by volume preference: Volume-Haters versus Volume-Lovers or -Indifferents. For each subgroup, the bars report the fraction of avoiders under the *Few Avoided* (blue) and *Many Avoided* (orange) treatments. Error bars represent 95% confidence intervals.

Table B.6 shows our main regression results if we exclude Scream-Lovers. Table B.7 shows our main regression results if we exclude Volume-Lovers. In both subsamples, the magnitude of the estimated treatment effect increases, compared to the full sample (reported in Table B.5), for all the reported specifications. The treatment coefficient becomes significant at 1% for all specifications.

	Dependent variable: Avoided information				
	(1)	(2)	(3)	(4)	(5)
Many Avoided	0.128*** (0.045)	0.162*** (0.052)	0.157*** (0.052)	0.169*** (0.052)	0.170*** (0.053)
Information Preferences Scale			-0.126** (0.056)	-0.124** (0.059)	-0.114* (0.060)
<i>Social preferences</i>					
Altruism				-0.049 (0.031)	-0.072 (0.047)
× Many Avoided					0.050 (0.064)
Positive reciprocity				-0.030 (0.032)	-0.061 (0.046)
× Many Avoided					0.064 (0.066)
Negative reciprocity				0.031 (0.028)	0.037 (0.037)
× Many Avoided					-0.008 (0.054)
Constant	0.194*** (0.031)	-0.041 (0.240)	0.188 (0.324)	0.123 (0.340)	0.112 (0.352)
Strata FEs		✓	✓	✓	✓
Field of study FEs		✓	✓	✓	✓
Individual characteristics			✓	✓	✓
Observations	375	375	371	371	371

* p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses.

Notes: The table reports estimates from a linear probability model regressing information avoidance on the treatment dummy, restricted to the subsample of Scream-Haters. See notes in Table B.5.

Table B.6: Treatment effect on information avoidance (subsample of Scream-Haters)

	Dependent variable: Avoided information				
	(1)	(2)	(3)	(4)	(5)
Many Avoided	0.134*** (0.044)	0.159*** (0.050)	0.161*** (0.049)	0.166*** (0.050)	0.164*** (0.051)
Information Preferences Scale			-0.133** (0.054)	-0.132** (0.057)	-0.127** (0.058)
<i>Social preferences</i>					
Altruism				-0.062** (0.030)	-0.076* (0.045)
× Many Avoided					0.033 (0.061)
Positive reciprocity				-0.023 (0.030)	-0.037 (0.041)
× Many Avoided					0.032 (0.063)
Negative reciprocity				0.021 (0.027)	0.007 (0.035)
× Many Avoided					0.024 (0.054)
Constant	0.183*** (0.028)	-0.136 (0.162)	0.015 (0.270)	-0.088 (0.284)	-0.094 (0.297)
Strata FEs		✓	✓	✓	✓
Field of study FEs		✓	✓	✓	✓
Individual characteristics			✓	✓	✓
Observations	383	383	380	380	380

* p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses.

Notes: The table reports estimates from a linear probability model regressing information avoidance on the treatment dummy, restricted to the subsample of Volume Haters. See notes in Table B.5

Table B.7: Treatment effect on information avoidance (subsample of Volume-Haters)

B.7 Robustness to alternative specifications

Specification curve analyses To assess the robustness of our main treatment effect, we conducted a specification curve analysis (Simonsohn et al., 2020) by systematically varying sample restrictions, control sets, and model specifications. Specifically, we estimated the treatment effect across all combinations of four subsamples (full sample, excluding Scream-Lovers, excluding Volume-Lovers, excluding either) and twelve covariate sets (ranging from no controls to different combinations of demographics, field-of-study fixed effects, GPS and IPS measures). The results are presented in Figure B.5. Each point in the top panel represents the estimated effect size under an alternative specification, sorted by magnitude. Black dots indicate that the estimate is statistically significant at 5% level. The lower panels display the corresponding sample, control, and model choices. The main specification is highlighted with a red square. Figure B.6 show a similar plot but at level $\alpha = 0.01$.

Across specifications, the estimated average treatment effect remains consistently positive and is statistically significant in the majority of cases. Excluding subgroups (e.g., Volume-Lovers, Scream-Lovers) or adding or removing controls (demographics, IPS, GPS scales) does not qualitatively change the result. These findings indicate that the estimated treatment effect remains robust to a broad set of alternative specifications.

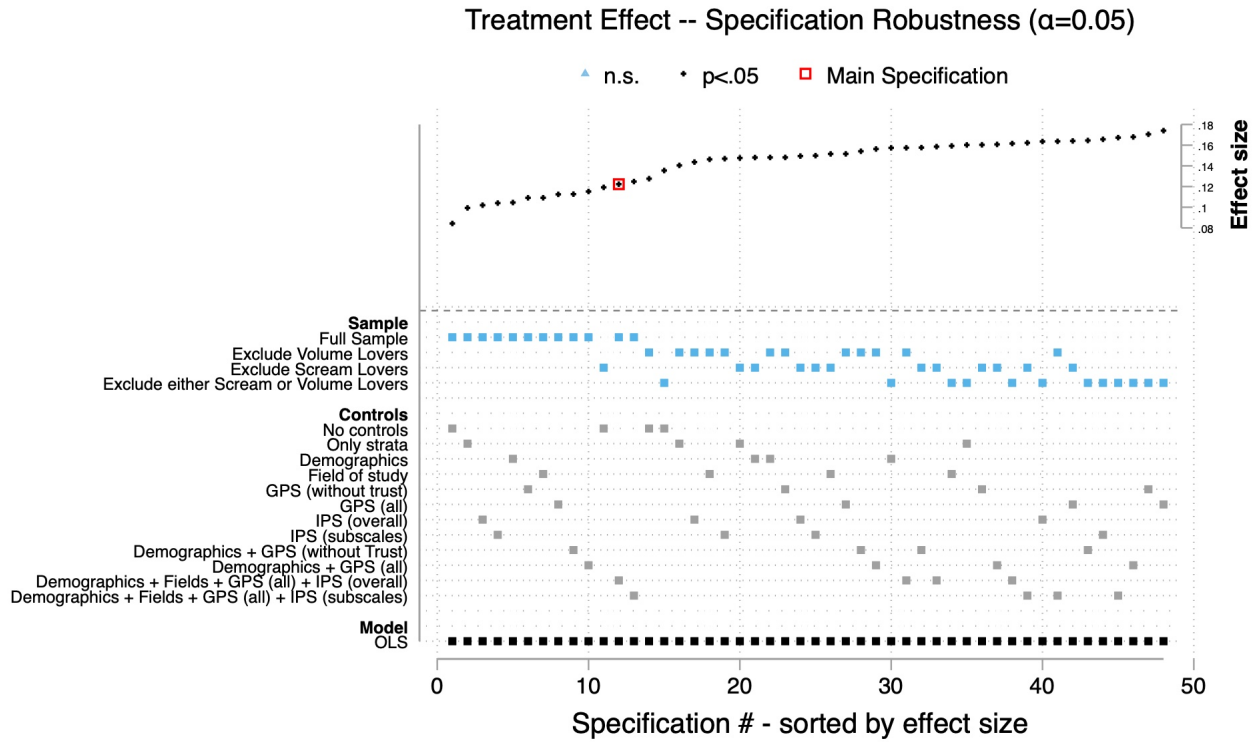
Post-double selection The specification curve analyses above rely on pre-defined subsets of covariates that are either included or excluded jointly as blocks. This design tests robustness across broad groups of control variables but lacks some flexibility: for instance, social preferences may be either included or excluded as a block, without considering combinations where altruism and reciprocity enter separately. To allow for such flexibility, we employ the post-double selection procedure (Belloni et al., 2014), a machine-learning-based, data-driven method for selecting controls from a potentially large set of covariates with the goal of causal estimation of parameters.

Results are reported in Table B.8. We estimate the treatment effect on information avoidance using both the plug-in method (columns 1–4) and cross-validation (columns 5–8). All specifications include strata fixed effects, and we progressively add four sets of controls: (i) baseline controls (economic and social preferences; IPS scales, subscales, and individual questions; field of study dummies; age; gender; and work experience), (ii) experimental context and comprehension controls (group size, perceived ease of instructions, and quiz difficulty), (iii) quiz performance until the information decision (questions answered and number correct), and (iv) quiz performance at the end of the experimental session (questions answered and number correct, performance level, and earnings).

Across all specifications, the estimated treatment effect remains positive and statistically significant, with magnitudes similar to those in the main regressions. These findings confirm that the

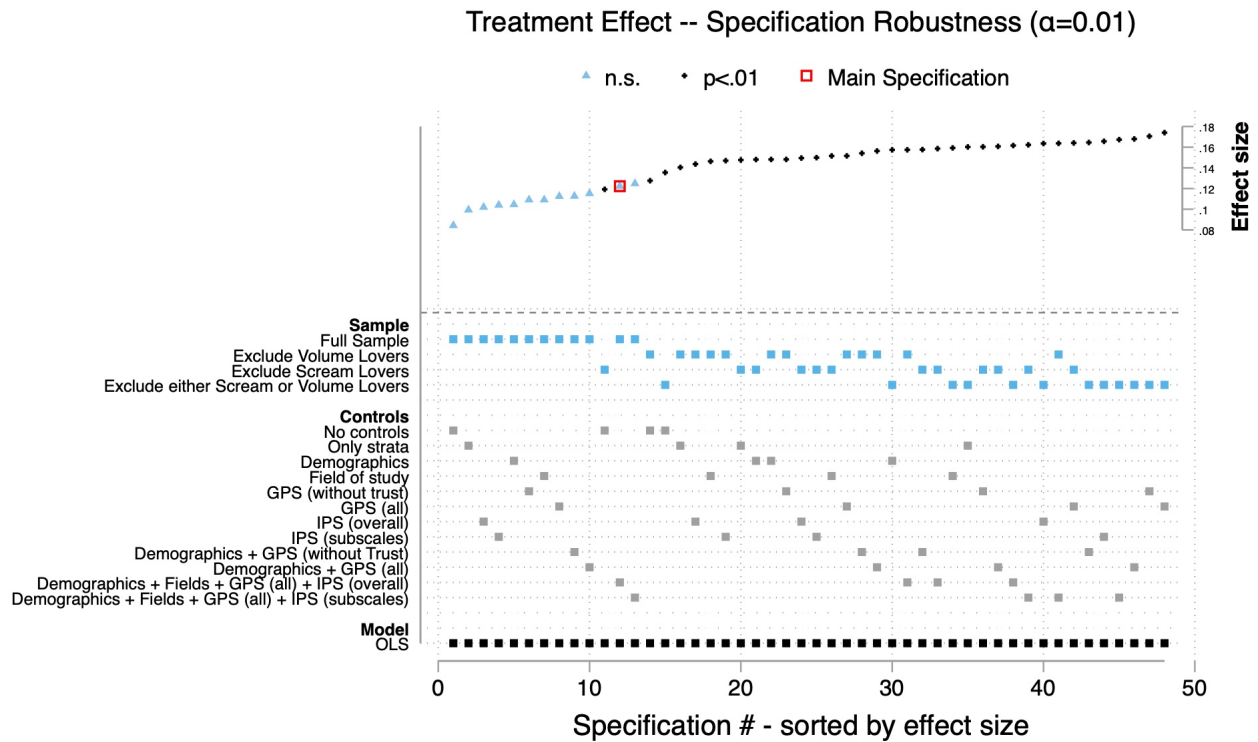
main result (higher information avoidance in the *Many Avoided* condition) is robust to alternative specifications where selection of controls is data-driven.

Figure B.5: Specification Curve Analysis ($\alpha = 0.05$)



Notes: The figure reports a specification curve analysis of the treatment effect. Each point represents the estimated effect under an alternative model specification, sorted by effect size. The top panel plots estimated effect sizes, with black dots indicating specifications significant at the 5% level, blue triangles indicating non-significant estimates, and the red square denoting the main specification reported in Section 5.2. The lower panels display the sample restrictions, control sets, and model choices corresponding to each specification. Effect sizes remain consistently positive across specifications, indicating robustness of the main result.

Figure B.6: Specification Curve Analysis ($\alpha = 0.01$)



Notes: The figure reports a specification curve analysis of the treatment effect. Each point represents the estimated effect under an alternative model specification, sorted by effect size. The top panel plots estimated effect sizes, with black dots indicating specifications significant at the 1% level, blue triangles indicating non-significant estimates, and the red square denoting the main specification reported in Section 5.2. The lower panels display the sample restrictions, control sets, and model choices corresponding to each specification. Effect sizes remain consistently positive across specifications, indicating robustness of the main result.

	Method: Plug-in				Cross-Validation			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Many Avoided	0.095* (0.050)	0.101* (0.056)	0.101* (0.056)	0.101* (0.056)	0.123** (0.049)	0.109* (0.056)	0.110* (0.056)	0.110* (0.057)
Strata FEs	✓	✓	✓	✓	✓	✓	✓	✓
Set 1: Baseline controls	✓	✓	✓	✓	✓	✓	✓	✓
Set 2: Context & comprehension		✓	✓	✓		✓	✓	✓
Set 3: Quiz performance A			✓	✓			✓	✓
Set 3: Quiz performance B				✓				✓

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Notes: The table reports post-double selection estimates of the effect of being in the Many Avoided condition on information avoidance. Columns (1)–(4) present results using the plug-in method, while columns (5)–(8) use cross-validation. All specifications include strata fixed effects and progressively add four sets of controls to the covariate selection algorithm: (i) baseline controls (age, gender, field of study dummies, social and economic preferences, Information Preferences Scale (IPS), and work experience), (ii) context and comprehension controls (group size, perceived ease of instructions, and quiz difficulty), (iii) quiz performance before making information decision (number of questions answered and number correct), and (iv) quiz performance at the end of experiment (number of questions answered and number correct, performance level, and earnings).

Table B.8: Post-double selection estimates

B.8 Alternative outcome: continuous measure of avoidance (IA Scale)

As a robustness check, we replicate our main analysis using the continuous measure of information avoidance (the “IA Scale”) rather than the binary indicator. Recall that the IA Scale captures the strength of preference for the chosen information decision, ranging from -10 (strong preference for acquiring information) to $+10$ (strong preference for avoiding).

Table B.9 reports the results. The overall pattern closely mirrors that of the main regressions with the binary outcome (Table B.5): signs of coefficients and statistical significance remain unchanged. Across specifications, the treatment effect of *Many Avoided* remains positive and significant. The coefficient is stable with the inclusion of controls and fixed effects, indicating that our main finding —that information avoidance increases under *Many Avoided*— remains robust to the choice of outcome measure.

Moreover, the results with IA Scale also echo the findings on social preferences documented earlier: altruism significantly predicts lower avoidance, while reciprocity remains directionally similar as before, but without though reaching statistical significance. The IPS remains negatively and significantly associated with strength of preference for avoiding information.

	Dependent variable: IA Scale				
	(1)	(2)	(3)	(4)	(5)
Many Avoided	1.271** (0.624)	1.610** (0.746)	1.677** (0.740)	1.846** (0.743)	1.838** (0.753)
Information Preferences Scale			-1.441* (0.772)	-1.487* (0.807)	-1.389* (0.834)
<i>Social preferences</i>					
Altruism				-1.094** (0.428)	-1.168* (0.638)
× Many Avoided					0.291 (0.873)
Positive reciprocity				-0.326 (0.444)	-0.855 (0.620)
× Many Avoided					1.098 (0.937)
Negative reciprocity				0.224 (0.405)	0.120 (0.586)
× Many Avoided					0.181 (0.820)
Constant	-3.821*** (0.431)	-4.525 (3.463)	-2.052 (4.706)	-2.815 (4.889)	-2.761 (5.065)
Strata FEs		✓	✓	✓	✓
Field of study FEs		✓	✓	✓	✓
Individual characteristics			✓	✓	✓
Observations	465	465	460	460	460

* p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses.

Notes: The table reports estimates from a linear regression of the IA Scale on the treatment dummy. The IA Scale captures the strength of preference for the chosen information decision, ranging from -10 (strong preference for acquiring information) to +10 (strong preference for avoiding). The treatment dummy equals one if the subject was in the *Many Avoided* condition and zero in the *Few Avoided* condition. Regressions include strata fixed effects, demographic controls, field of study, and measures of economic and information preferences (see notes in Table B.4). The number of observations drops in column (3) to (5) since five participants did not provide information about their age.

Table B.9: Treatment effect on IA Scale

B.9 Instrumental variable regressions (2SLS)

Table B.10 reports instrumental variable regressions (2SLS) of information avoidance on beliefs about the share of avoiders. Beliefs are instrumented with the random treatment assignment. The model specifications (columns 1-5) match those of the main regressions (Table B.5), except that the treatment dummy is replaced by the instrumented belief.

Once again, the overall pattern is very similar to that of the main regressions (Table B.5). Across specifications (columns 1–5), a higher belief that others avoid information significantly increases the likelihood of avoiding information. The estimated coefficient on beliefs is stable, positive, and statistically significant across specifications (ranging from $\hat{\beta}^{2SLS} = 0.384$ to 0.604).

The coefficients associated to social preference measures remain directionally similar. Altruism is associated with lower avoidance, while positive reciprocity reduces avoidance in the Few Avoided condition but reverses under the Many Avoided condition, although coefficients lose statistical significance relative to the main regressions.

Finally, the IPS is negatively associated with the decision to avoid information.

	Dependent variable: Avoided information				
	(1)	(2)	(3)	(4)	(5)
Belief about share of avoiders	0.384** (0.181)	0.504** (0.213)	0.523** (0.206)	0.556*** (0.201)	0.604** (0.239)
Information Preferences Scale			-0.123*** (0.045)	-0.122*** (0.046)	-0.065 (0.060)
<i>Social preferences</i>					
Altruism				-0.072*** (0.025)	-0.076 (0.144)
× Belief					0.030 (0.314)
Pos. reciprocity				-0.032 (0.028)	-0.296* (0.165)
× Belief					0.635 (0.386)
Neg. reciprocity				-0.000 (0.024)	-0.080 (0.124)
× Belief					0.091 (0.275)
Constant	0.100 (0.088)	-0.130 (0.232)	0.117 (0.282)	0.006 (0.288)	-0.209 (0.359)
Strata FEs		✓	✓	✓	✓
Field of study FEs		✓	✓	✓	✓
Individual characteristics			✓	✓	✓
Observations	465	465	460	460	460
First Stage F-stat	104.71	75.93	76.49	81.34	12.77

* p < 0.1, ** p < 0.05, *** p < 0.01. Robust standard errors in parentheses.

Notes: The table reports the second stage 2SLS estimates from regressing the information avoidance indicator (equal to one if the subject chose Later and zero otherwise) on the subject's reported belief about the share of information avoiders in his group, where the belief is instrumented by the treatment assignment dummy (equal to one if the subject was in the *Many Avoided* condition and zero otherwise). Depending on the specification, listed in columns 1 to 5, regressions include strata fixed effects, demographic controls, field of study, and measures of economic and information preferences (see notes in Table B.4). The number of observations drops in column (3) to (5) since five participants did not provide information about their age.

Table B.10: 2SLS estimates: effect of beliefs on information avoidance, using treatment assignment as instrument

B.10 Treatment effect on the distribution of strategy types (extensive margin)

	Dependent variable: Subject belongs to strategy type			
	(1) Always Getter	(2) Always Avoider	(3) Complement	(4) Substitute
Many Avoided	-0.065 (0.046)	0.014 (0.026)	0.009 (0.038)	0.036 (0.037)
Control group mean	0.457*** (0.034)	0.081*** (0.018)	0.202*** (0.027)	0.175*** (0.026)
Observations	465	465	465	465

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Notes: The table examines whether the treatment impacts the distribution of strategy types. It reports estimated coefficients from linear regressions where each column corresponds to a different strategy type. The dependent variable is a dummy equal to 1 if the subject's best-response schedule matches the given type and 0 otherwise. The independent variable is the treatment dummy (equal to 1 for the *Many Avoided* condition and 0 for the *Few Avoided* condition).

Table B.11: Treatment effect on distribution of strategy types (extensive margin)

B.11 Mechanisms: anticipatory utility (KP questions)

Table B.12 assesses the predictive validity and mediation properties of our Kreps-Porteus (KP) measures of anticipatory utility. Panel A examines predictive power; Panel B examines whether the treatment effect operates through anticipatory utility.

Panel A: Predictive power. Column (1) shows that controls alone explain only 6% of variation in information avoidance (adjusted $R^2 = 0.059$). Column (2) adds the Information Preferences Scale (IPS), a validated measure of general preferences for information. The IPS significantly predicts avoidance (individuals with higher willingness to know are less likely to avoid), but only modestly improves explanatory power ($R^2 = 0.071$).

Columns (3)-(4) replace the IPS with our KP measures of net anticipatory utility—continuous and dichotomous versions, respectively. Both measures are highly significant and positively associated with avoidance: higher net utility from avoiding predicts higher avoidance rates. Including the KP measures increases the R^2 to 0.185 (continuous) and 0.211 (dichotomous), nearly three times the explanatory power of the IPS. This suggests the KP measures capture decision-relevant aspects specific to our environment.

Columns (5)-(6) include both the IPS and KP measures. The coefficients on both remain stable, and the KP measures explain variance beyond the IPS. This pattern supports both the predictive and discriminant validity of our KP measures: they predict behavior independently of general information preferences and are not mere proxies for the IPS.

Panel B: Treatment and mediation. Panel B adds the treatment indicator. Column (1) shows the baseline treatment effect ($\beta = 0.122$, $p < 0.01$). Adding the IPS (column 2) leaves the treatment effect nearly unchanged, suggesting the IPS does not likely mediate the treatment effect. In contrast, adding the KP measures (columns 3-4) reduces the treatment coefficient by approximately 50% and renders it marginally significant or insignificant, suggesting the treatment operates through shifting anticipatory utility. Columns (5)-(6) confirm this pattern holds when controlling for both the IPS and KP measures.

	Dependent variable: Avoided information					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Predictive power						
Information Preferences Scale (IPS)		-0.129** (0.052)			-0.136*** (0.047)	-0.148*** (0.049)
Net utility of avoidance (continuous)			0.023*** (0.003)		0.023*** (0.003)	
Net utility of avoidance (dichotomous)				0.372*** (0.046)		0.379*** (0.045)
Controls and fixed effects	✓	✓	✓	✓	✓	✓
Observations	460	460	460	448	460	448
Adjusted R^2	0.059	0.071	0.185	0.211	0.199	0.226
Panel B: Treatment and utility						
Many Avoided	0.122** (0.049)	0.123** (0.049)	0.061 (0.047)	0.081* (0.045)	0.062 (0.046)	0.080* (0.045)
Information Preferences Scale (IPS)		-0.130** (0.052)			-0.136*** (0.047)	-0.147*** (0.048)
Net utility of avoidance (continuous)			0.022*** (0.003)		0.022*** (0.003)	
Net utility of avoidance (dichotomous)				0.365*** (0.047)		0.372*** (0.046)
Controls and fixed effects	✓	✓	✓	✓	✓	✓
Observations	460	460	460	448	460	448
Adjusted R^2	0.070	0.082	0.187	0.214	0.200	0.230

Notes: The table presents OLS estimates analyzing the predictive power and potential mediating role of the KP measures on actual information decisions. The dependent variable is an indicator equal to 1 if the subject chose to avoid information about the state (option Later) and 0 otherwise. Panel A assesses the predictive power of a number of measures. “Information Preferences Scale (IPS)” (Ho et al., 2021) measures overall preference to know potentially undesirable information; higher values indicate stronger preference to know. “Net utility of avoidance (continuous)”, $\hat{\phi}^i$, is the empirical measure of the net expected utility of avoiding information (relative to acquiring it), derived from the KP questions. “Net utility of avoidance (dichotomous)” is an indicator equal to 1 if the empirical net utility of is higher than zero, $\hat{\phi}^i > 0$. In Panel A, coefficients are standardized for comparability across measures. Panel B focuses on how the treatment dummy (“Many Avoided”) changes upon inclusion of the predictive measures; coefficients are not standardized for comparability with Table B.5. All regressions control for demographic characteristics (age and gender), economic and social preferences (patience, risk-seeking, trust, altruism, and positive and negative reciprocity), and fixed effects by strata and field of study of the participant. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B.12: Anticipatory Utility (KP Measures) and Information Avoidance

B.12 Mechanisms: supplementary tables

	Dependent variable: Avoided information				
	(1)	(2)	(3)	(4)	(5)
Belief about share of avoiders	0.090 (0.173)	0.122 (0.175)	0.098 (0.176)	0.085 (0.173)	0.063 (0.169)
Pr(screams)	-0.194 (0.150)	-0.176 (0.148)	-0.151 (0.148)	-0.156 (0.144)	-0.208 (0.145)
Belief about share of avoiders $\times$ Pr(screams)	0.852*** (0.294)	0.800*** (0.294)	0.787*** (0.297)	0.826*** (0.293)	0.909*** (0.292)
Constant	0.111 (0.076)	0.136 (0.166)	-0.116 (0.215)	0.125 (0.279)	0.029 (0.285)
Strata FEs		✓	✓	✓	✓
Field of study FEs			✓	✓	✓
Individual characteristics				✓	✓
Information preferences				✓	✓
Social and economic preferences					✓
Observations	465	465	465	460	460

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

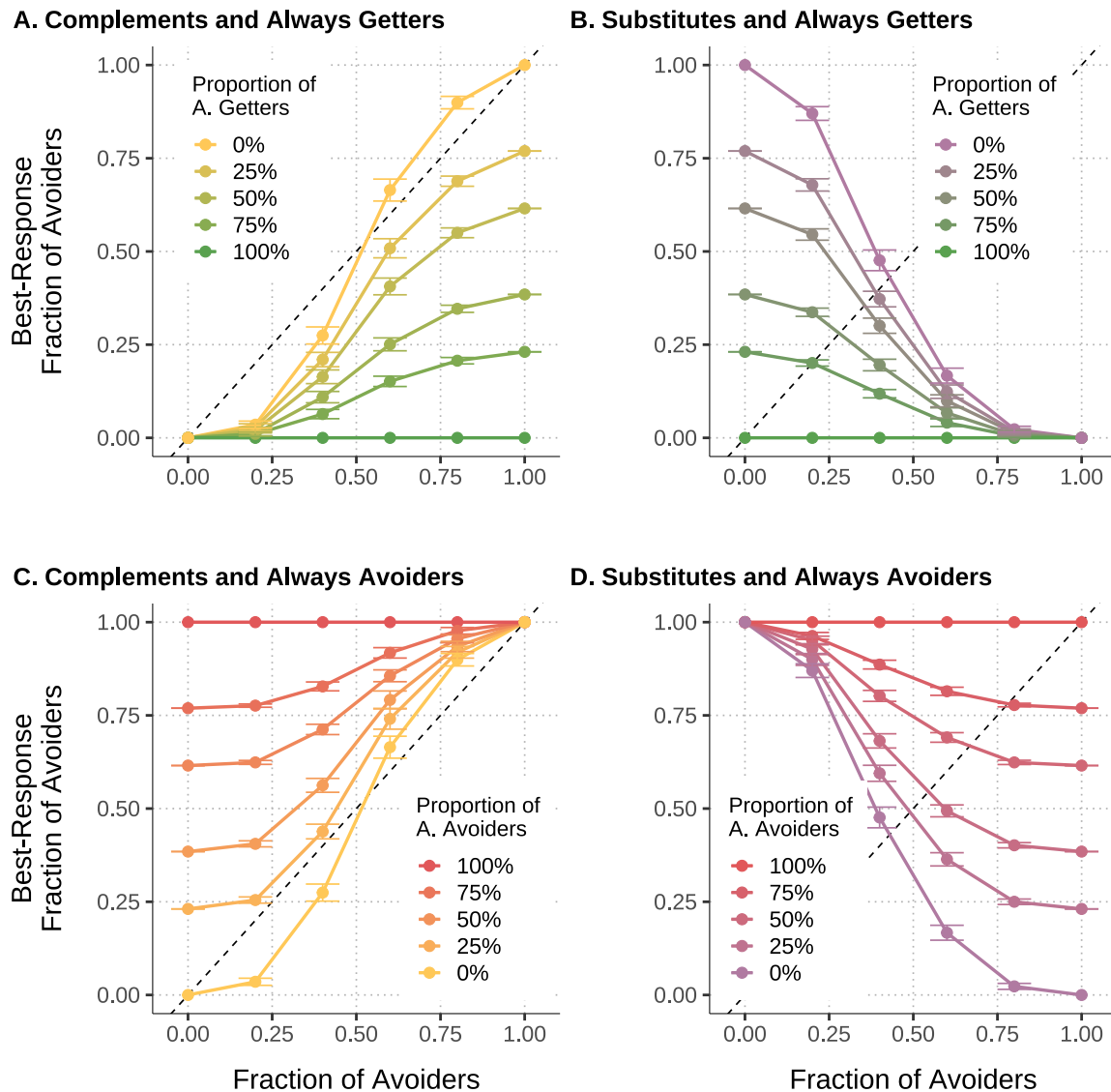
Notes: The table reports estimates from a linear probability model regressing information avoidance on the subject's belief about the share of information avoiders among others (i.e., who would choose Later), the subjective belief that the state is Screams ("Pr(screams)"), and their interaction. The regressions include a variety of controls; see notes in Table B.5.

Table B.13: Interaction of beliefs and information avoidance

C Group composition and equilibrium — additional analyses

C.1 Best-response schedules

Figure C.1: Effect of group composition on aggregate best-response schedules



Notes: The figure shows aggregate best-response schedules by group composition. Each panel focuses on mixtures of two strategy types. Lines show the average (over bootstrapped groups) aggregate best-response schedule of a given composition. Groups composed entirely of conditional types are shown in yellow (Strategic Complements) and purple (Strategic Substitutes). Within each panel, additional lines represent increasing shares of unconditional types—Always Getters (green) or Always Avoiders (red). Error bars denote 95% confidence intervals.

C.2 Higher-dimensional group compositions

In the main analysis, we focused on “restricted” sequences of group compositions, where only two strategy types were mixed. We now turn to richer cases involving more than two types. To visualize these, it is possible to map the richer composition space onto a unit probability triangle. Since Substitutes are highly likely to produce unstable equilibria, we omit them in this analysis. This allows redefining compositions as 3-dimensional probability vectors, which can be fully represented in the triangle.

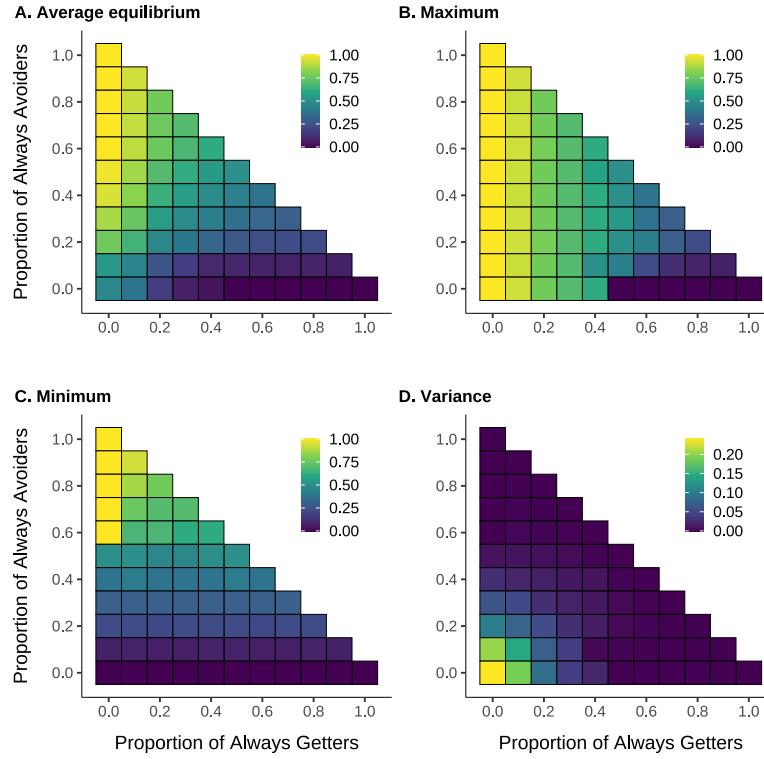
Figure C.2 illustrates this representation. The proportion of Always Getters is represented on the x -axis; and Always avoiders on the y -axis.⁴² Each cell corresponds to a unique group composition, associated with a distribution of fixed points (with examples visualized in the histograms above). Panel A presents, for each cell, the average of the set of stable equilibria. A group of pure Complements (the cell at $(0, 0)$) has an average close to 0.5. As we increase the share of Always Getters (moving horizontally along the x -axis) the average equilibrium share of avoiders decreases smoothly towards zero. Conversely, increasing the share of Always Avoiders (moving upward from $(0, 0)$ along the y -axis) pushes the average smoothly toward one. Overall, across the triangle, average equilibrium outcomes respond smoothly—although nonlinearly—to group composition.

Panel B shows the maximum equilibrium in the distribution, for each cell. Intuitively, if a policy-maker regards avoidance of information as an undesirable behavior, panel B shows the “worst” equilibrium in the distribution, for each cell. For groups composed entirely of Complements, the worst equilibrium corresponds to full avoidance, and the addition of Always Avoiders does not change this. By contrast, introducing Always Getters improves the worse-case outcome: the maximum equilibrium share of avoiders declines monotonically. Notably, a strong discontinuity emerges around $p_{AG} = 0.5$. More generally, such discontinuities—where a fixed amount of change in composition produces large, abrupt shifts in the worst-case equilibrium—appear across the composition space, and these are stronger the higher the proportion of Complements. These discontinuities cluster around $p_{AG} = 0.5$: when the proportion of Always Getters is sufficiently high, the aggregate best-response function shifts downward enough to eliminate any stable interior equilibria, generating the jump in the maximum. The discontinuities weaken as the share of Always Avoiders increases, as their presence helps sustain interior equilibria.

Panel C shows the minimum among the equilibria (the “best” equilibrium in the distribution), with patterns analogous to those for the maximum. Here, the relevant probability is p_{AA} . Finally, panel D exhibits the variance of the equilibrium set, which may matter for a risk-averse policy-maker. Naturally, the variance is highest in groups consisting of pure Complements, where multiple equilibria exist. Adding unconditional types flattens the aggregate schedule, progressively collapsing the distribution of equilibria (as shown in Figure 11) and hence reducing variance.

⁴²The proportion of Complements is implied by the complementary probability and visible as the horizontal (or ver-

Figure C.2: Group composition and summary features of equilibrium distributions



Notes: The figure illustrates how group composition affects summary features of the distribution of information equilibria. Each cell in the unit probability triangle represents a distinct group composition, defined by the proportion of Always Getters (x -axis), Always Avoiders (y -axis), and Strategic Complements (implied by the vertical, or horizontal, distance to the hypotenuse). Each panel reports a different feature of the equilibrium distribution—the average (A), maximum (B), minimum (C), or variance (D)—with values represented by the color gradient. Only stable equilibria are considered; Strategic Substitutes are excluded ($p_{SS} = 0$), as their equilibria are typically unstable.

tical) distance from the cell to the hypotenuse of the triangle.

D Instructions and other screens

Preliminary instructions

At the start of the session, all participants received a printed copy of the instructions below, which were also read aloud by the experimenter.

Welcome to the CeDEx Lab!

Thank you very much for participating in this scientific experiment.

This is an experiment on decision-making. By carefully reading the instructions that will be shown on your computer, you can —depending on the decisions you make— earn an appreciable amount of money. It is therefore important that you pay close attention to the instructions.

We are interested in your individual choices. Therefore, communication is not allowed during the experiment. If you have a question, please just raise your hand.

In addition, there are some ground rules that need to be complied with to ensure fair conditions for all participants:

- **Focus on your own screen.** You should neither interfere with other participants nor look at their workstations (including their screens). Please do not lean backwards and always keep your chair close to your desk.
- **Be patient.** There will be some periods in the experiment in which you will be asked to wait for some time (for example, you might finish your tasks earlier than other people; or you might be asked to just plainly wait). In any case, please remain in your workstation, waiting in silence, without disturbing other participants.
- **No phones.** Please, put your phone(s) in silence mode and put them away (for instance, put them in your coat or bag): do not leave them on your desk. Please do it now. During this session, you should not check or use your phone at any time. If you check or use your phone, you will be asked to leave the session. This is to ensure fair play conditions for all participants.
- **Headphones at all times.** In this experiment, you are asked to wear the headphones provided next to your computer at all times. This is because sounds may be played to you as we will explain later. If you don't properly wear your headphones, you will be asked to leave the experiment.

If you do not comply with any of these ground rules, you will be asked to leave the experiment. Throughout the experiment, some of us, experimenters, will be walking around invigilating for these conditions to be met.

ou will soon be given instructions about today's experiment. All instructions will be provided on your computer screen.

Note that participation in this study is entirely voluntary. You can leave the experiment at any time you want. If you leave, we will take into account your progress up to that point to calculate your payment and will pay you accordingly.

In case you have any questions, feel free to ask the experimenter at any time, in a very quiet voice so that no one else is disturbed.

Now, please put on your headphones and then we will be ready to start the experiment.

After these instructions were read aloud, subsequent instructions were presented on participants' computer stations, as described in the next subsection.

Main instructions and decision screens

The following text reproduces the instructions displayed to participants via the computer stations. Each new screen in the instructions is marked below by a horizontal line. We provide clarifying notes in square brackets in italics.

General instructions for participants

Welcome and thank you for participating in this scientific experiment.

First, you will be asked for some contact information. Then, you will be given the instructions on how to earn money in this experiment.

Contact Information

In the following, we will ask for the email address associated with your PayPal account, and your university email address.

This information will be stored only for payment and research purposes.

Please, **be sure that your email addresses are typed correctly. Otherwise, we may not be able to make the payment.**

Please input the email address associated to your PayPal account. [Open text box]

Please type it again. [Open text box]

Now input your university email address. [Open text box]

Input it again. [Open text box]

Please input your **seat number** (this is the number written in the plastic card that you have drawn when you entered into the lab).

[Open text box]

Your consent to participate

Please be informed that the present study involves the possibility of listening to sounds that have been rated as aversive (i.e., unpleasant or disturbing) by participants in other studies. The safety of the participants is a priority for the CeDEx research centre. Therefore, technical measures have been put in place to ensure the hearing safety of the participants.

In order to participate in today's study, your explicit consent to participate is needed. Please note that participation in this study is entirely voluntary.

If you do not give your explicit consent, you will withdraw from this study. There will be no further consequences, and we will keep inviting you to other studies. We will give you £3 as a token of appreciation for coming today.

- ☐ "I do NOT GIVE my explicit consent to participate"
 - ☐ "I GIVE my explicit consent to participate"
-

In this experiment, you and other participants in the room will be part of the same group.

Groups are typically composed of around 30 participants.

Introduction

In today's experiment, you will be asked to do some tasks, which will be explained as you progress through the session. Depending on your performance (and your performance alone) in those tasks, your earnings may increase.

For your participation in today's session, you will receive £5, plus additional earnings based on your performance in the tasks.

Instructions: The Quiz (1/2)

In this experiment, you will be participating in a Quiz. The Quiz questions cover different topics (for instance, sports, geography, history, arts, music, etc.).

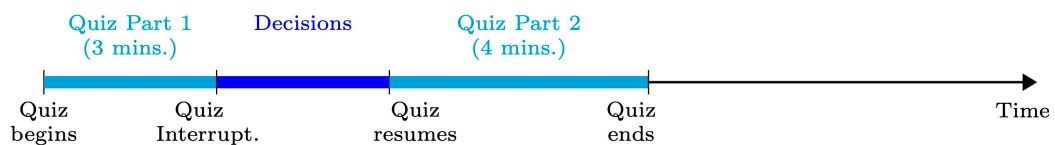
For each question, you will be provided with 4 possible answers, out of which exactly one will be correct.

In this Quiz task, **the more questions you answer correctly, the more you earn.**

The Quiz consists of **two parts**, namely Part 1 and Part 2.

- **Quiz Part 1:** You start with Part 1 of the quiz. You have 3 minutes to answer as many quiz questions as possible.
- **Quiz Interruption and Decisions:** After those 3 minutes, the Quiz will be briefly interrupted. During this interruption, you will be asked to make some decisions, as will be explained on the following screens.
- **Quiz Part 2:** After the interruption, you will answer Part 2 of the quiz, for 4 minutes.

The following figure illustrates the timeline:



The Quiz (2/2)

Below we explain how your earnings are calculated. The key feature is that **you should try to maximise the number of questions answered correctly**.

The number of correct answers in Parts 1 and 2 will be added up to calculate your earnings.

Your earnings increase in accordance with the level you reach. You start at Level 1. To reach Level 2, you need to answer 10 questions correctly (you do not need to correctly answer 10 questions in a row, what counts is the total number of correct answers). Then, to reach each additional level you need to answer correctly 20 extra questions per level. Thus, your earnings from the quiz are determined as follows:

- Level 1 (0-9 correct answers) = £5
- Level 2 (10-29 correct answers) = £6
- Level 3 (30-49 correct answers) = £7
- Level 4 (50-69 correct answers) = £9
- Level 5 (70-89 correct answers) = £13
- Level 6 (90+ correct answers) = £20

Your earnings from the quiz will be added to the £5 you will earn just for participating in this experiment.

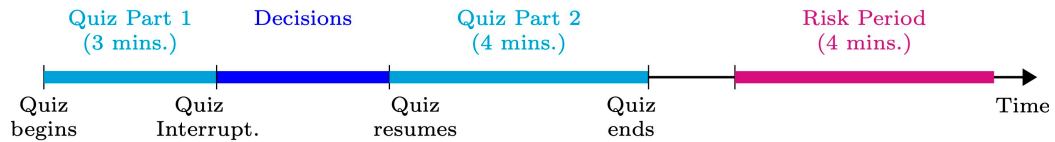
Example: suppose you answer correctly 25 questions in Part 1 and 35 in Part 2. Then the total number of correct answers is 60 ($=25+35$), so the level you reach is 4. Therefore, your earnings from the quiz would be £9. Adding up the £5 from your participation in this experiment, your total earnings would be £14 ($=£9 + £5$).

You can see that your earnings increase substantially, the higher the level that you reach.

The Risk Period

After the quiz finishes, a period called the “Risk Period” will start. The Risk Period lasts for 4 minutes.

During the Risk Period, a series of events may or may not happen as explained on the next screens.



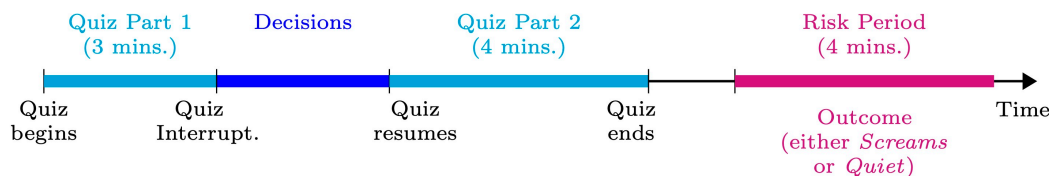
The Outcome

In the Risk Period, one of two outcomes will happen:

- ***Screams*** outcome: During the Risk Period, your group (including you) will hear a series of **unpleasant distress screams** from your headphones. These will happen at *unpredictable* moments during the Risk Period.

Previous research has shown that people consider similar screams as more aversive (disturbing) than the sound of nails sliding on a chalkboard.

- ***Quiet*** outcome: you and your group will not hear any screams during the whole experiment.



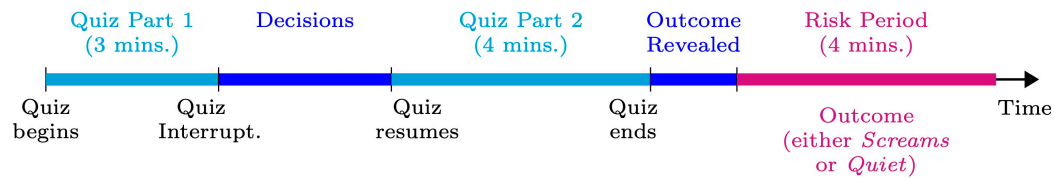
The Lottery

The Outcome (*Screams* or *Quiet*) in the Risk Period is the **same for everyone** in your group, and will be randomly determined by a computerised lottery.

Revelation of Outcome

The Outcome (i.e., *Screams* or *Quiet*) will be **revealed to everyone**, just a few screens **before the Risk Period**.

Therefore, everybody will already know the Outcome before the Risk Period starts.



Your Decision (1/3)

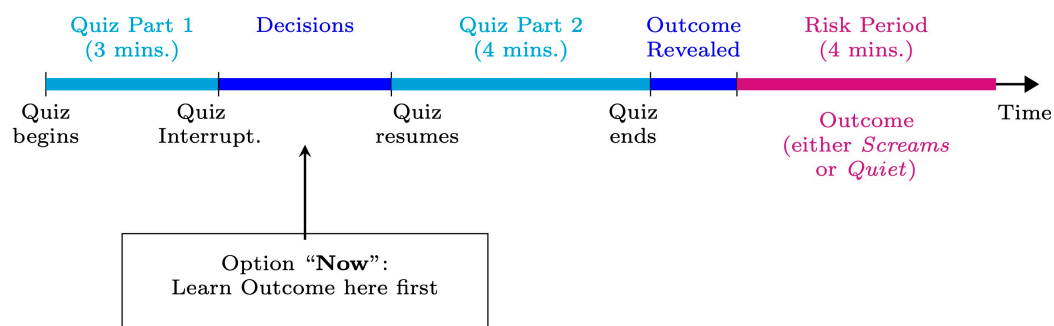
However, you can choose whether you want to know about the Outcome (*Screams* or *Quiet*) earlier or whether you prefer not to be aware of the Outcome until the beginning of the Risk Period.

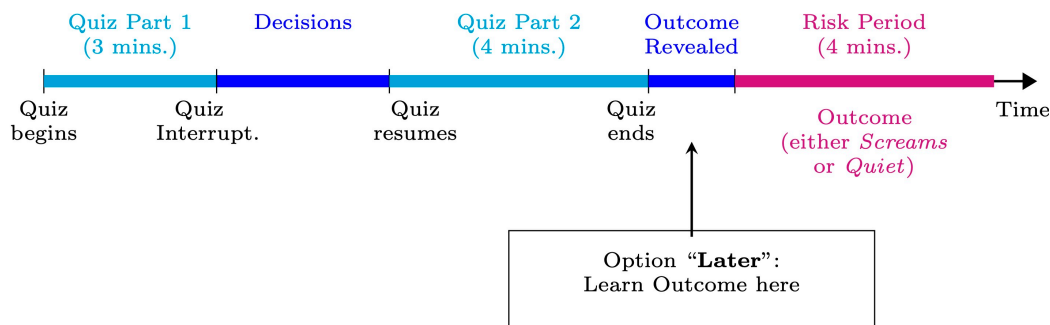
At the end of Part 1 of the quiz, you will choose one of two options:

- Option “**Now**” implies that you will learn the Outcome (*Screams* or *Quiet*) **immediately** (i.e., during the interruption of the quiz).
- Option “**Later**” implies that you will **not** learn about the Outcome (*Screams* or *Quiet*) immediately, and rather you will learn it just before the Risk Period.

This means that, during Part 2 of the quiz, you will not know whether in the forthcoming Risk Period you will hear the screams or not.

So the difference between “Now” and “Later” is **whether in Part 2 of the Quiz you are aware** of the Outcome or not. However, the Outcome is not affected by the option you choose.





The Volume of Screams

Although the Outcome is not affected by the option you choose, the **volume** of the screams (in case the Outcome is *Screams*) is affected by the choices that you and everyone else make:

- Each group member who chooses "Later" **increases the volume level by about 2 points out of 100.**
- Hence, if everybody chooses "Later", then the screams will be played at maximum volume (we call this level of volume "level 100").
- If nobody chooses "Later", then the screams will be played at volume "level 50".

This means that, if the Outcome is *Screams*, your choice between "Now" and "Later" would generate a difference in volume of about **2 points, out of 100.**

If instead the Outcome is *Quiet*, your choice has no implications beyond the timing in which you will find out the Outcome.

Check your Understanding

To check your understanding of the instructions, please imagine you are in each of the situations below and answer the following questions.

Situation 1a) Suppose that the Outcome is *Screams*. In addition, suppose that **everybody** in your group (including you) chooses "Later". What would be the volume of the screams?

- ☐ Level 50
- ☐ Level 52
- ☐ Level 98
- ☐ Level 100
- ☐ There would be no screams

Situation 1b) Suppose that the Outcome is *Screams*. In addition, suppose that **everybody else** in your group chooses “**Later**”, but **you** choose “**Now**”. What would roughly be the volume of the screams?

- ☐ Level 50
- ☐ Level 52
- ☐ Level 98
- ☐ Level 100
- ☐ There would be no screams

Situation 2a) Suppose that the Outcome is *Screams*. In addition, suppose that **everybody** in your group (including you) chooses “**Now**”. What would be the volume of the screams?

- ☐ Level 50
- ☐ Level 52
- ☐ Level 98
- ☐ Level 100
- ☐ There would be no screams

Situation 2b) Suppose that the Outcome is *Screams*. In addition, suppose that **everybody else** in your group chooses “**Now**”, but **you** choose “**Later**”. What would roughly be the volume of the screams?

- ☐ Level 50
- ☐ Level 52
- ☐ Level 98
- ☐ Level 100
- ☐ There would be no screams

Situation 3) Suppose that the Outcome is *Screams*. In that case:

- Your decision between “Now” and “Later” will...
 - ☐ Only affect the volume of the screams to be heard by you
 - ☐ Only affect the volume of the screams to be heard by participants other than you
 - ☐ Affect the volume of the screams to be heard by everyone (including you)
- The decisions between “Now” and “Later” **by participants other than you** will...
 - ☐ Only affect the volume of the screams to be heard by you
 - ☐ Only affect the volume of the screams to be heard by participants other than you
 - ☐ Affect the volume of the screams to be heard by everyone (including you)

Situation 4) Suppose that the Outcome is *Quiet*. In addition, suppose that **everybody** in your group (including you) chooses “Later”. What would roughly be the volume of the screams?

- ☐ Level 50
- ☐ Level 52
- ☐ Level 98
- ☐ Level 100
- ☐ There would be no screams at all

When you are ready, press the “Continue” button below.

(In case you are not taken to the next screen: some of your answers to the questions above might be wrong, please review them).

Instructions: Your Decision (2/3)

As mentioned, the difference between “Now” and “Later” is whether in Part 2 of the Quiz you are aware of the Outcome or not, and whether the volume of the screams (if the Outcome is Screams) changes by 2 points out of 100.

However, your choice between “Now” and “Later” does **not** change any of the following:

- the duration of the session;

- the sequence of the quiz;
- the amount of time you can spend on the quiz;
- when or whether you may potentially hear the screams.

Finally, note that for this choice between “Now” and “Later”, there is no right or wrong choice. You should simply decide based on your own preferences.

Instructions: Your Decision (3/3)

Recall that if you choose “Now”, then you will be shown a screen that informs you about the Outcome.

Upon being informed about the Outcome, you will be asked to select a button that correctly reflects it. You will need to do it to proceed to the next screens, which include Part 2 of the Quiz.

Differently, if you choose “Later”, then you will be shown a screen that will ask you to wait, and you will not learn the Outcome until Quiz Part 2 is finished, just before the Risk Period.

Control Questions

To briefly check that you understood the instructions correctly, please answer the following questions.

- If you choose option “Now”...
 - ☐ You will learn the Outcome immediately
 - ☐ You will not learn the Outcome immediately, but rather just before the Risk Period
 - ☐ You will never learn the Outcome
- If you choose option “Later”...
 - ☐ You will learn the Outcome immediately
 - ☐ You will not learn the Outcome immediately, but rather just before the Risk Period

- You will never learn the Outcome
- If the Outcome is *Screams*, the **timing** in which you will hear the screams...
 - will be earlier if you choose option “Now”
 - will be earlier if you choose option “Later”
 - will be the same independently of whether you choose “Now” or “Later”
- If the Outcome is *Screams*, the **volume** of the screams...
 - will be 2 points higher if you choose option “Later”
 - will be 2 points higher if you choose option “Now”
 - will be the same independently of whether you choose “Now” or “Later”
- Just before the Risk Period starts, you will be informed about the Outcome...
 - Only if you have chosen “Now”
 - Only if you have chosen “Later”
 - In any case, independently of whether you have chosen “Now” or “Later”
 - You will not be informed about the Outcome at that point

When you are ready, press the “Continue” button below.

(In case you are not taken to the next screen: some of your answers to the questions above might be wrong, please review them).

A Pilot Study

[Note: The text below was shown to participants in the Many Avoided condition. The version for the Few Avoided condition differed, with deviations indicated in square brackets.]

The information below gives you an idea of the choices of other people in a similar situation.

We ran a pilot study similar to this experiment, and found that, out of a set of 10 University of Nottingham students, **8 [2] chose the option “Later”**, preferring not to find out the Outcome until the end of Part 2. Therefore, in that group, **the overwhelming majority [minority] chose “Later”**.

If people in your group behave similarly to those in the pilot study, you can expect around 23 [6] people other than you to choose Later, and around 6 [23] to choose Now. In that case, the resulting volume level would roughly be 89 [60] if you choose Now, and 91 [62] if you choose Later.

Out of any 10 members of your group (*other than you*), how many do you believe will choose “Later”?

[Slider ranging from “0 out of 10 will choose “Later”” to “10 out of 10 will choose “Later””]

You are now ready to start the quiz

From the next screen, the quiz will start.

When you feel ready to start, press “Continue”, and both the quiz and the 3-minute timer will start at once.

Quiz

Please, answer the following questions to achieve a higher level and earn a higher payment.

[List of multiple choice quiz questions is displayed.]

Your quiz time has ended.

Please, press the Continue button.

The quiz has been interrupted. You will now be asked to make a series of decisions, including your preferred choice between knowing your lottery Outcome “Now” and “Later”.

What do you think the Outcome of your group's lottery will be? (*Screams* or *Quiet*)

[Radiobuttons (10 levels), with extremes labeled as "Definitely Quiet" and "Definitely Screams".]

[Note: this screen elicits the Kreps-Porteus questions (KP questions)]

Before choosing between "Now" and "Later", please answer the following questions about how happy you would be in different (hypothetical) situations.

- Imagine that you choose "Now", and then a moment later discover that the Outcome is Screams. How happy would you be in that moment *thinking about the Outcome ahead*?

[Slider ranging from -10="Very unhappy" to 10="Very happy".]

- Imagine that you choose "Now", and then a moment later discover that the Outcome is Quiet. How happy would you be in that moment *thinking about the Outcome ahead*?

[Slider ranging from -10="Very unhappy" to 10="Very happy".]

- Imagine that you choose "Later", and therefore you do not learn the Outcome until later. How happy would you be in that situation *thinking about the Outcome ahead*?

[Slider ranging from -10="Very unhappy" to 10="Very happy".]

Do you want to know the Outcome "Now" or "Later"?

Choose whether you want to know "Now" or "Later" whether the screams are upcoming.

- ☐ Later
- ☐ Now

How strong is your preference for your selected option, over the other?

[Radiobuttons (10), ranging from “I am **indifferent** between the two options” to “I have a **very strong** preference for the selected option”.]

A quick question

You have chosen the option [“Now”/“Later”]. Why did you prefer this option?

Please take a moment to answer this question.

[Open response textbox.]

Instructions: The Decisions Table

One participant will have the opportunity to **condition their decision on what the other participants chose** in the previous screen. This participant will be selected randomly by the computer.

Suppose you are that randomly selected participant. Then, the decision between “Now” and “Later” that you complete in the table on the following screen will be implemented for you.

See the image below for an example. In the table, you have the possible choices that other participants have made in the previous screen. Each row represents a different situation. **For each row, you can decide which option you would prefer.** Then, according to what the other players have *actually* chosen in the previous screen, the relevant row of the table will be determined, and the option selected in that row of the table will actually be implemented for you.

You have a positive chance of being selected as this player. Therefore, you must complete the table carefully according to your preferences.

(Please scroll down to see the ‘Continue’ button)

Out of 10 other participants, those who chose <i>Later</i> are...	In that situation, you prefer to know the Outcome...	Volume of screams if you choose Now	Volume of screams if you choose Later
0	Now Later	50	52
2	Now Later	60	62
4	Now Later	69	71
6	Now Later	79	81
8	Now Later	89	91
10	Now Later	98	100

Please, select your preferred option for each of the following situations.

Out of 10 other participants, those who chose <i>Later</i> are...	In that situation, you prefer to know the Outcome...	Volume of screams if you choose Now	Volume of screams if you choose Later
0	Now Later	50	52
2	Now Later	60	62
4	Now Later	69	71
6	Now Later	79	81
8	Now Later	89	91
10	Now Later	98	100

You will now be asked a series of questions. To answer them, please imagine you are in each of the situations presented.

Questions (1/5)

[Note: This and following screens elicit the Information Preferences Scale (IPS; [Ho et al., 2021](#))]

Please, imagine you are in each of the following situations, and answer the questions:

1. As part of a medical checkup, your doctor asks you a series of questions. The answers to these questions can be used to estimate your life expectancy (the age you are predicted to live to). Do you want to know how long you can expect to live?
 - ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
2. You provide some genetic material to a testing service to learn more about your ancestors. You are then told that the same test can, at no additional cost, tell you whether you have an elevated risk of developing Alzheimer's (a brain disorder that slowly destroys memory and thinking skills and, eventually, the ability to carry out the simplest tasks). Do you want to know whether you have a high risk of developing Alzheimer's?
 - ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
3. At your medical checkup, you are given the option to see the results of a diagnostic test, which can identify the extent to which your body has suffered long-term effects from stress. Do you want to know how much lasting damage your body has suffered from stress?
 - ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know

Questions (2/5)

4. Ten years ago, you had the opportunity to invest in two retirement funds: Fund A and Fund B. For the past 10 years, you have invested all your retirement savings in Fund A. Do you want to know the balance you would have if you had invested in Fund B instead?

- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
5. You decide to go to the theater for your birthday and give your close friend (or partner) your credit card so they can purchase tickets for the two of you, which they do. You aren't sure but suspect that the tickets may have been expensive. Do you want to know how much the tickets cost?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
6. You bought an electronic device at a store at what seemed like a reasonable, though not particularly low, price. A month has passed, and the item is no longer returnable. You see the same device displayed in another store with a sign announcing "SALE." Do you want to know the price you could have bought it for?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know

Questions (3/5)

7. You gave a close friend one of your favorite books for her birthday. Visiting her apartment a couple of months later, you notice the book on her shelf. She never said anything about it; do you want to know if she liked the book?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
8. Someone has described you as quirky, which could be interpreted in a positive or negative sense. Do you want to know which interpretation he intended?

- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
9. You gave a toast at your best friend's wedding. Your best friend says you did a good job, but you aren't sure if he or she meant it. Later, you overhear people discussing the toasts. Do you want to know what people really thought of your toast?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
10. As part of a fundraising event, you agree to post a picture of yourself and have people guess your age (the closer they get, the more they win). At the end of the event, you have the option to see people's guesses. Do you want to learn how old people guessed that you are?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
11. You have just participated in a psychological study in which all of the participants rate others' attractiveness. The experimenter gives you an option to see the results for how people rated you. Do you want to know how attractive other people think you are?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know

Questions (4/5)

12. Some people seek out information even when it might be painful. Others avoid getting information that they suspect might be painful, even if it could be useful. How would you describe yourself?

- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
13. If people know bad things about my life that I don't know, I would prefer not to be told.
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know

Questions (5/5)

The questions below ask about what you would decide in the workplace. If you have previously worked in a company, imagine yourself in it and answer the questions. If you had not worked for a company, read the following questions as imagining you are working in a company.

14. Employers commonly encourage workers to complete health risk assessments that gauge their health characteristics and risk. These are usually used to estimate workplace-wide risks the company may face. If you had the opportunity, would you like to get your individual risk assessment?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know
 - ☐ Probably want to know
 - ☐ Definitely want to know
15. Most people spend some time at work on activities they are not proud of (e.g., browsing social media). Suppose you could compose a list of these activities and have your computer track how much time you spend doing them. The information would not be shared with your employer. Would you like to know what fraction of each hour at work you spend on these activities?
- ☐ Definitely don't want to know
 - ☐ Probably don't want to know

- Probably want to know
 - Definitely want to know
16. As part of a unit-wide evaluation, you and your coworkers are asked how easy it is to get along with each other and what your strengths and weaknesses are. Would you like to know how your coworkers rated you?
- Definitely don't want to know
 - Probably don't want to know
 - Probably want to know
 - Definitely want to know
17. The rise of artificial intelligence is likely to lead to job losses in a wide range of occupations, affecting workers in industries from long-distance trucking to healthcare. Would you like to know whether your job is at risk owing to automation in the next 10 years?
- Definitely don't want to know
 - Probably don't want to know
 - Probably want to know
 - Definitely want to know
18. One option in your employer's retirement plan allows you to compare your investment return with that of your coworkers. Would you like to know how your investment return compares with the average and highest investment returns earned by employees at your firm?
- Definitely don't want to know
 - Probably don't want to know
 - Probably want to know
 - Definitely want to know

Please answer the following questions about how happy you would be in different (hypothetical) situations.

- Imagine that the Outcome is *Screams*, and that the **volume of the screams** will be **level 50**. How happy would you thinking about the Outcome ahead?
[Slider from -10="Very unhappy" to 10="Very happy"]

- Imagine that the Outcome is *Screams*, and that the **volume of the screams** will be **level 100**. How happy would you thinking about the Outcome ahead?
[Slider from -10=“Very unhappy” to 10=“Very happy”]
-

Thank you for answering the questions.

In the next screen, you will get to know the Outcome if you selected option Now, or remain without knowing it, if you selected option Later. If you were selected as the random player, then the option you selected in the relevant row of the table will be implemented.

*[Note: this screen was displayed to participants whose implemented option was **Now**]*

Based on your previous decisions, the choice implemented for you is “Now”.

The Outcome is **Screams** [Quiet].

Therefore, you **will hear the screams** [not hear any screams] during the Risk Period.

Please, indicate below which is the Outcome.

- ☐ Screams
 - ☐ Quiet
-

*[Note: this screen was displayed to participants whose implemented option was **Later**]*

Based on your previous decisions, the choice implemented for you is “Later”.

You may or may not hear the screams during the Risk Period. You will be informed about the Outcome later, just before the Risk Period starts.

Press Continue to proceed to Quiz Part 2.

Quiz

Please, answer the following questions to achieve a higher level and earn a higher payment.

[The same list of multiple choice quiz questions is displayed.]

The Quiz has ended.

You will be asked a series of questions, and then move to the Risk Period.

Questions (1/5)

[Note: This and following screens elicit social and economic preferences from the Global Preferences Survey (GPS; [Falk et al., 2018](#))]

We now ask for your **willingness to act** in a certain way in four different areas.

Please indicate your answer on a scale from 0 to 10.

How willing are you to give up something that is beneficial for you today in order to benefit more from that in the future? [Slider: 0=“Completely unwilling to do so”, 10=“Very willing to do so”.]

In general, how willing or unwilling are you to take risks? [Slider: 0=“Completely unwilling to take risks”, 10=“Very willing to take risks”.]

Questions (2/5)

1. When someone does me a favor I am willing to return it.

[Slider: 0=“Does not describe me at all”, 10=“Describes me perfectly”.]

2. Please think about what you would do in the following situation:

You are in an area you are not familiar with, and you realize you lost your way. You ask a stranger for directions. The stranger offers to take you to your destination. Helping you costs the stranger about £20 in total. However, the stranger says he or she does not want any money from you. You have six presents with you. The cheapest present costs £5, the most expensive one costs £30. Do you give one of the presents to the stranger as a "thank-you" gift? If so, which present do you give to the stranger?

- ☐ No present
- ☐ Present worth £5
- ☐ Present worth £10
- ☐ Present worth £15
- ☐ Present worth £20
- ☐ Present worth £25
- ☐ Present worth £30

Questions (3/5)

If I am treated very unjustly, I will take revenge at the first occasion, even if there is a cost to do so. [Slider: 0="Does not describe me at all", 10="Describes me perfectly".]

How willing are you to punish someone who treats others unfairly, even if there may be costs for you? [Slider: 0="Completely unwilling to do so", 10="Very willing to do so".]

Questions (4/5)

Imagine the following situation: Today you unexpectedly received £1,000. How much of this amount would you donate to a good cause? [Open text box; inputs restricted to numbers between 0 and 1000]

How willing are you to give to good causes without expecting anything in return? [Slider: 0="Completely unwilling to do so", 10="Very willing to do so".]

Questions (5/5)

"I assume that people have only the best intentions." [Slider: 0="Does not describe me at all", 10="Describes me perfectly".]

Instructions: the Risk Period (1/2)

Thanks for kindly answering the questions.

You are now about to enter the Risk Period.

Before the Risk Period starts, two things will happen:

1. In the next screen, you will be provided with a sample scream, to get an idea of the kind of screams that may happen during the Risk Period.
2. After that, everybody will be informed about: a) which is the Outcome (*Screams* or *Quiet*); and b) the volume of the screams as the result of the group's choices, if the Outcome is *Screams*.

After that, the Risk Period will start. It will last 4 minutes, and you will be provided with the answers to the quiz to review if you want to.

Pre-experiencing a sample scream (1/2)

To give you an idea of what the screams sound like, on the next screen, you will hear a sample scream, similar to the screams you would hear if your lottery outcome were *Screams*.

On the next screen, you will hear only 1 scream, whereas during the Risk Period you may hear more than 1 scream if your lottery Outcome is *Screams*, and 0 if your lottery outcome is *Quiet*.

To ensure your safety, the volumes of the screams have been calibrated to comply with the safe hearing levels recommended in the UK.

Please, keep your headphones on at all times. You are also reminded you may leave the session at any time you would like.

Please, try to remain silent throughout.

Now, if you wish to proceed, press Continue.

Pre-experiencing a sample scream (2/2)

To hear the sample scream, press the “Play” button. From the moment you press the “Play” button below, the sample scream will be played at a random moment within 30 seconds.

You will be able to hear the sample scream only once. In other words, the button will not do anything from the second time you press it.

Please, try to remain silent throughout.

After the 30 seconds, the Continue button will appear. You will be able to proceed by pressing it.

[Play button]

The Outcome, revealed

[If the lottery Outcome was Screams, the following text was displayed:]

Your group’s lottery Outcome is **Screams**.

Therefore, your group **will hear the screams** during the Risk Period.

The volume level of the screams is [**volume level**]. This is the result of the choices of all participants.

[If the lottery Outcome was Quiet, the following text was displayed:]

Your lottery outcome is **Quiet**.

Therefore, you will **not hear any screams** during the Risk Period.

Risk Period

The Risk Period has started. Please, remain in silence until it ends.

In the meantime, you can read the list of quiz questions and answers below.

At the end of the Risk Period, a “Continue” button will appear, at the bottom. (You may need to scroll down)

Quiz Answers

Below is the list of quiz questions. The correct answers to these are underlined.

[List of all quiz questions and their answers is displayed]

Thank you for your patience. The Risk Period has ended.

You will now be asked a few final questions.

Please, press the Continue button to proceed.

Final Questions (1/4)

Please, kindly answer the following questions.

- How easy to understand were the instructions? [Radiobuttons (5): “Very easy” to “Very hard”]
 - How hard to answer were the quiz questions? [Radiobuttons (5): “Very easy” to “Very hard”]
 - Do you have any experience at work, even casually? [“Yes” / “No”]
-

Final Questions (2/4)

- All in all, how do you feel about this study? [Open text box]

- Is there anything else you would want to tell us? [Open text box]
-

Final Questions (4/4)

- What is your age? [Open text box, restricted to numbers]
 - What is your gender? [Female/Male/Other]
-

You have answered $[Q]$ questions correctly. Therefore, you have reached performance level $[P]$. In consequence, you have earned $\pounds[e]$ from the quiz.

Adding the $\pounds5$ for your participation, your **total earnings** from the experiment are $\pounds[e + 5]$.

You will be contacted about your payment in the next couple of days.

Thanks for your participation

We appreciate that you took the time to come and participate in today's experiment. Thanks a lot, and hope to see you around soon!

Your session is ending now. You can now leave the lab. Please, do it in silence, so other participants are not disturbed.